

Sudden Stops and Wealth Inequality: The Severity of Downturns*

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Abstract

Wealth inequality deepens the contractions in GDP, domestic absorption, and asset prices that accompany sudden stops, which are recessions marked by excessive capital outflows. Cross-country data show that outflows of the same size generate larger contractions during sudden stops in countries with greater wealth inequality. European household survey data show that household leverage rises with wealth inequality and that only the leveraged households cut debt and assets together. I build a tractable model in which heterogeneous households face borrowing limits tied to the market value of their assets. The model rationalizes the empirical facts as greater inequality places more households at the borrowing limit and deepens the declines in asset prices and aggregate consumption following an adverse aggregate shock. A quantitative model that extends the tractable model to infinite horizon demonstrates that a 0.01 increase in the wealth Gini produces additional GDP and consumption declines of 0.07 and 0.11 percentage points per percentage point of capital outflow during a sudden stop in the model, compared with 0.09 and 0.07 percentage points in the data. Using the quantitative model, I show that an optimal taxation on wealth and external debt with commitment improves social welfare by 17.38% relative to the equilibrium without taxes.

Keywords: Capital flows, Fisherian debt deflation, Optimal taxation, Sudden stops, Wealth inequality

JEL Classification Codes: E21; E44; E63; F38; F41; F44

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1 Introduction

Deep recessions in open economies often coincide with capital outflows. These episodes are also known as sudden stops, which are crises marked by abrupt reversals of the current account, collapsing domestic absorption, and falling asset prices (Calvo (1998); Mendoza (2010)). Although these movements are aggregate, the adjustment is carried out through individual balance sheets. Some households must repay debt and liquidate assets, while others are in a position to absorb those sales. This paper studies how the distribution of wealth across these balance sheets shapes the severity of the aggregate downturn.

The canonical explanation for why recessions are extremely deep in sudden stops is the Fisherian debt deflation mechanism. A tightening of borrowing limits forces indebted households to sell assets. The resulting decline in asset prices erodes collateral values, tightens borrowing limits further, and sets off additional deleveraging. Wealth inequality can either strengthen or dampen the decline in asset prices. Greater wealth inequality can enlarge the mass of households at the borrowing limit and intensify the selling pressure on one hand. On the other hand, it can also concentrate resources among unconstrained households and increase their capacity to absorb the sales. Which force dominates determines whether inequality amplifies or dampens the contraction generated by the same capital outflow.

This paper shows that the amplification force dominates. Greater wealth inequality expands the leveraged side of the wealth distribution and makes aggregate asset demand less responsive to falling prices. A given capital outflow consequently requires a larger decline in asset prices, which tightens borrowing limits for a larger fraction of households and deepens the contraction in aggregate consumption. I establish this mechanism using cross-country and household data, characterize it analytically in a tractable model, and quantify its aggregate implications in an infinite-horizon economy. I then use the quantitative model to study the optimal taxation of wealth and external debt under commitment.

The cross-country evidence comes from an annual panel of 54 countries from 1979 to 2016. I identify sudden stops as sharp current-account reversals that coincide with elevated global financial stress, following Korinek and Mendoza (2014). I measure wealth inequality using household net wealth Gini coefficients from the World Inequality Database. I regress the growth of GDP,

consumption, investment, and stock prices on capital outflows during sudden stops and their interaction with wealth inequality. Outflows of the same size generate substantially larger contractions when wealth is more unequally distributed. Moving from the median to the 75th percentile of wealth inequality approximately triples the GDP response and doubles the responses of consumption, investment, and stock prices. The result is specific to the distribution of wealth, and income inequality does not amplify contractions in GDP or consumption.

European household data identify the balance-sheet adjustment behind this cross-country pattern. I use the Household Finance and Consumption Survey (HFCS), which covers fifteen countries before and after the sudden stops in Greece, Spain, Italy, and Portugal in 2012. The survey reveals two further facts. Household debt relative to assets rises with wealth inequality, both at the country level and across individual households. The contraction in asset demand during the crisis is concentrated among the households that carry this debt.

Indebted households reduce asset demand when a sudden stop occurs. Mortgagors have leverage ratios two to four times those of households without a mortgage in the HFCS data, so I use mortgage status as a proxy for household indebtedness. Relative to mortgagors in countries without a sudden stop, mortgagors in the crisis countries reduce outstanding debt by roughly one quarter. Their holdings of real estate fall by 23% to 27%, and their financial assets fall by 15% to 28%. They also stop acquiring assets. The share that originated a mortgage within the preceding three years falls by 9 to 13 percentage points, while the share that recently purchased its main residence falls by 5 to 8 percentage points. Outright owners display no comparable changes in their holdings or purchases.

The cross-country panel identifies where sudden-stop contractions are largest, while the household data identify whose balance sheets contract. Together, they imply that wealth inequality affects more than the concentration of assets at the top. It is associated with a larger leveraged side of the distribution, whose asset demand depends on continued access to credit. When external financing contracts, a larger fraction of households must deleverage, and the same capital outflow withdraws asset demand from a larger part of the economy.

I formalize this mechanism in a tractable three-period small open economy. Households differ in initial wealth, trade domestic capital and foreign bonds, and face borrowing constraints that limit debt to a fraction of the market value of their assets. Wealth determines who is constrained.

There is an endogenous threshold of net wealth below which households borrow up to the limit. A negative productivity shock lowers household income and pushes households near the threshold into the constrained region. These households reduce asset demand to deleverage. The asset price falls to clear the market, eroding collateral values and forcing further deleveraging.

Wealth inequality governs the strength of this feedback through both sides of the asset market. A mean-preserving spread of the wealth distribution places more households below the wealth threshold before the shock arrives. More households therefore reduce asset demand when the shock hits, while fewer unconstrained households remain to absorb their sales. Constrained households also have less price-elastic asset demand than unconstrained households. Greater inequality thus increases the direct selling pressure and reduces the elasticity of aggregate asset demand at the same time. I show analytically that the responses of both the asset price and aggregate consumption to a productivity shock are larger under a more unequal wealth distribution. The larger price decline further tightens borrowing limits and transmits the shock to a larger fraction of households. The mechanism also applies to a tightening of the borrowing limit, and the numerical analysis produces similar amplification following an increase in the world interest rate.

The tractable model takes the initial wealth distribution as given in order to characterize the inequality comparative statics analytically. I endogenize the distribution in an infinite-horizon model with aggregate uncertainty. Households differ permanently in productivity, while representative capital producers transform aggregate investment into capital. The economy is subject to shocks to aggregate productivity, the world interest rate, and the tightness of the borrowing constraint. I simulate the model economy to study macroeconomic dynamics in sudden stops. In a typical sudden stop, aggregate demand and asset price all plummet. To quantify the role of wealth inequality, I simulate economies with different degrees of permanent productivity dispersion under the same aggregate shocks. In regressions with the simulated data, a 0.01 increase in the wealth Gini adds 0.074 and 0.105 percentage points to the GDP and consumption contractions generated by a one-percentage-point capital outflow. The corresponding estimates in the data are 0.086 and 0.070 percentage points.

Household heterogeneity also allows the model to reproduce the countercyclicality of external debt relative to GDP. The correlation between net external debt relative to GDP and output is -0.141 in the data and -0.281 in the heterogeneous-household model. A representative-

household version of the same model instead produces a correlation of 0.125. Productive households hold most external debt and can finance investment partly from their current income and accumulated capital. Their debt positions therefore move less than aggregate output. Homogeneous households adjust their debt more strongly with aggregate investment, causing debt to move more than output.

The model's distributional dynamics are also consistent with the household evidence. Net wealth falls more for households at the bottom of the distribution during a sudden stop. These households have few asset buffers and borrow up to the limit to smooth the decline in income. Households at the top are better insured by their assets and reduce debt when borrowing costs rise. Their deleveraging prevents net wealth from falling as much as it does for poorer households. The model thereby reproduces the changes in net assets across wealth groups observed in the European survey.

Using the quantitative model, I study optimal taxation in the presence of pecuniary externalities generated by market-determined collateral values. Individual households take the capital price as given and therefore do not internalize how their asset choices jointly affect collateral values, the borrowing capacity of other households, and the distribution of resources across heterogeneous balance sheets. I consider a utilitarian government that chooses taxes on wealth and external debt under commitment. The two instruments operate on distinct margins. The debt tax discourages borrowing that must be rolled over when borrowing constraints tighten. The wealth tax redistributes resources across capital positions but also lowers the after-tax return on investment. Optimal policy, therefore, balances the redistributive motive against the incentive to carry capital across states.

Around a sudden stop, the two optimal taxes move in opposite directions. The debt tax rises sharply, whereas the wealth tax falls. By committing to a high debt tax in states with tight borrowing limits, the government makes debt costly to roll over in those states and induces households to enter them with less external debt. At the same time, it lowers the wealth tax to preserve the return to capital when productivity and collateral capacity are already depressed. Relative to the competitive equilibrium without taxes, joint taxation lowers external debt one period before the sudden stop from 47.3% to 27.8% of output. The consumption contraction falls from 24.2% to 18.2%, while the fraction of constrained households rises by only 2.4 percentage points rather

than 61.4 percentage points.

These gains do not arise from stabilizing the capital price. The price falls by 28.7% without taxes and by 32.3% under the optimal taxes on both wealth and external debt. The government allows for the larger price decline because cheaper capital redistributes resources toward households for which additional wealth has greater social value. The optimal taxation nevertheless raises social welfare by 17.38%. Optimal capital controls without a wealth tax stabilize consumption and the capital price more strongly but raise welfare by only 10.03%. Adding the wealth tax produces substantially larger gains among low-wealth households at the cost of larger losses among wealthy households. In summary, the welfare gain from joint taxation reflects the redistribution of resources to poor households, the reduction of external debt before the sudden stop, and the containment of the increase in constrained households rather than the stabilization of the asset price drop in sudden stops.

Outline This paper is organized as follows. I review the literature in Section 2. I present the empirical relationship between wealth inequality and sudden stops in Section 3 and rationalize the facts using the tractable model in Section 4. I illustrate the quantitative model in Section 5 and conduct the quantitative analysis of the role of wealth inequality in sudden stops in Section 6. I proceed to study the government's optimal taxation in Section 7 and conclude in Section 8.

2 Literature Review

This paper relates to a broad literature on the macroeconomic consequences of inequality. Galor and Zeira (1993) show that the initial distribution of wealth affects human-capital investment and long-run output when credit markets are imperfect, while Persson and Tabellini (1994) link inequality to growth through the political economy of redistribution. More recently, Mian, Straub, and Sufi (2021) show how rising income inequality increases household indebtedness and depresses aggregate demand. I contribute to this literature by establishing wealth inequality as a key determinant of macroeconomic vulnerability to capital outflows.

The paper is closely related to the literature on inequality and the propagation of aggregate shocks. Beginning with Krusell and Smith (1998), quantitative heterogeneous-agent models have

examined how household heterogeneity matters for aggregate dynamics. Kaplan, Moll, and Violante (2018) and Auclert (2019) show that heterogeneity in liquidity, balance-sheet exposures, and marginal propensities to consume changes the transmission of monetary policy, while Bayer, Born, and Luetticke (2024) quantify the interaction between inequality and the business cycle. At the intersection of inequality and financial crises, Kumhof, Rancière, and Winant (2015) show how rising income inequality can generate household leverage and crisis risk, and Paul (2023) finds that rising top-income inequality predicts financial crises and deeper subsequent recessions. These two papers primarily study how inequality contributes to the incidence of crises. This paper instead studies how wealth inequality affects their severity, showing that capital outflows generate larger contractions when wealth is more unequally distributed.

The paper also contributes to the literature on sudden stops. Calvo (1998) studies how an abrupt withdrawal of international credit can produce financial and balance-of-payments crises. Mendoza (2010) develops a model in which collateral constraints generate Fisherian debt deflation, and Bianchi (2011) shows that market-determined collateral values create a pecuniary externality and excessive external borrowing. These papers abstract from the distribution of wealth across domestic households. Recent work instead emphasizes firm dynamics and exchange rate adjustment (Castillo-Martinez (2025)) or the role of production networks in propagating financial crises (Miranda-Pinto, Rojas, Saffie, and Silva (2025)). Among studies of household heterogeneity, Villalvazo (2026) is closest to this paper. Villalvazo (2026) identifies the opposing effects of asset sales by leveraged households and asset purchases by wealthy households and finds that lower inequality produces less severe sudden stops. I provide cross-country evidence on how wealth inequality affects macroeconomic dynamics in sudden stops and also present European household evidence underlying the relationship. My model rationalizes how a mean-preserving spread of wealth magnifies the declines in asset prices and aggregate demand.

The normative analysis builds on the literature on macroprudential regulation of capital flows. Bianchi (2011), Bianchi and Mendoza (2018), and Korinek (2018) show that private agents fail to internalize the effect of their borrowing on market prices and collateral values, providing a rationale for the prudential regulation of external debt. Biljanovska and Vardoulakis (2024) introduce redistribution between workers and entrepreneurs into a model of sudden stops and studies how this motive changes optimal policy. I allow for a full distribution of household wealth and

jointly characterize taxes on wealth and external debt. The two instruments separate redistribution across capital positions from the management of external debt.

Finally, the paper relates to the literature on household heterogeneity in open economies. de Ferra, Mitman, and Romei (2020) show that household leverage, portfolio composition, and foreign-currency debt determine the consumption effects of a current-account reversal. Auclert, Rognlie, Souchier, and Straub (2021) study how household heterogeneity amplifies the real-income channel of exchange-rate movements, while Oskolkov (2023) examines how wealth and sectoral exposure shape the distributional effects of exchange-rate policy. Ferrante and Gornemann (2025) study how deposit dollarization exposes heterogeneous households to devaluations through bank balance sheets. On the empirical side, Guntin, Ottonello, and Perez (2023) document consumption adjustment across the income distribution during aggregate crises, and Verner and Gyöngyösi (2020) show that the revaluation of household foreign-currency debt depresses spending and local economic activity. I contribute to this strand of studies by demonstrating household-level evidence of the adjustment of debt and assets during sudden stops and connecting the evidence to the amplification of sudden stops under greater wealth inequality.

3 Empirical Analysis

I document three facts associated with wealth inequality and the vulnerability of an economy when a sudden stop occurs. First, recessions in sudden stops are deeper in more unequal countries. Second, households in more unequal countries carry more debt relative to their wealth, both in aggregate and individually. Third, the decline in asset demand during sudden stops is concentrated among indebted households. These findings together suggest that greater wealth inequality increases the share of indebted households and amplifies contractions in asset demand and domestic absorption in sudden stops.

3.1 Data

I use two different sources of data for empirical analysis. First, I use an annual panel of 54 countries during the period from 1979 to 2016 to analyze the differential effects of wealth inequality on sudden stop dynamics. To be specific, I combine the data from Bianchi and Mendoza (2020),

which consists of GDP, consumption, investment, the current account, the stock market index, and the real exchange rate of the countries. I combine this with the wealth inequality data from the World Inequality Database (WID), where wealth inequality is measured as the household net wealth Gini coefficient. As additional country-level control variables, I also use trade openness measured as a fraction of trade in GDP, collected from the World Bank World Development Indicators, the IMF financial development index, debt-to-GDP and reserves-to-GDP ratios calculated by Lane and Milesi-Ferretti (2018), and the financial account openness index constructed by Chinn and Ito (2006).

I follow the convention in the literature (Korinek and Mendoza (2014); Bianchi and Mendoza (2020)) to define a sudden stop episode. A sudden stop episode is a country-year observation with (i) the year-on-year growth rate of the current account to GDP ratio being higher by two standard deviations than the mean in each country's longitudinal distribution and (ii) VIX and JP Morgan EMBI aggregate spreads being higher by two standard deviations than the mean for advanced economies and emerging market economies, respectively. I classify 47 countries as having experienced sudden stops in the sample period and provide a full list of sudden stop episodes in Table A1 of Appendix A.1.

In addition, I use data from the Household Finance and Consumption Survey (HFCS) conducted by the European Central Bank (ECB) to study household-level asset and debt holdings. The data consists of household-level disaggregated information on each household's income from different sources and various types of assets. To be specific, it is a repeated cross-sectional survey of households from 15 countries in 2011 and 2014.¹ Among the 15 countries in the sample, sudden stops occurred in Greece, Italy, Portugal, and Spain in 2012 during the European debt crisis. Thus, the sample captures snapshots of households before and after the crisis.

3.2 Fact 1: Wealth Inequality Amplifies Contractions in Sudden Stops

I first document that sudden stops incur larger drops in aggregate demand when wealth is disproportionately distributed among households. To provide a motivating example, I compare the cyclical movements of GDP and consumption per capita around a sudden stop in countries with

¹I exclude data from countries that were collected only in 2014. The nominal values of household wealth and income are adjusted to constant 2014 values, taking into account the differences in the cost of living across countries and inflation. I describe the adjustment procedure in detail in Appendix A.2.

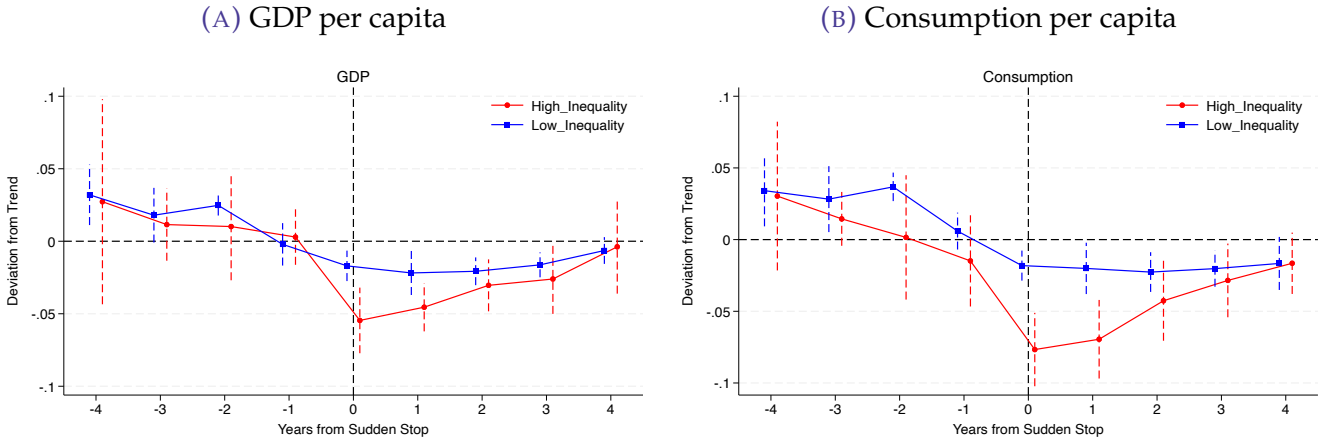


FIGURE 1. Dynamics around Sudden Stops

Notes: Cyclical components of GDP and consumption per capita are estimated using HP filter by each country. Dotted lines are 90% confidence intervals computed using Cameron et al. (2011) two-way clustered standard errors.

high and low degrees of wealth inequality by running a distributed lag regression specified as

$$y_{it} = \sum_{h=-4}^4 \beta^h SS_{i,t+h} + \alpha_i + \eta_t + \varepsilon_{it}, \quad (1)$$

separately for countries with high and low degrees of inequality. High inequality corresponds to the case when a country i 's wealth Gini is higher by one standard deviation than the cross-country average in year t , while low inequality is the case when a country i 's wealth Gini is lower by one standard deviation compared to the cross-country average in that year. The dummy variable $SS_{i,t+h}$ is 1 if a country i experienced a sudden stop in the year $t + h$. Estimates of β^h are plotted in Figure 1.

Countries with higher wealth Gini coefficients experience larger drops in GDP and consumption per capita during sudden stops, as shown in Figure 1. The drops are more persistent than those in countries with a lower wealth Gini coefficient. GDP per capita falls by roughly 2% in a low-inequality country during sudden stops, whereas it falls by 5% in a high-inequality country. Differences in consumption per capita are more pronounced, with consumption falling by 2% in a low-inequality country, while it drops by 7% in a high-inequality country.

I proceed to analyze the differential effects of wealth inequality on sudden stop dynamics through a series of regressions. To be specific, I study the effect of capital outflows during sudden stops on aggregate demand and asset prices. While regression (1) treats all sudden stops identically as a dummy variable, I now take into account that each crisis can have different magnitudes by multiplying the ratio of capital net outflow to GDP with the sudden stop dummy. Capital

net outflow is measured as the difference between capital inflow and outflow data of Forbes and Warnock (2021). I denote this variable as Net Flow^{SS} henceforth. Using this variable, I examine the effect of a one percentage-point increase in the capital outflows-to-GDP ratio during a sudden stop on GDP, consumption, investment, and stock price growth rate.

My main focus is on the differential effects of wealth inequality on recessions in sudden stops. I first construct a measure of wealth inequality to make the results interpretable. To be specific, I follow a similar approach in Auerbach and Gorodnichenko (2017) and Lakdawala, Moreland, and Schaffer (2021). I first standardize all countries' wealth Gini coefficients by subtracting the mean and dividing by the standard deviation. Then, I apply the logistic transformation to the standardized Gini coefficients, which I denote as $\ell_{i,t-1}$. I normalize the transformed values by the difference between the median value and the 75% percentile value of the Gini coefficients, denoting it as $v_{i,t-1} = \frac{\ell_{i,t-1} - \ell_{50\%}}{\ell_{75\%} - \ell_{50\%}}$. Using this measure, I estimate regressions on GDP, consumption, investment, and stock price growth rates specified as

$$y_{it} = \alpha + \beta \text{Net Flow}_{it}^{SS} + \gamma \text{Net Flow}_{it}^{SS} \times v_{i,t-1} + \Gamma X_{it} + \mu_i + \eta_t + \varepsilon_{it}, \quad (2)$$

$$\Delta \log q_{it} = \alpha + \beta \text{Net Flow}_{it}^{SS} + \gamma \text{Net Flow}_{it}^{SS} \times v_{i,t-1} + \Gamma X_{it} + \mu_i + \varepsilon_{it}, \quad (3)$$

where X_{it} denotes a vector of lagged values of the following variables of country i in year t : the cyclical component of GDP per capita, the lagged value of v , the real exchange rate, trade openness, the financial development index, financial openness, the debt-to-GDP ratio, and the foreign reserves-to-GDP ratio.

In regressions (2) and (3), coefficient β denotes the effects of capital outflows on aggregate demand and stock market indices when a sudden stop occurs in a country with median degree of wealth inequality. The interaction term of $v_{i,t-1}$ and $\text{Net Flow}_{it}^{SS}$ captures the differential effects of wealth inequality on contractions in sudden stops. Its coefficient γ can be interpreted as the additional effect of capital outflows during a sudden stop in a country at the 75% percentile level of wealth inequality compared to a country at the median level of wealth inequality. It is the main coefficient of interest in the regressions.

Table 1 shows that wealth inequality amplifies the contractionary effects of capital outflows in sudden stops. A one percentage-point increase in capital outflows during a sudden stop reduces GDP per capita by 0.154% relative to trend in a country with the median degree of wealth

TABLE 1: Aggregate Demand and Stock Prices in Sudden Stops

	GDP	Consumption	Investment	Stock Price
Net Flow $_t^{SS}$	-0.154*** (0.047)	-0.314*** (0.031)	-1.084*** (0.170)	-1.877*** (0.511)
Net Flow $_t^{SS} \times \nu_{t-1}$	-0.309*** (0.064)	-0.252*** (0.045)	-1.414*** (0.175)	-1.821** (0.816)
ν_{t-1}	0.001 (0.002)	-0.002 (0.003)	-0.002 (0.009)	-0.035 (0.025)
GDP $_{t-1}$	0.690*** (0.045)	0.636*** (0.063)	2.038*** (0.179)	-1.278 (0.977)
Real Exchange Rate $_{t-1}$	0.002 (0.012)	0.044** (0.018)	0.063 (0.063)	-0.182 (0.130)
Trade Openness $_{t-1}$	0.000** (0.000)	0.000 (0.000)	0.000 (0.001)	-0.000 (0.001)
Financial Development $_{t-1}$	0.022 (0.022)	0.050* (0.025)	0.107 (0.112)	-0.786*** (0.258)
Financial Openness $_{t-1}$	-0.001 (0.001)	-0.002 (0.002)	0.003 (0.012)	0.040 (0.029)
Debt/GDP $_{t-1}$	-0.001 (0.001)	-0.000 (0.002)	-0.004 (0.005)	-0.009 (0.015)
Reserves/GDP $_{t-1}$	0.011 (0.019)	0.016 (0.025)	0.040 (0.099)	0.489** (0.239)
Observations	864	864	847	782
Within R ²	0.531	0.432	0.357	0.085

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

inequality. There is an additional decline of 0.309 percentage points when the country's inequality measure is at the 75th percentile. The corresponding contraction in consumption rises from 0.314% to 0.566%, and the contraction in investment rises from 1.084% to 2.498%. Stock prices fall by 1.877% in the median-inequality country and by 3.698% in the 75th-percentile country. Thus, moving from the median to the 75th percentile approximately doubles the response of consumption, investment, and stock prices, and triples the response of GDP to an outflow of a given size. Wealth inequality also has prolonged differential effects on recessions and asset price drops, as shown in Appendix B.2.

Robustness Checks I run a battery of robustness checks for stylized fact 1. I obtain consistent results when I use the log wealth Gini instead of $\nu_{i,t-1}$ in Appendix B.3 and the wealth ratio of the

top 10% to the bottom 50% in Appendix B.4. The inequality interaction also explains variation not accounted for by the other controls, as shown in Appendix B.5. Because wealth inequality is persistent, I split the sample by countries' average wealth Gini and estimate the crisis response separately in Appendix B.6. The contractions remain larger in the high-inequality group. The result holds in emerging and advanced economies in Appendix B.7, survives interactions with other country characteristics in Appendix B.8, and is concentrated in indirect-investment outflows in Appendix B.9. It is also robust to controlling for private rather than total debt before the crisis.

Capital Outflows in Normal Times A surge in capital outflows in normal times generates contractionary dynamics but does not lead to asset market corrections, which is one of the distinct features of sudden stop crises. When I replace the Net Flow^{SS} variable with the ratio of capital net outflows to GDP in regressions (2) and (3), I find that GDP, consumption, and investment per capita all decrease when capital flows out of a country. The contraction is larger in unequal countries. However, stock market prices do not necessarily fall when there is a simple capital outflow, as shown in Table 2. In Table 2, I estimate regressions specified as

$$y_{it} = \alpha + \beta \text{Net Flow}_{it}^{\text{raw}} + \gamma \text{Net Flow}_{it}^{\text{raw}} \times v_{i,t-1} + \Gamma X_{it} + FE + \varepsilon_{it}, \quad (4)$$

$$y_{it} = \alpha + \sum_{k \in \{\text{SS}, \text{Non-SS}\}} \left(\beta^k \text{Net Flow}_{it}^k + \gamma^k \text{Net Flow}_{it}^k \times v_{i,t-1} \right) + \Gamma X_{it} + FE + \varepsilon_{it}, \quad (5)$$

where Net Flow^{raw}_{it} are the raw values of the capital net outflows to GDP ratio and Net Flow^{Non-SS}_{it} are the raw values of the capital net outflows to GDP ratio in normal times.

Aggregate demand falls more when capital flows out of a country in normal times. Wealth inequality implies a larger dispersion of the marginal propensity to consume within an economy, as studied by Hong (2023). While this may have played a role in enlarging the drops in aggregate demand when resources are pulled out of a country, the differential effect of wealth inequality is much more muted in normal times without severe asset market corrections. This signals that the interaction between asset market corrections and wealth inequality plays a significant role in deepening the crisis.²

²Nonetheless, capital outflows play a major role in generating contractionary dynamics during a crisis. Crises that do not necessarily involve capital outflows do not exhibit contractionary dynamics, as shown in Appendix B.10.

TABLE 2: Effects of Capital Net Outflows

	GDP		Consumption		Investment		Stock Price	
Net Flow _t	-0.051*** (0.015)		-0.071** (0.027)		-0.435*** (0.081)		0.731 (0.399)	
Net Flow _t × ν_{t-1}	-0.114*** (0.024)		-0.088*** (0.024)		-0.477*** (0.135)		-0.535 (0.475)	
Net Flow _t ^{Non-SS}		-0.038*** (0.013)		-0.040* (0.024)		-0.355*** (0.071)		1.062** (0.480)
Net Flow _t ^{Non-SS} × ν_{t-1}		-0.095*** (0.017)		-0.073*** (0.023)		-0.377*** (0.098)		-0.399 (0.329)
Net Flow _t ^{SS}		-0.148*** (0.051)		-0.308*** (0.037)		-1.044*** (0.177)		-1.927*** (0.443)
Net Flow _t ^{SS} × ν_{t-1}		-0.319*** (0.052)		-0.259*** (0.035)		-1.455*** (0.132)		-1.810** (0.819)
ν_{t-1}	-0.001 (0.002)	0.000 (0.002)	-0.004 (0.003)	-0.003 (0.003)	-0.010 (0.009)	-0.005 (0.008)	-0.056** (0.026)	-0.043* (0.022)
GDP _{t-1}	0.655*** (0.049)	0.672*** (0.045)	0.591*** (0.069)	0.619*** (0.062)	1.826*** (0.187)	1.916*** (0.170)	-1.123 (0.918)	-1.065 (0.892)
Real Exchange Rate _{t-1}	0.002 (0.015)	-0.002 (0.013)	0.044** (0.020)	0.040** (0.018)	0.055 (0.065)	0.037 (0.059)	-0.106 (0.109)	-0.107 (0.109)
Trade Openness _{t-1}	0.000** (0.000)	0.000** (0.000)	0.000 (0.000)	0.000 (0.000)	0.001 (0.001)	0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
Financial Development _{t-1}	0.019 (0.019)	0.018 (0.020)	0.046* (0.024)	0.046* (0.025)	0.079 (0.105)	0.074 (0.108)	-0.732*** (0.237)	-0.705*** (0.220)
Financial Openness _{t-1}	-0.001 (0.001)	-0.001 (0.001)	-0.002 (0.002)	-0.002 (0.002)	0.001 (0.011)	0.003 (0.011)	0.043 (0.028)	0.042 (0.028)
Debt/GDP _{t-1}	-0.003*** (0.001)	-0.002** (0.001)	-0.002 (0.002)	-0.001 (0.001)	-0.013** (0.005)	-0.010** (0.005)	-0.014 (0.015)	-0.007 (0.013)
Reserves/GDP _{t-1}	0.018 (0.021)	0.016 (0.018)	0.021 (0.028)	0.019 (0.024)	0.071 (0.119)	0.057 (0.105)	0.497* (0.253)	0.458* (0.247)
Observations	864	864	864	864	847	847	782	782
Within R ²	0.543	0.560	0.420	0.447	0.378	0.407	0.084	0.117

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

Comparison to Income Inequality Income inequality does not have any significant differential effects on the recessions that occur during sudden stop episodes. To be specific, I construct the income inequality measure as in regression (2) using the previous year's income Gini coefficients collected by the World Bank. Table 3 shows that the additional decreases in GDP and consumption attributable to income inequality are insignificant. Investment and stock prices fall during crises, but there is no additional effect due to income inequality. This implies that the distribution of assets that can be used as capital buffers in a financial crisis matters for the magnitude of the recession, while the income distribution by itself does not have a significant effect on recessionary dynamics.

This result shows that the distribution of asset buffers across households matters more than

TABLE 3: Sudden Stops and Income Inequality

	GDP	Consumption	Investment	Stock Price
Net Flow $_t^{SS}$	-0.205** (0.093)	-0.276** (0.112)	-1.323*** (0.364)	-2.351*** (0.725)
Net Flow $_t^{SS} \times \nu_{t-1}^{Income}$	-0.034 (0.045)	0.068 (0.089)	-0.186 (0.115)	-0.323 (0.308)
ν_{t-1}^{Income}	0.001 (0.001)	-0.001 (0.001)	0.002 (0.005)	0.033* (0.017)
GDP $_{t-1}$	0.690*** (0.037)	0.638*** (0.056)	1.924*** (0.175)	-1.189 (0.776)
Real Exchange Rate $_{t-1}$	0.004 (0.007)	0.042*** (0.011)	0.060* (0.031)	-0.270** (0.119)
Trade Openness $_{t-1}$	0.000*** (0.000)	0.000 (0.000)	0.001 (0.000)	-0.000 (0.001)
Financial Development $_{t-1}$	0.015 (0.012)	0.028** (0.014)	0.030 (0.063)	-0.512*** (0.166)
Financial Openness $_{t-1}$	-0.000 (0.001)	-0.002 (0.001)	0.003 (0.007)	0.017 (0.019)
Debt/GDP $_{t-1}$	-0.000 (0.001)	0.001 (0.001)	-0.003 (0.004)	-0.010 (0.012)
Reserves/GDP $_{t-1}$	0.013 (0.014)	0.030 (0.021)	0.026 (0.077)	0.321 (0.191)
Observations	1229	1229	1207	981
Within R ²	0.501	0.400	0.298	0.064

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

the income distribution for amplifying crisis dynamics. While recent works by Liu, Spiegel, and Zhang (2023) and Villalvazo (2026) focus on the relationship between capital outflows and income inequality, Table 3 shows that the wealth distribution remains the relevant margin after controlling for country characteristics. Income differences contribute to wealth accumulation over time, but current income alone does not measure the stock of resources that a household can use when income and borrowing capacity fall together. Assets provide that buffer, while debt determines how much of the buffer must instead be used for repayment. This distinction explains why wealth inequality predicts the severity of the balance-sheet adjustment, while income inequality does not. Appendix B.11 confirms that the interaction term with wealth inequality remains significant when both inequality measures enter the same regression.

TABLE 4: Wealth Inequality and Net Foreign Debt

	Wealth Gini		Top 10/Bottom 50	
	(1)	(2)	(3)	(4)
$\log(\text{Inequality Measure})_t$	3.412** (1.633)	2.741 (1.624)	0.210*** (0.058)	0.219*** (0.064)
Observations	996	996	896	896
Within R^2	0.023	0.067	0.025	0.048
Country FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes

Notes: The dependent variable is the log of net foreign debt relative to GDP. *** and ** denote significance at 1% and 5% levels using the Cameron et al. (2011) two-way clustered standard errors.

3.3 Fact 2: More Unequal Economies Are More Indebted Relative to Wealth

I relate this to the fact that households are more heavily indebted in more unequal countries. Table 4 first establishes the relationship in the country panel. The dependent variable is net foreign debt relative to GDP, and the regressions use changes within a country. A 1% increase in the wealth Gini is associated with a 3.412% increase in net external debt relative to GDP, obtained from the data of Lane and Milesi-Ferretti (2018). The positive association is stronger for the relationship between net external debt and the wealth ratio of the top 10% to the bottom 50% of households. This is consistent with a recent finding by de Ferra, Mitman, and Romei (2025), which documents that capital flows from equal countries to unequal countries.

Using the HFCS data, I also observe that the total debt relative to the assets of households is larger in countries with higher net wealth Gini coefficients, as shown in panel (A) of Figure 2. Moreover, a larger fraction of households is heavily indebted in more unequal countries when I calculate the fraction of households with a leverage ratio higher than 30% in panel (B). Hence, wealth inequality does not only mean that assets are concentrated at the top of the wealth distribution. It is also associated with greater indebtedness among the rest of the households.³

Household debt is primarily mortgage debt, as real estate takes a large portion of households' assets. Hence, households with high leverage ratios are mostly indebted in terms of mortgages,

³This is consistent with the Shapley value decomposition of the cumulative growth of net wealth Gini in the country-level aggregate panel data in Appendix A.3. I observe that the top 10%, the middle 40%, and the bottom 50% of households contributed to the cumulative change in net wealth Gini by 32%, 40.3%, and 25%, respectively.

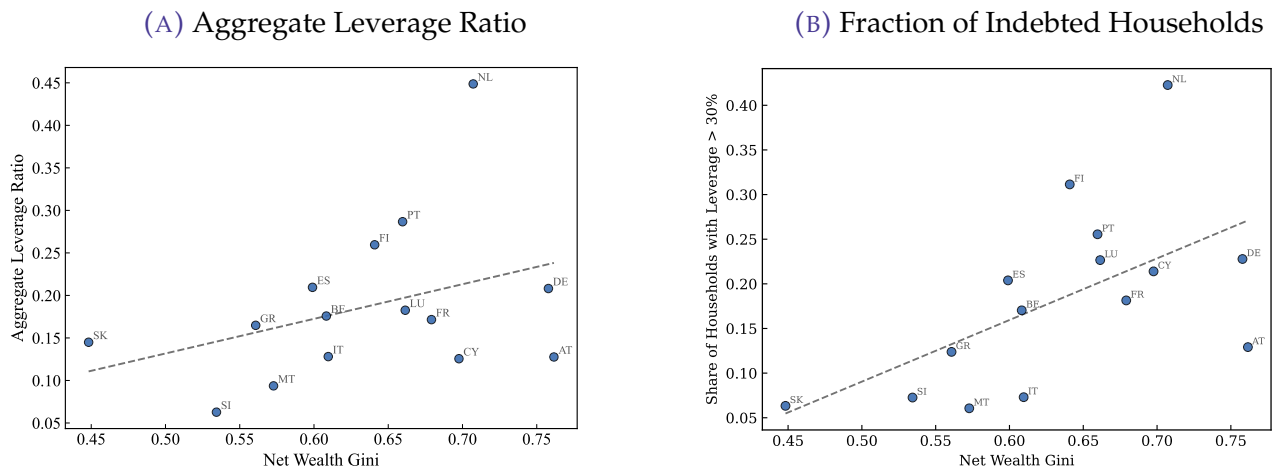


FIGURE 2. Household Leverage Ratio

Notes: Leverage ratio is defined as the ratio of total debt (DL1000) to total assets (DA3001). Panel (A) plots ratios of sum of debt to sum of assets of all households in each country in 2011. Panel (B) plots the fraction of households with leverage ratio higher than 30% in each country in 2011.

except for those at the bottom of the wealth distribution who have few assets.⁴ Households without a mortgage have a smaller leverage ratio than those with a mortgage. Utilizing this fact, I proxy a household's indebtedness by whether a household has a mortgage or not for further analysis, as will be discussed in the next section.

3.4 Fact 3: Indebted Households Reduce Asset Demand in Sudden Stops

I analyze how indebtedness changes a household's adjustment in assets and debt when a sudden stop occurs. As mentioned earlier, I consider households with mortgages as indebted households for analysis and focus on how the wealth of indebted households in countries with sudden stops changed in comparison to the same group of households in countries without a sudden stop.⁵ In a sudden stop, income falls during the recession while aggregate deleveraging makes external financing harder to obtain. A mortgagor must continue servicing debt from lower income and may be unable to roll over the loan. An outright owner who does not have a mortgage holds the same type of real asset without this payment or refinancing need.

I classify households by housing tenure.⁶ Renters and non-owners do not hold real estate.

⁴Liabilities and assets of firms owned by households are excluded from the data, so the leverage ratio in the data should be considered a lower bound.

⁵The repeated cross-sectional structure of the HFCS prevents me from following the same household but allows me to compare the adjustments of these groups before and after the crisis.

⁶I restrict the sample to reference persons aged 25–59 as of 2011.

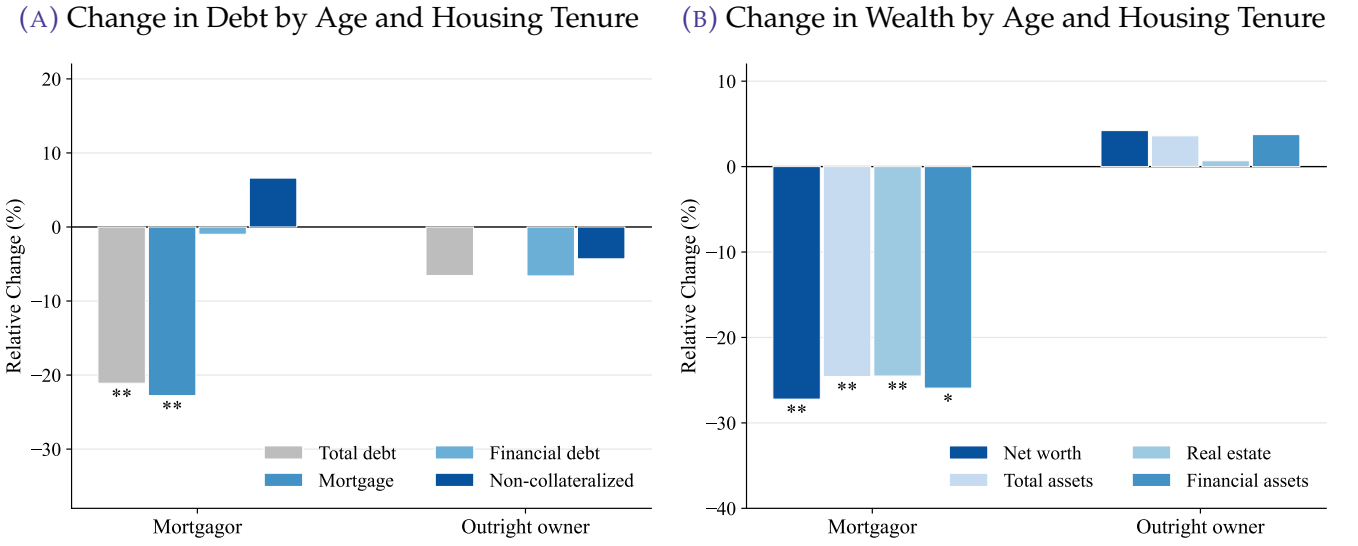


FIGURE 3. Balance Sheet Change by Age and Housing Tenure in Sudden Stop Countries

Notes: Each panel plots the difference-in-differences estimate in (6) for the subcategories as follows. Panel (A) plots total debt (DL1000), mortgage debt (DL1100), financial debt (DL1200), and non-collateralized loans (DL1230). Panel (B) plots net assets (DN3001), gross assets (DA3001), real estate (DA1400), and financial assets (DA2100). ** and * denote significance at the 5% and 10% levels.

They hold 3% of total household assets and are concentrated in the bottom two deciles of the wealth distribution, as shown in Tables A3 and A4. Outright owners own real estate without a mortgage. They represent 46% of households and hold 59% of gross assets. Mortgagors own real estate financed with a mortgage. They make up 26% of households and hold 38% of gross assets. Because renters hold only a small fraction of total assets, I focus on the changes in the wealth of mortgagors and outright owners.

I compare how the subcategories of assets and debts of households changed for these groups of households in countries with sudden stops. For each asset and debt subcategory X , country c , and group g , I compute the mean of each balance-sheet component. I take its log change between 2011 and 2014, denoted D_{cg}^X , and form

$$DD_g^X = \frac{1}{|\mathcal{S}|} \sum_{c \in \mathcal{S}} D_{cg}^X - \frac{1}{|\mathcal{N}|} \sum_{c \in \mathcal{N}} D_{cg}^X \quad (6)$$

where $\mathcal{S}$ denotes the set of four sudden stop countries and $\mathcal{N}$ denotes the set of comparison countries. Equation (6) compares the change over time in the sudden stop countries with the change over time in the comparison countries, separately for each group.⁷

⁷With only four treated countries, I conduct exact permutation inference. I reassign the sudden stop label to all possible subsets of four countries, recompute the statistic, and locate the actual estimate in the resulting placebo distribution. Because the direction of the adjustment is predicted, I report p -values for a predicted contraction, $p = \frac{1}{1001} \sum_{\tilde{\mathcal{S}}} \mathbf{1}\{DD(\tilde{\mathcal{S}}) \leq DD(\mathcal{S})\}$, where $\tilde{\mathcal{S}}$ runs over all placebo assignments.

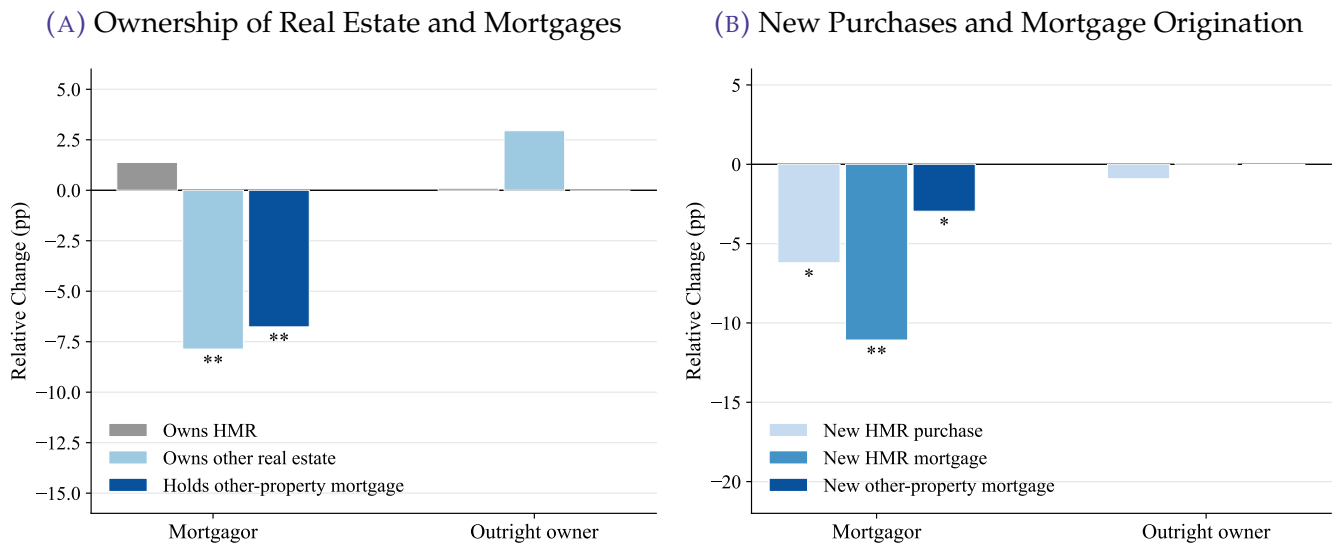


FIGURE 4. Real Estate Ownership Change in Sudden Stop Countries

Notes: Panel (B) shows the percentage-point change in the share of households that purchased their main residence (HB0700) or originated a mortgage on the main residence (HB1301–HB1303) or on other property (HB3301–HB3303) within the three years preceding the survey. HMR stands for household main residence. ** and * denote significance at the 5% and 10% levels.

Panel (A) of Figure 3 shows that the outstanding debt of mortgagors fell by more than 20% in the sudden stop countries compared to the comparison countries, driven by the fall in outstanding mortgage debt. In contrast, the debt growth of outright owners does not show a statistically significant difference relative to the comparison countries. Non-mortgage financial debt and non-collateralized loans, which include auto, consumer, and personal loans, show no significant changes in either group.

A fall in mortgage debt can be interpreted as a combination of two effects. First, amortization occurs on a fixed schedule over time through mortgage repayments. Second, existing borrowers' loans are not replaced by newly issued mortgages as new lending stops. This halt in new lending is more evident when I separately look at the change in new debt issuance rates. Panel (B) of Figure 4 shows that, relative to the same group in the comparison countries, the share of mortgagors whose mortgages were originated within the previous three years falls by more than 10 percentage points. The share of mortgagors who purchased their main residence within the previous three years falls by more than 5 percentage points. Outright owners who could buy a house without a loan show no comparable decline in recent purchases. New borrowing stopped, and home purchases that depended on new borrowing ceased with it, while purchases that did not require a loan did not decline.

The asset-side changes are consistent with the debt side. Panel (A) of Figure 4 shows that mortgagors held less real estate other than their main residence. The share of mortgagors owning other real estate fell by 7.5 percentage points more in sudden stop countries than in comparison countries. While the data does not provide direct information about asset sales due to debt foreclosures, mortgagors reduced real estate other than their main residences more in sudden stop countries. This is consistent with the changes in values of real estate subcategories in Figure A1, and I observe that mortgagors' holdings of real estate and associated mortgages significantly dropped more in sudden stop countries.

Changes in asset components of households in panel (B) of Figure 3 are consistent with the adjustments mentioned above. The real estate value of mortgagors fell by 27% more in the sudden stop countries. Their financial assets also fell relatively more, consistent with households drawing down savings to continue servicing their mortgages while incomes fell. Their debt also fell, but the decline in debt is smaller than the decline in assets. Therefore, net worth, which is assets minus debt, fell more in sudden stop countries. For outright owners, every component of wealth shows a small and insignificant change compared to that of the comparison countries.

The three empirical facts hint at the mechanism by which wealth inequality amplifies contractions in aggregate demand and asset prices in sudden stops, as shown in Table 1 (Fact 1). In a more unequal country, more households are indebted and each leveraged household borrows more (Fact 2). Therefore, a capital outflow of the same size induces deleveraging in a larger share of households and reduces aggregate net worth by a greater amount in a more unequal economy. The deleveraging pressure is concentrated on heavily indebted households since new lending is halted, preventing households from buying assets as they would in normal times (Fact 3). Hence, asset prices drop, decreasing aggregate demand through wealth effects. The price drop exacerbates this feedback, as the decline in asset prices reduces the amount of loans that households can raise for asset purchases. I formalize these empirical facts into a concrete mechanism with a tractable model in the next section.

4 Tractable Model

I now build a tractable model that connects the empirical facts. The empirical analysis establishes that contractions in sudden stops are larger where wealth is more unequally distributed and that the balance sheet adjustments are concentrated among leveraged households. The model rationalizes the set of facts observed from different sources of data.

I construct a model in which the wealth distribution shapes the response of asset prices and aggregate demand to a negative aggregate TFP shock. The shock lowers consumption and asset prices and generates capital outflows when the economy enters the period of sufficiently low productivity. I consider a unit measure of households that live for three periods $t \in \{0, 1, 2\}$ and are born with different levels of initial wealth drawn from an exogenous distribution. Households hold domestic productive capital and foreign assets and face collateral constraints when borrowing. Aggregate productivity in period 1 is the only source of uncertainty. An adverse realization depresses domestic income and asset prices, which tightens the constraints tying debt to the market value of capital. Wealth inequality raises the fraction of households that become constrained when the shock hits. Their joint withdrawal from the asset market produces a larger price decline that feeds back into every household's collateral value.

4.1 Environment

4.1.1 Household

An individual household i endowed with an initial stock of domestic capital k_{i0} maximizes life-time utility by choosing consumption c_{it} , domestic assets $k_{i,t+1}$, and foreign bonds $b_{i,t+1}$ while being subject to budget and borrowing constraints. The household's problem can be written as

$$\max_{\{c_{it}, b_{i,t+1}, k_{i,t+1}\}_{t=0,1}, c_{i2}} \mathbb{E}_0 \left[u(c_{i0}) + \beta u(c_{i1}) + \beta^2 c_{i2} \right], \quad (7)$$

$$\text{s.t. } c_{i0} + q_0 k_{i1} + \frac{b_{i1}}{R} = f(k_{i0}) + q_0 k_{i0}, \quad (8)$$

$$c_{i1} + q_1 k_{i2} + \frac{b_{i2}}{R} = z_1 f(k_{i1}) + q_1 k_{i1} + b_{i1}, \quad (9)$$

$$c_{i2} = f(k_{i2}) + k_{i2} + b_{i2}, \quad (10)$$

$$\frac{b_{i,t+1}}{R} \geq -\theta q_t k_{i,t+1}; \quad \forall t < 2, \quad (11)$$

where $u(\cdot)$ is a strictly increasing and concave function. I impose the linear utility function at period 2 because there is no uncertainty once z_1 is realized at the beginning of period 1.⁸ q_t and R denote the prices of domestic assets and the international interest rate. I assume that the international interest rate is constant over time. Domestic assets bought today can be used for production in the future, and I impose the production function as $f(k) = k^\alpha$ with $\alpha < 1$ for simplicity. Production in period 1 is scaled by aggregate productivity z_1 , so that period 1 output is $z_1 f(k_{i1})$, while I normalize productivity to $z_0 = z_2 = 1$ in periods 0 and 2. For simplicity, I abstract from capital depreciation over time, and households can trade their assets after production at price q_t in periods 0 and 1. In the final period, they simply consume their assets after production.

The household should pledge a θ fraction of the present market value of its domestic assets as collateral in the spirit of Jermann and Quadrini (2012) and Bianchi and Mendoza (2018).⁹ I assume that the collateral value is evaluated using today's market price q_t . Setting the collateral value equal to the present market value creates a source of pecuniary externality, as studied by Ottonello, Perez, and Varraso (2022), because households do not internalize the fact that their asset choices collectively affect the asset price and change the borrowing limit. For tractability, I assume that the only source of uncertainty lies in aggregate productivity z_1 in period 1 and it is stochastic from the households' perspective at period 0.

The first-order conditions of the household's problem regarding asset and bond choices are obtained as

$$k_{i1} :: q_0 u'(c_{i0}) = \beta \mathbb{E}_0 [u'(c_{i1})(z_1 f'(k_{i1}) + q_1)] + \mu_{i0} \theta q_0, \quad (12)$$

$$k_{i2} :: q_1 u'(c_{i1}) = \beta (f'(k_{i2}) + 1) + \mu_{i1} \theta q_1, \quad (13)$$

$$b_{i1} :: u'(c_{i0}) = \beta R \mathbb{E}_0 u'(c_{i1}) + \mu_{i0}, \quad (14)$$

$$b_{i2} :: u'(c_{i1}) = \beta R + \mu_{i1}, \quad (15)$$

and the complementary slackness conditions hold as

$$\mu_{it} \left(\frac{b_{i,t+1}}{R} + \theta q_t k_{i,t+1} \right) = 0; \quad \mu_{it} \geq 0, \quad \forall t. \quad (16)$$

⁸This can be considered as taking the certainty equivalence value of CARA utility as the utility function at period 2.

⁹Previous studies, such as Bianchi (2011), Seoane and Yurdagul (2019), and Rojas and Saffie (2022), address the real exchange rate depreciation in Sudden Stops by specifying a nontradable good sector and a borrowing constraint in which debt is limited to a fraction of the sum of tradable and nontradable income. However, I abstract from including the real exchange rate in the model because I do not observe differential effects of wealth inequality on the real exchange rate depreciation in a typical sudden stop in my sample.

4.1.2 Market Clearing

The balance of payments in this economy is represented as

$$\int \left(c_{it} + \frac{b_{i,t+1}}{R} \right) di = \int (z_t f(k_{it}) + b_{it}) di. \quad (17)$$

I assume that the domestic assets are of fixed supply and that the asset price q_t clears the asset market in each period. The asset market clearing condition is defined as

$$\int k_{i,t+1} di = \int k_{i0} di = \bar{K}, \quad \forall t. \quad (18)$$

The total amount of wealth is the total supply of assets traded in periods 0 and 1.

4.1.3 Competitive Equilibrium

The competitive equilibrium of this economy is defined as follows.

Definition 1. Given the international interest rate R and initial wealth $\{w_i\}_i$, the competitive equilibrium is defined as allocations $\{c_{it}, k_{i,t+1}, b_{i,t+1}\}_{i,t}$ and prices $\{q_t\}_t$ such that

1. Given $\{q_t\}$ and w_i , $\{c_{it}, k_{i,t+1}, b_{i,t+1}\}_t$ maximizes an individual household's utility (7).
2. The domestic asset market is cleared in each period.

4.2 Effect of a Negative TFP Shock

I analyze the effects of a negative shock to aggregate productivity z_1 in period 1. A negative productivity shock is widely known to trigger sudden stops. In my model, it operates through the balance sheets of financially constrained households. I first characterize an endogenous threshold level of wealth in period 1 that distinguishes between constrained and unconstrained households. I then analytically examine how a negative shock to z_1 moves the threshold and the asset price q_1 in period 1. I finally show that greater wealth inequality, which I consider a mean-preserving spread of the initial wealth distribution, amplifies the effect of the productivity shock on the asset price q_1 by raising the fraction of constrained households before the shock. Note that the economy becomes deterministic once z_1 is realized at the beginning of period 1.

4.2.1 Wealth Inequality and Productivity Shock in Period 1

I denote an individual household's net wealth at the beginning of period 1 as $\tilde{w}_i = z_1 f(k_{i1}) + b_{i1} + q_1 k_{i1}$, and let G be the joint distribution of (k_{i1}, b_{i1}) determined in period 0. While G itself is independent of the price q_1 and the realization of z_1 , each household's period-1 wealth depends on both. In the absence of idiosyncratic shocks throughout the periods, $\tilde{w}_i$ is increasing in the household's initial wealth at the beginning of period 0. Throughout this section, I assume $u(c) = \log c$ and $\beta R = 1$. For analytical tractability, I also impose the assumption that

$$\bar{K}^{1-\alpha} \geq 1 + (1-\alpha) \left(3 + \frac{4}{1-\theta} \right), \quad (19)$$

which bounds the price elasticity of asset demand, and that the gross international interest rate satisfies $R \leq 2$.¹⁰

A household is either constrained or unconstrained in equilibrium at period 1. I assume that the distribution G is such that not all households are unconstrained under a given realization of z_1 , which generally holds when wealth is continuously distributed across households or when a non-zero measure of households is near the borrowing limit. Across the wealth distribution G , there is a cutoff level of net wealth that separates constrained from unconstrained households.

Lemma 1. *The unconstrained households demand an identical amount of capital $\bar{k}_2$ where*

$$\bar{k}_2 = (f')^{-1}(q_1 R - 1) \quad \text{and} \quad \frac{\partial \bar{k}_2}{\partial q_1} < 0. \quad (20)$$

Households with wealth $\tilde{w}_i < \bar{w}$ are constrained, where $\bar{w}$ satisfies

$$\bar{w} = 1 + (1-\theta)q_1 \bar{k}_2, \quad (21)$$

and I have

$$\frac{\partial \bar{w}}{\partial z_1} = 0 \quad \text{and} \quad \frac{\partial \bar{w}}{\partial q_1} < 0. \quad (22)$$

Proof. See Appendix C.1. □

Lemma 1 characterizes how the productivity shock reaches households in period 1. Because the shock does not alter the tightness of the borrowing constraint, it leaves the threshold $\bar{w}$ un-

¹⁰Assumption (19) is used only for the inequality comparative statics of Proposition 2. By Bernoulli's inequality, (19) implies $\bar{K} \geq 4 + \frac{4}{1-\theta}$ for every $\alpha \in (0, 1)$. The bound $R \leq 2$ is innocuous for a world interest rate.

affected at a given price as $\partial \bar{w} / \partial z_1 = 0$. Instead, a lower z_1 moves the wealth distribution itself downward and also changes the equilibrium price of assets. I characterize how households' demand for assets changes under the productivity shock to study what determines the price change.

For unconstrained households, the productivity shock has indirect effects through the asset price but no direct effects. For the constrained households with $\tilde{w}_i < \bar{w}$, the binding collateral constraint implies $b_{i2} = -R\theta q_1 k_{i2}$. Combining the first-order conditions with respect to k_{i2} and b_{i2} yields the consolidated Euler equation

$$(1 - \theta)q_1 u'(\tilde{w}_i - (1 - \theta)q_1 k_{i2}) = \beta (f'(k_{i2}) + 1 - \theta q_1 R), \quad (23)$$

where the asset demand k_{i2} increases with wealth $\tilde{w}_i$.¹¹ Totally differentiating (23) with respect to z_1 , q_1 , and k_{i2} , the total response of the constrained household's asset demand to a productivity shock can be written as

$$\frac{dk_{i2}}{dz_1} = \underbrace{\frac{(1 - \theta)q_1 u''(c_{i1}) f(k_{i1})}{\beta f''(k_{i2}) + (1 - \theta)^2 q_1^2 u''(c_{i1})}}_{\text{Direct effect} \equiv \frac{\partial k_{i2}}{\partial z_1} > 0} + \underbrace{\frac{\theta + (1 - \theta)(q_1 k_{i1} - (1 - \theta)q_1 k_{i2} - c_{i1}) u''(c_{i1})}{\beta f''(k_{i2}) + (1 - \theta)^2 q_1^2 u''(c_{i1})}}_{\text{Price effect} \equiv \frac{\partial k_{i2}}{\partial q_1} < 0} \frac{dq_1}{dz_1}, \quad (24)$$

which is obtained from the total differentiation of (23) with respect to z_1 , q_1 , and k_{i2} .

When the asset price falls as z_1 declines, constrained households buy more assets at the cheaper price, but they cannot buy as much as unconstrained households because they are up against their borrowing limit and must finance any additional purchases from their own scarce resources.¹² This makes constrained households' asset demand less elastic to price changes than that of unconstrained households, which gives us Lemma 2.

Lemma 2. *The price elasticity of asset demand is smaller for constrained households, i.e.,*

$$\left| \frac{\partial k_{i2}}{\partial q_1} \frac{q_1}{k_{i2}} \right| < \left| \frac{\partial \bar{k}_2}{\partial q_1} \frac{q_1}{\bar{k}_2} \right|, \quad (25)$$

where household i is a constrained household.

Proof. See Appendix C.2. □

¹¹It is immediate that $\lim_{\tilde{w}_i \rightarrow \bar{w}} k_{i2}(\tilde{w}_i) = \bar{k}_2$, which implies $k_{i2} < \bar{k}_2$ for the constrained households.

¹²The term associated with k_{i1} in the numerator of the price effect captures the valuation effect of a higher asset price on period-1 wealth $\tilde{w}_i$. I maintain throughout that this valuation effect does not dominate so that $\partial k_{i2} / \partial q_1 < 0$ holds for every constrained household at every price; the condition is stated in primitive form in Appendix C.2.

In equilibrium, the asset price q_1 clears the domestic asset market and satisfies

$$\int_{i \in \mathcal{C}} k_{i2} di + \int_{i \in \mathcal{U}} \bar{k}_2 di = \bar{K}, \quad (26)$$

where $\mathcal{C}$ and $\mathcal{U}$ denote the sets of constrained and unconstrained households, respectively. When z_1 falls, constrained households reduce their asset demand, while the shift in the wealth distribution enlarges the mass of constrained households. Both forces exert fire sale pressure on the asset market. Totally differentiating the market clearing condition with respect to z_1 and q_1 gives

$$\frac{dq_1}{dz_1} = \frac{\overbrace{\int_{i \in \mathcal{C}} \left| \frac{\partial k_{i2}}{\partial z_1} \right| di}^{\text{Average direct effect}}}{\underbrace{\int_{i \in \mathcal{C}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| di + \int_{i \in \mathcal{U}} \left| \frac{\partial \bar{k}_2}{\partial q_1} \right| di}_{\text{Average elasticity of asset demand}}} > 0. \quad (27)$$

Hence, a negative productivity shock lowers the asset price, and the size of the price drop depends on the average direct effect in the numerator and the average price elasticity of asset demand in the denominator. The price falls less when there are more unconstrained households with more elastic asset demand because these households absorb the assets at a cheaper price and dampen the fire sales.

The productivity shock also raises the fraction of constrained households. Whether a household is constrained is determined by its wealth relative to the threshold, $\tilde{w}_i - \bar{w}$, and the shock affects both terms: every household's distance to the threshold changes as

$$\frac{d(\tilde{w}_i - \bar{w})}{dz_1} = f(k_{i1}) + \left[k_{i1} - \frac{\partial \bar{w}}{\partial q_1} \right] \frac{dq_1}{dz_1}, \quad (28)$$

which is strictly positive. Two forces reinforce each other. First, the distribution of $\tilde{w}_i$ shifts to the left as output falls, which is the partial equilibrium channel. Second, Fisherian deflation increases the threshold $\bar{w}$ in general equilibrium as the price drop feeds into the borrowing limit. Since (28) is positive for every household, wealth falls relative to the threshold as z_1 falls. Households can only cross from the unconstrained region into the constrained region when z_1 falls, so the fraction of constrained households rises.

Now turning to the change in aggregate consumption due to a productivity shock, only constrained households change their consumption because, by assumption, the consumption of unconstrained households in period 1 is constant at 1. The bond Euler equation of an unconstrained

household implies $u'(c_{i1}) = 1$, so unconstrained households simply reduce their bond holdings to buy more capital without changing their consumption. The partial equilibrium effect of the shock on a constrained household's consumption is

$$\left. \frac{\partial c_{i1}}{\partial z_1} \right|_{q_1} = f(k_{i1}) \frac{\beta f''(k_{i2})}{\beta f''(k_{i2}) + (1 - \theta)^2 q_1^2 u''(c_{i1})} > 0, \quad (29)$$

In partial equilibrium with a fixed asset price, a lower z_1 directly reduces the household's output and thus its consumption. Combining this with the general equilibrium effect from the price drop, aggregate consumption changes as

$$\frac{dC_1}{dz_1} = \underbrace{\int_{i \in \mathcal{C}} \left. \frac{\partial c_{i1}}{\partial z_1} \right|_{q_1} di}_{\text{Partial equilibrium effect}} + \underbrace{\int_{i \in \mathcal{C}} \frac{\partial c_{i1}}{\partial q_1} \frac{dq_1}{dz_1} di}_{\text{General equilibrium effect}}. \quad (30)$$

A negative shock to z_1 lowers aggregate consumption in partial equilibrium. Moreover, combining the budget constraint with the consolidated Euler equation shows that $\partial c_{i1} / \partial q_1 > 0$ applies to every constrained household. Hence, the general equilibrium term in (30) reinforces the partial equilibrium effect rather than offsetting it. As a result, aggregate consumption decreases when productivity falls. I establish the effects of a negative TFP shock as follows.

Proposition 1. Let $\bar{F} \equiv \int f(k_{i1}) di$. A negative shock to z_1

1. lowers the asset price q_1 ;
2. lowers aggregate consumption C_1 ;
3. generates a capital outflow if $z_1 < z^*$ such that z^* satisfies $dC_1 / dz_1|_{z^*} = \bar{F}$.

Proof. See Appendix C.3. □

Note that any negative TFP shock decreases asset prices and aggregate consumption, but the economy must already be vulnerable, with low productivity for capital outflows to occur. When productivity is high, few households are constrained, and the consumption-smoothing motive against a temporary output loss dominates. In that case, households borrow, and capital flows in. When productivity is already low such that $z_1 < z^*$, a large fraction of households is constrained. A further negative shock forces them to deleverage as the falling asset price erodes the value

of their collateral. Their consumption then falls by more than output, and capital flows out.¹³ The threshold z^* is the productivity level at which these two forces balance. I demonstrate in Appendix D.1 that a negative shock on the borrowing limit θ has the same effect on asset prices and aggregate consumption through a similar mechanism while unconditionally incurring capital outflows.

I now turn to the role of wealth inequality in transmitting the productivity shock to asset price and aggregate consumption. The empirical analysis in Section 3 showed that asset prices fall more during sudden stops when wealth is distributed more unequally across households. I now show that the model reproduces this pattern through the channels described above. Recall that the distribution G is a bivariate distribution of (k_{i1}, b_{i1}) . Denote the distribution of $\tilde{w}_i$ that corresponds to G under a given price q_1 as $\Phi(q_1, G)$, and let $\Phi(q_1, F)$ correspond to a more unequal distribution F for any level of q_1 .¹⁴

A more dispersed wealth distribution makes more households constrained, which raises the overall direct effect in the numerator of (27) and leaves fewer unconstrained households whose elastic asset demand would dampen the price drop. Since all constrained households are net sellers and all unconstrained households are net buyers of assets in this economy, both forces intensify the fire sale pressure, and inequality does not dampen the drop in the asset price. I establish this formally in Proposition 2.

To analytically characterize the differential effects of greater inequality on the drops in asset prices and aggregate consumption, I consider a specific family of inequality-increasing perturbations under which moving from G to F raises the fraction of households below every wealth level lying beneath the mean by a common proportional factor, while holding the mean fixed at every asset price between the two equilibrium prices. The numerical analysis in the next subsection imposes no such restriction on the wealth distribution and shows that the main result continues to hold for a general mean-preserving spread of the initial wealth distribution.

¹³Endogenous tightening of the borrowing limit is crucial in generating the capital outflow. Under a constant borrowing limit, as in Aiyagari (1994), more households become constrained and borrow up to the limit when z_1 falls, which always increases aggregate debt and capital inflow.

¹⁴Conceptualizing an increase in inequality as a larger wealth dispersion at both ends of the distribution is consistent with the empirical fact that both the upper and lower wealth decile groups contributed to the change in wealth by comparable amounts, as shown in Appendix A.3.

Proposition 2. Let F be a more unequal initial distribution connected to G by a family of continuous distributions $\{\Phi_\lambda\}_{\lambda \in [1, \lambda_F]}$ with $\lambda_F > 1$, $\Phi_1 = G$ and $\Phi_{\lambda_F} = F$, such that at every price q_1 between q_1^F and q_1^G , the induced wealth distributions satisfy $\Phi(x; q_1, \Phi_\lambda) = \lambda \Phi(x; q_1, G)$ for all x smaller than the mean of $\Phi(q_1, G)$, with the mean held fixed. Suppose the wealth threshold $\bar{w}$ evaluated at q_1^F lies below the median and the mean of $\Phi(q_1^F, F)$. Then a larger wealth inequality amplifies responses of both the asset price and aggregate consumption to a shock on z_1 with

$$\left. \frac{dq_1}{dz_1} \right|_{\Phi(q_1^G, G)} < \left. \frac{dq_1}{dz_1} \right|_{\Phi(q_1^F, F)}, \quad (31)$$

$$\left. \frac{dC_1}{dz_1} \right|_{\Phi(q_1^G, G)} < \left. \frac{dC_1}{dz_1} \right|_{\Phi(q_1^F, F)}, \quad (32)$$

where q_1^G and q_1^F are the equilibrium asset prices under initial distributions G and F , respectively.

Proof. See Appendix C.4. □

The household responses in the model are consistent with Fact 3 in Section 3. A constrained household loses current income when productivity falls and also loses borrowing capacity when the asset price declines. It cannot finance the shortfall by issuing more debt. It instead repays debt and reduces k_{i2} while withdrawing asset demand precisely when the economy needs buyers. An unconstrained household faces the same lower price but does not need to deleverage. It buys the assets released by constrained households. The HFCS evidence displays the same division. Mortgagors reduced debt, real estate, and financial assets together and stopped originating mortgages and purchasing housing. Outright owners, who held the same type of assets without debt, showed no comparable adjustment.

The wealth threshold connects this household response to Fact 2. Households below $\bar{w}$ borrow up to the limit, while those above can use bonds to separate their asset purchases from current resources. A mean-preserving spread places more households below the threshold. Hence, greater inequality changes more than just the concentration of asset ownership at the top. It expands the leveraged part of the distribution, which has asset demand tied to current income and collateral values. This is the model counterpart of the positive relationship between wealth inequality and indebtedness in Table 4 and Figure 2.

A larger fraction of constrained households under greater wealth inequality in the model ratio-

nalizes Fact 1. A more unequal economy brings more forced sellers and fewer elastic buyers to the same asset market. The numerator of (27) rises because a larger mass of constrained households cuts demand directly. Its denominator falls because the remaining buyers are fewer in number, while constrained demand responds less to a price discount than unconstrained demand. The price must consequently fall further to induce the remaining buyers to absorb the assets. This is not just a larger initial fire sale. The lower price erodes collateral and pushes additional households below the wealth threshold, forcing further deleveraging. Proposition 2 shows that this feedback produces both the larger asset-price decline and the larger consumption contraction observed in more unequal countries.

The same amplification mechanism operates when the economy is hit by a financial shock that tightens the borrowing constraint θ . Such a shock raises the fraction of constrained households and depresses asset prices. Greater wealth inequality amplifies the drop through the same fire sale and Fisherian debt-deflation channels. A financial shock differs only in that it acts through the households' debt capacity rather than their output, raising the threshold $\bar{w}$ directly through $\partial \bar{w} / \partial \theta < 0$ instead of shifting the wealth distribution. I state and prove the results for the financial and interest rate shocks in Appendices D.1 and D.2.

4.2.2 Numerical Example

I provide a numerical example of the productivity shock that parallels the analysis above. I keep the same calibration as in the baseline environment, imposing $u(c) = \log c$, $\beta R = 1$ with an international interest rate of 4%, and $\alpha = 0.66$. I keep the tightness of the borrowing constraint at $\theta = 0.8$ and assume that aggregate productivity z_1 takes the value of 0.95 with a probability of 0.1 and 1.0 with a probability of 0.9, corresponding to a 5% negative productivity shock. I compare the equilibria under two initial wealth distributions, $G = \text{LogNormal}(0, 1.2^2)$ and $F = \text{LogNormal}(-0.33125, 1.45^2)$, with wealth Gini coefficients of 0.604 and 0.695, respectively, where F is a mean-preserving spread of G .

The household's first-order conditions in period 1 imply that a household always consumes $1/\beta R$ if it is unconstrained under all possible realizations of z_1 . For such households with a

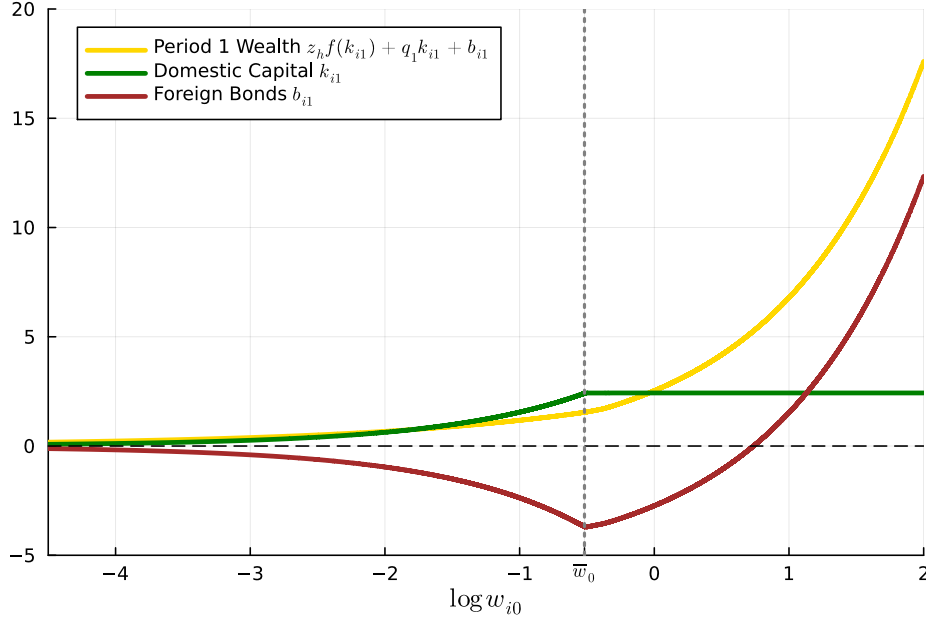


FIGURE 5. Policy Functions of k_{i1} and b_{i1} at Period 0

sufficient level of wealth, the first-order conditions in period 0 can be written as

$$q_0 R - \mathbb{E}_0[z_1]f'(k_{i1}) = \mathbb{E}_0 q_1 + \frac{\text{Cov}_i(u'(c_1), z_1 f'(k_{i1}) + q_1)}{\mathbb{E}_0 u'(c_{i1})} = \mathbb{E}_0 q_1, \quad (33)$$

and they buy the same amount of k_{i1} in period 0. Denote that level of k_{i1} as $\bar{k}_1$. Since the capital demand k_{i1} is weakly increasing in k_{i0} , this implies the existence of a threshold value of k_{i0} such that a household with an initial capital stock below that level buys strictly less than $\bar{k}_1$. Correspondingly, there is a wealth threshold $\bar{w}_0$ for the initial wealth such that households below the threshold borrow up to the limit in period 0, as shown in Figure 5. For bonds, constrained agents' debt increases with initial wealth because the larger capital stock raises the borrowing limit, whereas it decreases with wealth for the unconstrained agents. As a result, households' wealth at the beginning of period 1 increases with their initial wealth.

Table 5 summarizes the effects of a negative productivity shock on aggregate consumption and the price of capital. Here, wealth at period 1 is $q_1 k_{i1} + b_{i1}$. When aggregate productivity falls by 5%, aggregate consumption decreases by 1.62% and the price of capital falls by 0.30% in the baseline economy, as the fraction of constrained households rises from 37.69% to 39.54%. The contraction is amplified in the alternative economy with a more unequal distribution of k_0 at period 0. Aggregate consumption falls by 2.13% and the price of capital drops by 0.40%, while the constrained fraction rises from 49.43% to 51.56%. Consistent with Proposition 2, a mean-preserving

TABLE 5: Numerical Results under a Productivity Shock

	Baseline Distribution (G)	Alternative Distribution (F)	Counterfactual
Wealth Gini at Period 0	0.605	0.696	0.605
Mean Wealth at Period 1 ($z_1 = z_H$)	3.492	3.451	3.492
Wealth Gini at Period 1 ($z_1 = z_H$)	0.787	0.861	0.861
Constrained Fraction at Period 1 ($z_1 = z_H$)	37.69%	49.43%	37.69%
Constrained Fraction at Period 1 ($z_1 = z_L$)	39.54%	51.56%	46.31%
Change in Consumption C_1 (%)	-1.62%	-2.13%	-2.18%
Change in Asset Price q_1 (%)	-0.30%	-0.40%	-0.38%

spread of the initial wealth distribution increases the fraction of constrained households before the productivity shock and makes the aggregate variables respond more to the shock.

I also analyze the effect of greater inequality on the changes in aggregate consumption and asset prices while keeping the fraction of constrained households before the negative productivity shock fixed. To be specific, I construct a counterfactual distribution of wealth at period 1 by keeping the average wealth and the fraction of constrained households at period 1 fixed at the baseline level under the initial distribution G and increasing the wealth Gini coefficient to the level under the initial distribution F .¹⁵ The counterfactual distribution matches the Gini coefficient at period 1 under the initial distribution F , while matching the average wealth and the fraction of constrained households under the initial distribution G .

Under the counterfactual distribution, the fraction of constrained households rises to 46.31%, aggregate consumption falls by 2.18%, and the capital price falls by 0.38% when TFP z_1 at period 1 decreases by 5% from z_H to z_L . The drops in consumption and the price of capital are larger than under G because more households become constrained after the shock. This highlights that greater inequality can amplify the responses of aggregate variables to a shock, even when we control for the fraction of constrained households before the shock.

I conclude the numerical example with a decomposition of the response of each household's consumption to the productivity shock. A household's consumption change can be decomposed into four components: a partial equilibrium effect and three general equilibrium effects that operate through asset prices. Using the binding collateral constraint, a constrained household's

¹⁵I transform the wealth $q_1 k_{i1} + b_{i1}$ at period 1 as

$$q_1 k_{i1} + b_{i1} = -z_H f(k_{i1}) + \frac{\frac{1}{N} \sum_j [z_H f(k_{j1}) + q_1 k_{j1} + b_{j1}]}{\frac{1}{N} \sum_j [z_H f(k_{j1}) + q_1 k_{j1} + b_{j1}]^\gamma} [z_H f(k_{i1}) + q_1 k_{i1} + b_{i1}]^\gamma,$$

where $\gamma > 1$ to match the wealth Gini under the initial distribution F .

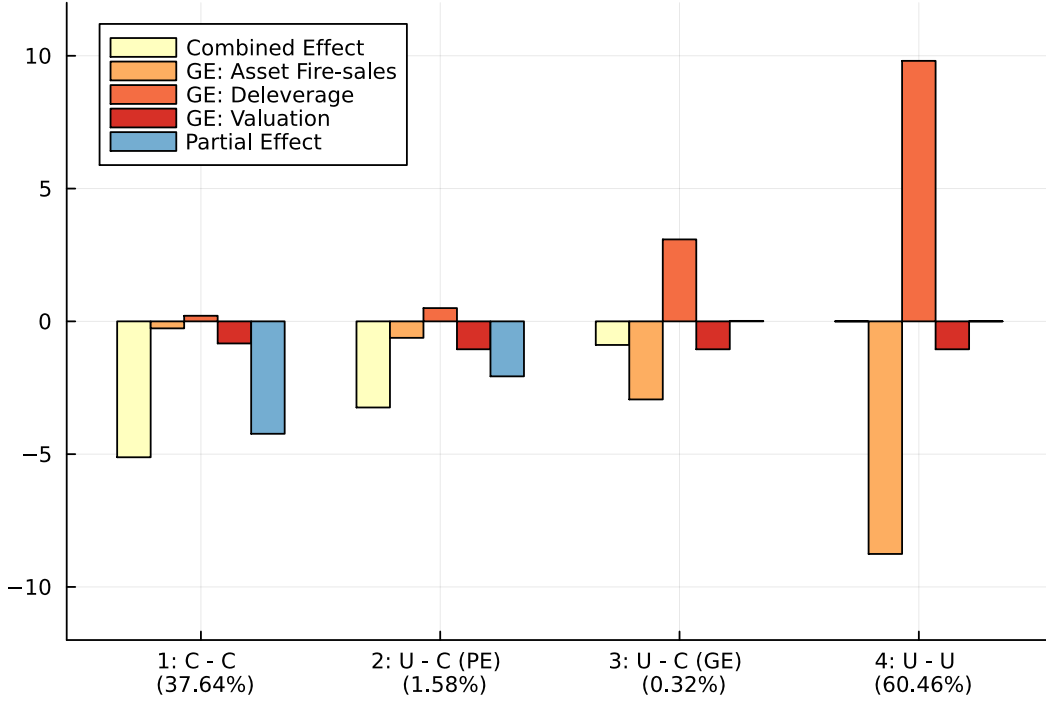


FIGURE 6. Change in Individual Consumption under a Productivity Shock

consumption change is

$$\begin{aligned}
 \frac{dc_{i1}}{dz_1} = & \underbrace{f(k_{i1}) - (1 - \theta)q_1 \frac{\partial k_{i2}}{\partial z_1}}_{\text{Partial Equilibrium Effect}} + \underbrace{k_{i1} \frac{dq_1}{dz_1}}_{\text{GE: Valuation Effect}} \\
 & - \underbrace{\left(k_{i2} + q_1 \frac{\partial k_{i2}}{\partial q_1} \right) \frac{dq_1}{dz_1}}_{\text{GE: fire sales Effect}} + \underbrace{\theta \left(k_{i2} + q_1 \frac{\partial k_{i2}}{\partial q_1} \right) \frac{dq_1}{dz_1}}_{\text{GE: Deleveraging Effect}}, \quad (34)
 \end{aligned}$$

where I group households into four categories: those that are always constrained ($\mathcal{C} \rightarrow \mathcal{C}$), always unconstrained ($\mathcal{U} \rightarrow \mathcal{U}$), and two groups of switchers that become constrained either in partial equilibrium as the wealth distribution shifts down or in general equilibrium as the asset price falls. Figure 6 reports the decomposition for the economy with an initial wealth distribution of $\text{LogNormal}(0, 1.2^2)$.

The always unconstrained households leave their consumption unchanged, as the reduction in their bond holdings offsets the additional capital purchases and the valuation effect. The always constrained households reduce their consumption the most, and the partial equilibrium effect, which scales with output, is the main driver of this change. The switchers in partial equilibrium behave like the always constrained households, whereas the switchers in general equilibrium reduce their consumption only slightly because they are not directly affected by the output loss

but merely become constrained as the asset price falls.¹⁶ This pattern parallels the one under the financial shock in Appendix D.1, except that here the partial equilibrium effect is proportional to each household's output rather than to its debt.

5 Quantitative Model

I now extend the three-period model to an infinite horizon environment with aggregate uncertainty. This allows me to analyze the dynamic behavior of aggregate output and debt prior to a sudden stop, especially depending on the degree of wealth inequality that is endogenously generated in the model. There are three sources of aggregate uncertainty in terms of aggregate TFP z , world interest rate R , and financial friction θ . In addition, I assume that individual households are heterogeneous *ex ante*. They are born with permanently different levels of productivity that follow an exogenous distribution. This allows me to generate wealth inequality dynamically because all households' choices of capital would converge to a single point had there been no such individual heterogeneity.¹⁷ I abstract from idiosyncratic income shocks to distinguish the role of idiosyncratic risk from that of inequality. Lastly, I assume that labor and capital are used for production and introduce homogeneous capital producers to allow for capital accumulation.

5.1 Households

I consider a unit measure of heterogeneous households with permanently different levels of productivity ω , so that every aggregate defined as an integral over households coincides with the corresponding cross-sectional mean. A household maximizes its lifetime utility by choosing consumption c , capital k' , bonds b' , and labor supply n , and I assume the GHH utility function. The household's recursive problem is written as

$$V(s, k, b; \omega) = \max_{c, n, k', b'} u(c - v(n)) + \beta \mathbb{E} V(s', k', b'; \omega), \quad (35)$$

$$\text{s.t. } c + qk' + \frac{\zeta}{2}(k' - k)^2 + \frac{b'}{R} = z\omega f(k, n) + q(1 - \delta)k + b + \pi, \quad (36)$$

¹⁶The fraction of switchers is small under this calibration, but it increases when I adopt log-utility across all three periods.

¹⁷Without individual heterogeneity, the capital distribution becomes degenerate as individual asset choices converge to a single point in the long-run. A similar argument was made by Caselli and Ventura (2000).

$$\frac{b'}{R} \geq -\theta q k', \quad (37)$$

where $s \equiv \{z, R, \theta\}$ denotes the vector of aggregate state variables, q is the price of capital that households take as given, and ω is the individual's productivity, which follows an exogenous distribution with a unit mean. I also assume that each household owns an equal share of the capital-producing firms and receives the lump-sum profit π rebated from the capital producers.

I abstract from idiosyncratic productivity shocks to distinguish the role of income risk from inequality. Previous works used the volatility of individual productivity to determine the degree of inequality in the economy. However, such an approach introduces greater income risk by design. I abstract from idiosyncratic productivity shocks to generate wealth inequality over time without changing the degree of income risk that individual households face throughout their lifetime.

The optimal conditions of the household are obtained as

$$b' :: u'(c - v(n)) = \beta R \mathbb{E} u'(c' - v(n')) + \mu, \quad (38)$$

$$\begin{aligned} k' :: (q + \zeta(k' - k))u'(c - v(n)) \\ = \beta \mathbb{E} u'(c' - v(n'))(z' \omega f_k(k', n') + q'(1 - \delta) + \zeta(k'' - k')) + \theta q \mu, \end{aligned} \quad (39)$$

$$n :: v'(n) = z \omega f_n(k, n), \quad (40)$$

where μ is the Lagrangian multiplier for the borrowing constraint (37). Note that there is no wealth effect on labor supply as shown in the first-order condition with respect to n .

5.2 Capital Producers

I introduce a unit mass of homogeneous capital producers who supply new capital in this economy. The capital producers buy consumption goods from the households to use as production inputs. They maximize static profits defined as

$$\max_i \pi = qi - i - \frac{\phi}{2} \frac{i^2}{K}, \quad (41)$$

where they are subject to a quadratic cost that depends on the change in investment relative to the previous period's aggregate investment. The capital producers rebate the profits to the

households. The first-order condition of the representative capital producer yields

$$q = 1 + \frac{\phi i}{K}, \quad (42)$$

since the capital producer takes the aggregate stock of capital as given. In equilibrium, the price of capital satisfies

$$q = 1 + \frac{\phi I}{K}, \quad (43)$$

since the capital producers are homogeneous.

5.3 Market Clearing

There are two markets in this economy. First, the domestic capital market is cleared each period. Hence, the asset market clearing condition satisfies

$$\underbrace{\int (1 - \delta) k_s ds}_{\equiv (1 - \delta)K} + \underbrace{\int i_j dj}_{\equiv I} = \underbrace{\int k'_s ds}_{\equiv K'}, \quad (44)$$

where s denotes each entrepreneur-household and j denotes each capital producer. Combining this with individual households' budget constraints, I obtain the market clearing condition in the goods market as

$$C + I + \frac{\zeta}{2} \int (k'_s - k_s)^2 ds + \frac{\phi}{2} \frac{I^2}{K} + \frac{B'}{R} = z \int \omega_s f(k_s, n_s) ds + B, \quad (45)$$

which is the resource constraint of the economy. i_s denotes each individual household's investment and is therefore heterogeneous across households.

5.4 Exogenous Processes

I assume three exogenous processes for the aggregate state variables. For the international interest rate R and aggregate productivity, I assume that the two variables follow AR(1) processes specified as

$$\log z_t = \rho_z \log z_{t-1} + \epsilon_z, \quad \epsilon_z \sim N(0, \sigma_z^2), \quad (46)$$

$$\log \frac{R_t}{R} = \rho_R \log \frac{R_{t-1}}{R} + \epsilon_R, \quad \epsilon_R \sim N(0, \sigma_R^2). \quad (47)$$

Finally, as in the previous section, the tightness of the households' borrowing constraints is a binary random variable with two possible values $\{\theta_H, \theta_L\}$ and follows a stationary Markov process.

5.5 Competitive Equilibrium

The recursive competitive equilibrium of this economy is defined as follows.

Definition 2. Given states $s \equiv \{z, R, \theta\}$ and productivity distribution $F(\omega)$, the recursive competitive equilibrium is defined as a set of policy functions $\{c(s, k, b; \omega), k'(s, k, b; \omega), b'(s, k, b; \omega)\}$, the value function $V(s, k, b; \omega)$, and the price function $q(s)$ such that

1. Given $q(s)$ and ω , the policy functions $\{c(s, k, b; \omega), k'(s, k, b; \omega), b'(s, k, b; \omega)\}$ solve a household's problem (35), and $V(s, k, b; \omega)$ corresponds to the associated value function.
2. Given s and the aggregate capital K , $q(s)$ clears the domestic capital market and satisfies

$$q(s) = 1 + \frac{\phi \left(\int k'(s, k, b; \omega) d(k, b, \omega) - (1 - \delta)K \right)}{K}. \quad (48)$$

6 Model Simulation and Analysis

6.1 Calibration

TABLE 6: Calibration of Parameters

Parameter	Value	Source
σ	2	Standard
β	0.91	Standard
δ	0.08	Standard
ψ	2	Standard
α	0.34	Labor income share of 66%
ϕ	2.42	Penn World Tables 11.0
ρ_z	0.84	Penn World Tables 11.0
σ_z	0.029	Penn World Tables 11.0
R	1.011	Bianchi and Mendoza (2018)
ρ_R	0.68	Bianchi and Mendoza (2018)
σ_R	0.0186	Bianchi and Mendoza (2018)
ζ	0.8363	Estimated
θ_H	0.291	Estimated
θ_L	0.095	Estimated
p_{HH}	0.899	Estimated
σ_ω	0.8636	Estimated

I assume that households' preferences follow the GHH utility function specified as

$$u(c - v(n)) = \frac{\left(c - \frac{n^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}}\right)^{1-\sigma}}{1-\sigma}, \quad (49)$$

where ψ is the Frisch elasticity of labor supply. The relative risk aversion coefficient is set to $\sigma = 2$ as in many small open economy models, and the Frisch elasticity is set to $\psi = 2$, which is close to the range of macro-elasticities measured by Rogerson and Wallenius (2009). Households' time discount factor is set to 0.91, and the annual capital depreciation rate is $\delta = 0.08$ as each period corresponds to a year. The Cobb-Douglas parameter for capital input is set to $\alpha = 0.34$, which implies a labor income share of 66%. The capital production cost parameter ψ is set to 2.42, which I estimate by regressing the price of capital on the ratio of investment to capital stock collected from the Penn World Tables.¹⁸ The TFP process is estimated using the annual TFP of countries from the Penn World Tables. I estimate the AR(1) process of the linear-detrended components of the 54 countries in my sample while controlling for country fixed effects. This gives me estimates of $\rho_z = 0.84$ and $\sigma_z = 0.029$. I impose the same exogenous process for the world interest rate R in Bianchi and Mendoza (2018), which is estimated with the annualized return of 90-day US T-bills.

I assume that the exogenous process of θ follows a two-state stationary Markov chain. Hence, $\theta \in \{\theta_H, \theta_L\}$ and its transition probability matrix is

$$\Pi_{\theta'|\theta} = \begin{pmatrix} p_{HH} & 1 - p_{HH} \\ 0.999 & 0.001 \end{pmatrix}, \quad (50)$$

where I set the probability of θ staying at θ_L as 0.1% to ensure that a sudden stop lasts for a maximum of one year in the data. I estimate the individual capital adjustment cost parameter ζ , the parameters $\{\theta_H, \theta_L, p_{HH}\}$ in the exogenous process of θ , and the standard deviation of ω , denoted as σ_ω , to jointly match six moments in the data using the method of simulated moments.¹⁹ All parameters used in the model are listed in Table 6, and the target moments with their sources are summarized in Table 7.

I solve the model using the method of Krusell and Smith (1998) to update the law of motion

¹⁸I define the price of capital as the relative price of new capital stock to the price of investment goods, following the definition of Tobin's q . I regress the relative price against the ratio of investment to capital stocks, controlling for country and year fixed effects to obtain an estimate of 2.42.

¹⁹When estimating σ_ω , I fix the mean of individual productivity at 1.

TABLE 7: Moments from the Model and Data

Moment	Model	Data	Source
Average debt/GDP	27.62%	27.67%	Lane and Milesi-Ferretti (2018)
Sudden stop occurrence rate	3.65%	3.57%	Regression sample in Table 1
Average wealth Gini	0.67	0.77	World Inequality Database
σ_C / σ_{GDP}	1.073	1.13	IMF International Financial Statistics
σ_I / σ_{GDP}	1.589	3.95	IMF International Financial Statistics
$\sigma_{CA} / \sigma_{GDP}$	5.22%	3.85%	IMF International Financial Statistics

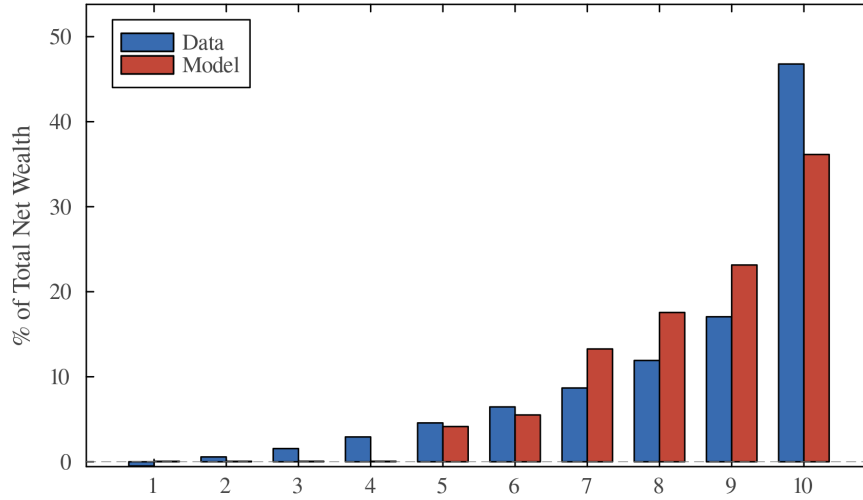


FIGURE 7. Net Wealth Share by Deciles (Untargeted)

of aggregate capital K . I generate 5,000 households and simulate the economy for 20,000 periods to solve the model. The evolution of aggregate capital follows

$$\log K' = 0.041 + 0.939 \log K + 0.114 \log z - 0.092 \log R - 0.052 \mathbb{I}(\theta = \theta_L), \quad (51)$$

with R^2 of 0.979. The estimated law of motion implies that investment increases with aggregate TFP and decreases with borrowing costs. The details of the computation algorithm are in Appendix E.1.

I further compare the wealth distribution obtained from the dynamic model to the HFCS data. In the model, I define each household's net wealth as the market value of the capital and bonds it holds at the end of each period. After classifying all households into decile groups based on net wealth, I calculate the average wealth share of each group in the total net wealth of the economy throughout the simulation. In the data, net wealth is defined as a household's gross assets net of gross liabilities. I also calculate the average wealth shares of net wealth decile groups in each

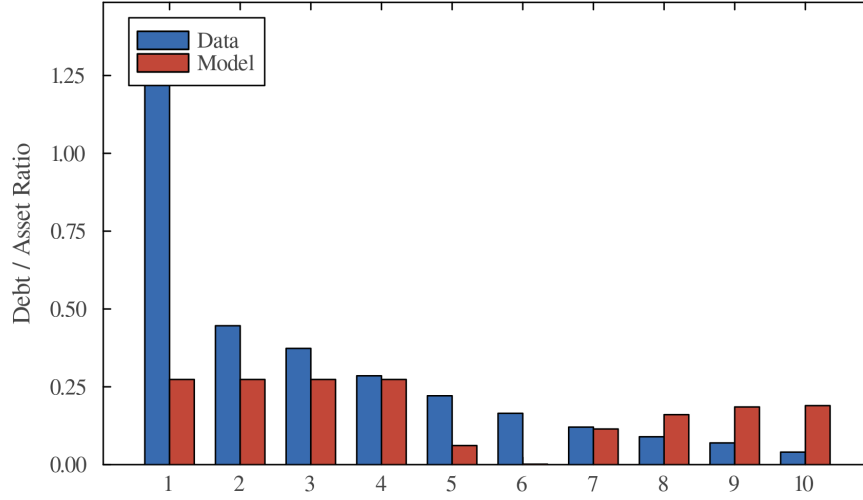


FIGURE 8. Leverage Ratio by Deciles (Untargeted)

Notes: The leverage ratio is the ratio of total debt to total assets within each net wealth decile. In the data it is the ratio of total debt (DL1000) to total assets (DA3001), averaged across the 15 European countries in 2011, with deciles taken from the ECB HFCS series DNTOP10. In the model it is the ratio of debt to the market value of assets within each net wealth decile, averaged over the simulation of 80,000 periods.

country in 2011 and compare them to the distribution generated by the model. As shown in Figure 7, the model generates a distribution that reasonably resembles the data. The model produces a lower concentration of wealth at the top than the data, but the discrepancy is small. Figure 8 compares the leverage ratio of households in the model and the data. The model shows that households with the lowest levels of TFP borrow up to the limit because their income and assets are too low to finance consumption and investment. The leverage ratio decreases for households in the middle deciles 5 and 6 but then increases in net wealth as these households are productive enough to borrow to finance investment.

6.2 Sudden Stop Dynamics and Wealth Inequality

I now conduct an event-study analysis of sudden stop episodes to study the aggregate dynamics around the crises by simulating the economy for 80,000 periods with 5,000 households. I use changes in the current account-to-GDP ratio to define a sudden stop and calculate the current account as $(B_{t+1} - B_t)$. A sudden stop is defined as an episode when the change in the current-account to GDP ratio is two standard-deviations larger than the mean, which is consistent with how I defined a sudden stop in the empirical analysis.

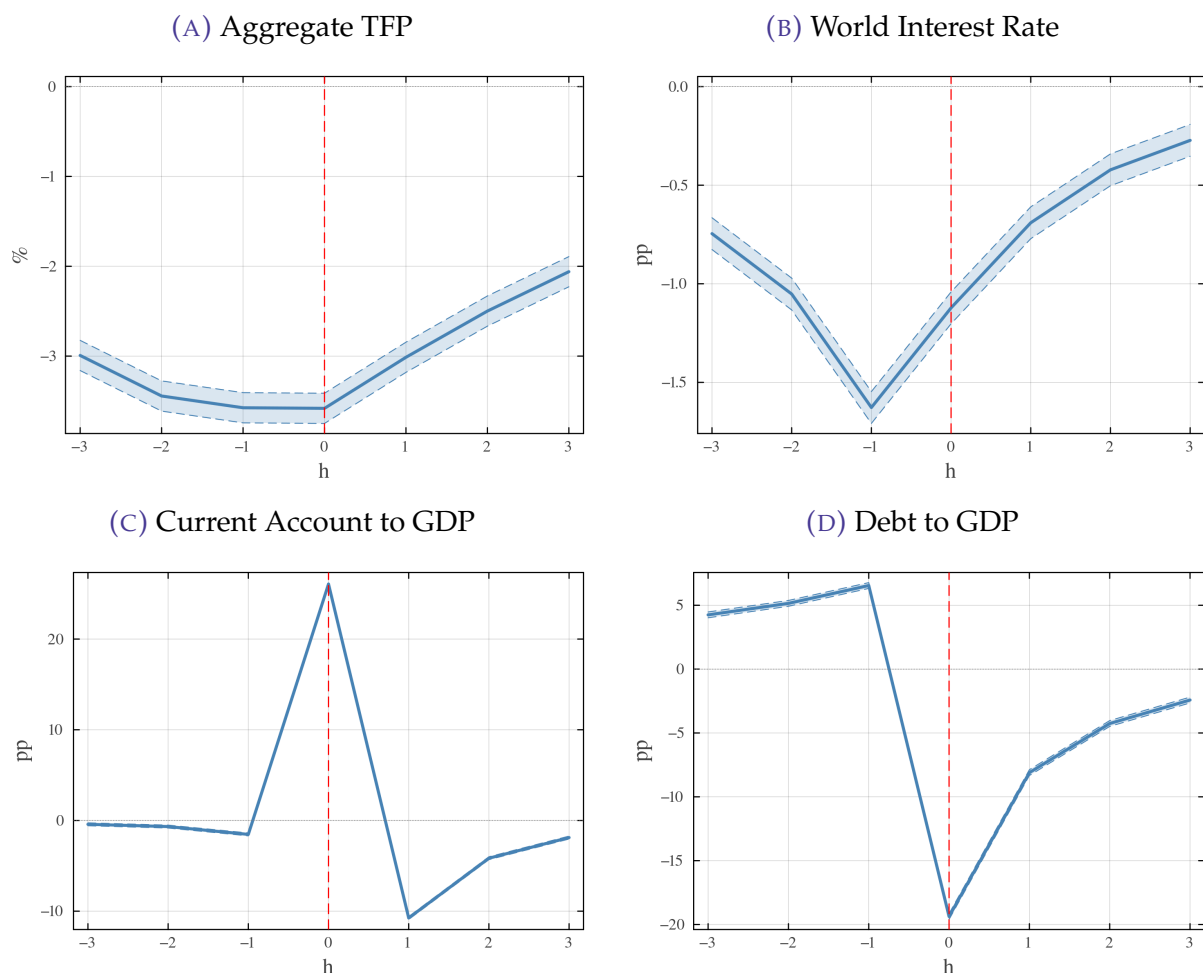


FIGURE 9. Current Account and Aggregate Shocks during Sudden Stops

Notes: The debt-to-GDP ratio in panel (D) is calculated as the end-of-period value $-B_{t+1}/GDP_t$. Current account to GDP ratio, debt to GDP, and the world interest rate are plotted as percentage-point deviations from the unconditional mean. Aggregate TFP is plotted as the percentage deviation from the unconditional mean. Dotted lines are 90% confidence intervals.

Figure 9 illustrates the dynamics of aggregate TFP, the world interest rate, the current account, and aggregate debt. In a sudden stop in the model, TFP remains low for consecutive periods in panel (A), leading to low capital accumulation before a sudden stop. External borrowing conditions become tighter as the world interest rate rises in panel (B). The borrowing constraint also tightens as θ decreases in a typical sudden stop episode. The current account shows a steep reversal during a sudden stop episode, moving from a persistently negative level before the crisis. Panel (D) of Figure 9 shows that the economy has been accumulating a positive level of debt until the crisis under lower interest rates.

Macroeconomic variables go through crisis dynamics similar to those studied by Mendoza (2010) and Bianchi and Mendoza (2018), as shown in Figure 10. Aggregate consumption, invest-

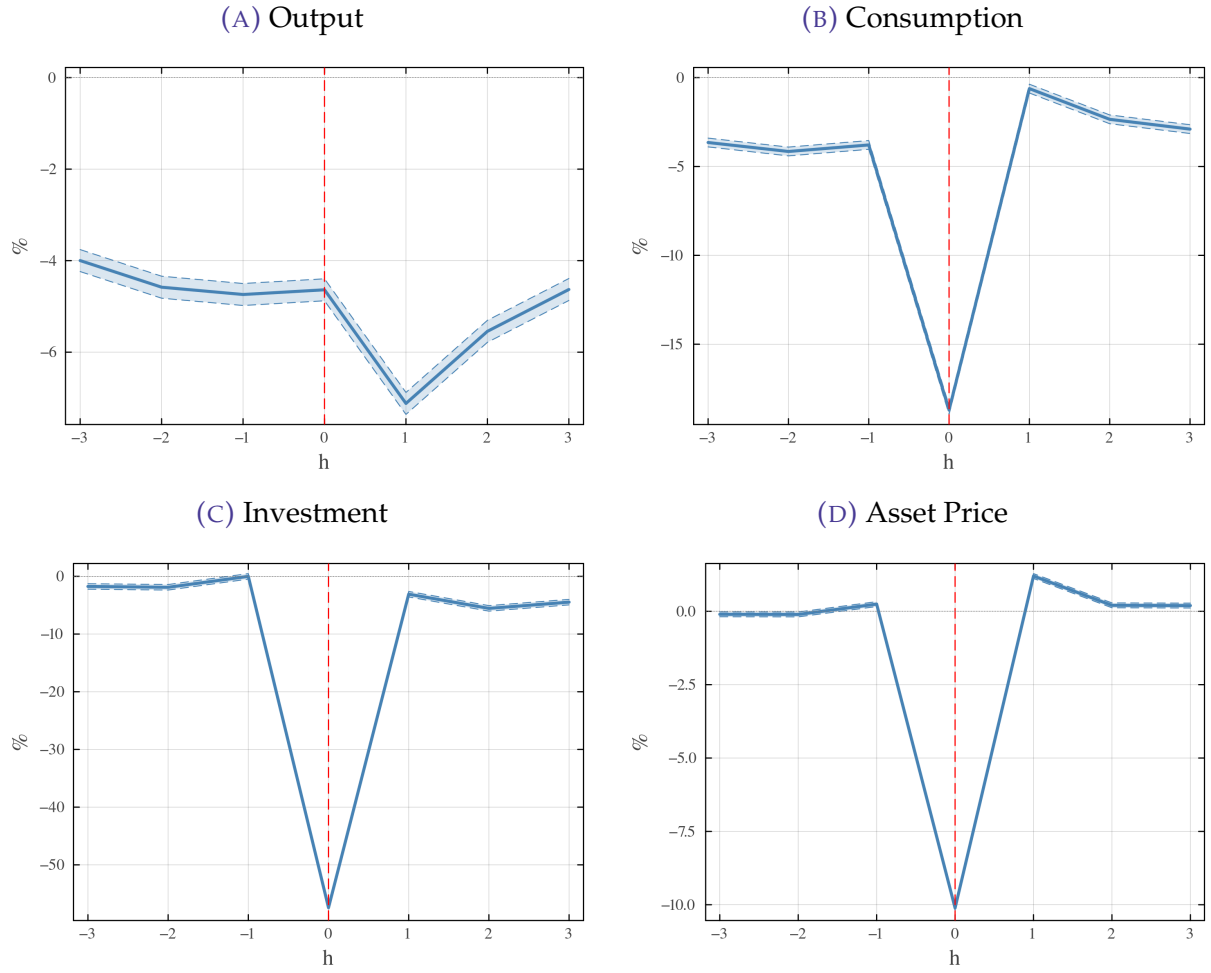


FIGURE 10. Macroeconomic Dynamics during Sudden Stops

Notes: All variables are plotted as the percentage deviations from the unconditional mean. Dotted lines are 90% confidence intervals.

ment, and asset prices all plummet in a sudden stop period. Consumption falls by more than 15%, and investment declines sharply by more than 50% of the long-run mean. The drop in investment is consistent with a sharp decline in the price of capital, which falls by 10% in a sudden stop episode. Output, however, does not drop immediately during the crisis period because aggregate capital is determined from the previous period. A drop in TFP and labor supply does not sufficiently decrease output at the moment of crisis, but reductions in domestic absorption incur massive deleveraging, as illustrated in the dynamics of the debt-to-GDP ratio. As investment drops today, output declines one period after the sudden stop and gradually recovers.

Incorporating household heterogeneity allows the model to reproduce the countercyclicality of the ratio of external debt to GDP observed in the data, which the representative agent model fails to match. After removing country-specific linear trends, the correlation between net foreign

debt relative to GDP and output is -0.141. The correlation is -0.281 in the baseline model with heterogeneous households, whereas the same model with homogeneous households generates the opposite sign of 0.125. The difference reflects how investment is financed in the two models. In the representative agent economy, a higher TFP raises both current income and the expected return to capital due to persistence. The household responds by increasing investment while smoothing consumption and financing the part of the investment boom that exceeds current resources through foreign borrowing. Higher investment also raises the price and collateral value of capital. Hence, both capital and leverage rise during an expansion, allowing external debt to grow faster than current output. Household heterogeneity weakens this link. The productive, capital-rich households that hold the majority of external debt can finance investment from their higher current income and accumulated net worth in times of expansion. The less productive, capital-poor households do not seek to increase as much and are credit-constrained, so they account for little in the change of aggregate debt. Hence, aggregate external debt adjusts more slowly than output, causing the debt-to-GDP ratio to be countercyclical, as it does in the data.

I now study the propagation effect of inequality on the sudden stop dynamics. I construct nine economies that differ only in permanent productivity dispersion σ_ω and simulate each economy for 80,000 periods under the same aggregate shocks. To be specific, I analyze the differential effects of a 0.01 increase in wealth Gini relative to the median of each economy in the simulated panel data and that of each country in the actual data. Table 8 summarizes the estimation results of the two sets of regressions. The interaction estimates are negative for all four outcomes in both panels. Comparing the coefficients of the interaction term in the two sets of regressions, the model generates similar amplification effects of wealth inequality on the contractions of all three real quantities in a sudden stop. A larger drop in investment under greater wealth inequality also implies that the price of capital falls more under greater inequality because the price of capital and aggregate investment have a one-to-one relationship.

Wealth inequality increases the probability of a sudden stop in the simulated data. When I compare the probability across the nine economies, it rises from 3.54% in the least unequal economy to 3.91% in the most unequal economy, as shown in Table 9. External debt carried into a sudden stop rises from 39.28% to 41.83% of GDP. A larger productivity dispersion increases the mass of households at both ends of the productivity distribution. Those households exhibit a high

TABLE 8: Regressions with Simulated and Actual Data

	Simulated Panel			Data		
	GDP	Consumption	Investment	GDP	Consumption	Investment
Net Flow $_t^{SS}$	-0.106*** (0.001)	-0.638*** (0.010)	-2.308*** (0.010)	-0.084 (0.095)	-0.256*** (0.064)	-0.796** (0.354)
Net Flow $_t^{SS} \times$ Inequality $_{t-1}$	-0.074*** (0.006)	-0.105*** (0.010)	-0.449*** (0.037)	-0.086* (0.051)	-0.070** (0.026)	-0.365** (0.155)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Within R ²	0.492	0.583	0.408	0.518	0.426	0.339

Notes: The dependent variables in the simulated data are percentage deviations from each economy's ergodic mean. I control for the capital stock at the beginning of period and the lagged value of debt-to-gdp ratio in the regressions with simulated data. For the regressions with real data, I use the same dependent variables and the same set of control variables used in regressions (2) and (3). I control for economy fixed effect for the simulated data and country and year fixed effects for the real data. Inequality $_{t-1}$ is deviation of the lagged wealth Gini from the median of each economy and country, divided by 0.01. The final row reports the ratio of the model and data interaction coefficients. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

leverage ratio because the bottom households behave like hand-to-mouth households, while the productive households at the top raise debt to finance investment, as shown in Figure 8. Larger debt accumulation before a sudden stop makes an economy more vulnerable, resulting in a higher probability of a crisis and amplified deleveraging when a crisis occurs. The average reversal in the current account increases with the level of wealth inequality, implying larger capital outflows in a typical sudden stop. Aggregate demand for consumption and investment also drops more on average when inequality is greater. However, the average contraction in output does not exhibit a monotonic relationship with wealth inequality because a larger productivity dispersion implies that there is a greater mass of productive households. The composition effect on the distribution of individual productivity may dampen the effect on GDP. In summary, wealth inequality incurs larger debt accumulation before a sudden stop and leads to a higher likelihood of a crisis, which is similar to what Kumhof et al. (2015) find in the U.S. data before the global financial crisis.

6.3 Distributional Effects of Sudden Stop

Sudden stops change households' asset and consumption choices unevenly across the wealth distribution. To measure the heterogeneity in the data, I sort households into four groups by net wealth: the bottom 50%, the 51–80%, the 81–95%, and the top 5%, and estimate the change in each

TABLE 9: Sudden Stops across Economies with Different Inequality Levels

	Extensive Margin		Intensive Margin				
Wealth Gini	Pr(SS) (%)	Debt/GDP (%)	$\Delta CA/GDP$ (pp)	ΔGDP (%)	ΔC (%)	ΔI (%)	Δq (%)
0.586	3.54	39.28	26.73	-4.74	-16.95	-54.16	-9.70
0.606	3.69	39.83	27.17	-4.57	-17.58	-54.90	-9.78
0.621	3.79	40.27	27.42	-4.42	-18.07	-55.45	-9.82
0.631	3.82	40.63	27.68	-4.42	-18.52	-56.05	-9.85
0.639	3.86	40.94	27.89	-4.32	-18.83	-56.64	-9.92
0.647	3.84	41.23	28.08	-4.41	-19.22	-57.46	-10.01
0.652	3.86	41.51	28.25	-4.39	-19.50	-57.96	-10.07
0.656	3.88	41.71	28.39	-4.34	-19.66	-58.29	-10.09
0.659	3.91	41.83	28.48	-4.30	-19.79	-58.58	-10.12

Notes: Probability of sudden stops are calculated as the fraction of sudden stop episodes during 80,000 periods in each economy. The debt-to-gdp ratio is measured at the beginning of a sudden stop episode. The intensive margin effects on GDP, consumption, investment, and asset price are average percentage deviations from the ergodic mean of each economy.

asset category between 2011 and 2014 with

$$y_{i,c,t} = \sum_{q=1}^4 (\beta^q + \gamma^q SS_c + \delta^q t + \kappa^q SS_c t) D_{i,c,t}^q + \Gamma X_{i,t} + \varepsilon_{i,c,t}, \quad (52)$$

where $D_{i,c,t}^q$ is the dummy variable for household i in net wealth group q in country c at year t , and SS_c is the dummy variable for the four countries that experienced sudden stops.²⁰ $X_{i,t}$ is a vector of control variables at the household-year level.²¹ The coefficient of interest is κ^q , the average change in the outcome of group q in the countries that underwent a sudden stop.

I define a household's net asset as its end-of-period net wealth $qk' + b'/R$, sort households into the same four groups in every period, and compute the log change in each group's mean net asset. The model counterpart of κ^q is the average of that change over sudden stop periods, less its average over normal periods, where periods within three periods of a sudden stop are excluded from the latter so that the comparison group is not contaminated by the crisis itself. I construct the consumption response in the same way. I first compare how the average net wealth of each group of households changes during a sudden stop period in my model and in the data.

Panel (A) of Figure 11 shows that net wealth decreased more for poor households than for

²⁰Household Finance and Consumption Network (2013, 2016) provide sampling weights for each household-year observation. I rescale the sampling weights to equalize the relative size of each country in the pooled sample and estimate a weighted least squares regression.

²¹I control for the education level of the primary earner, the household's total income from labor and self-employment, the size of the household, and the age of the primary earner along with its squared value.

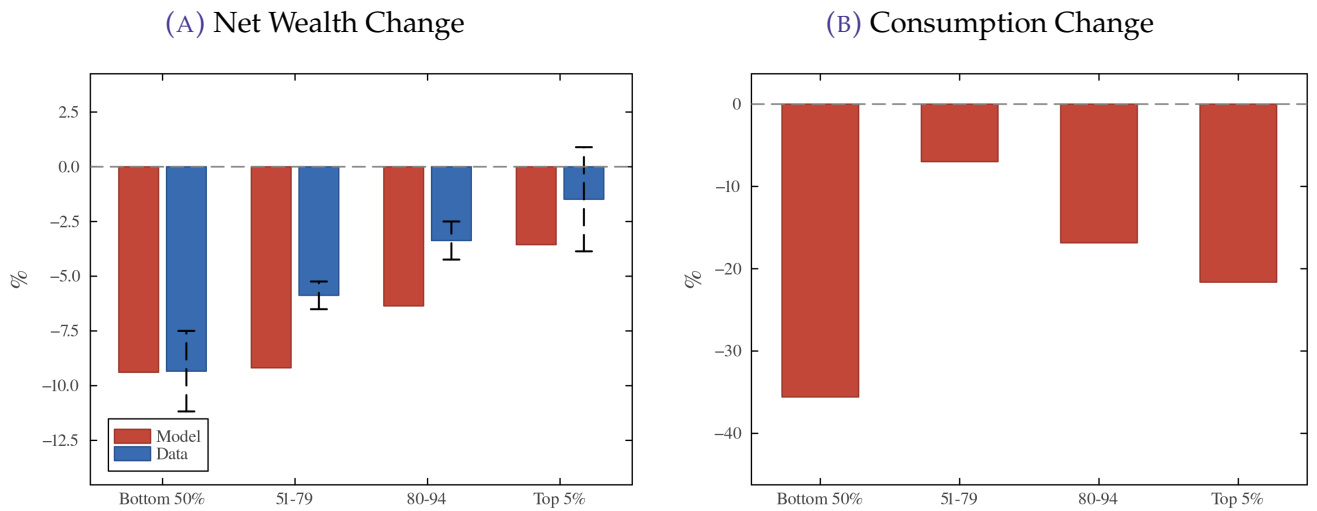


FIGURE 11. Distributional Effects of Sudden Stops

wealthy households. In the data, the top 5% virtually did not reduce their net assets, as the difference is statistically insignificant, while the drop in net wealth is monotonically larger for groups with smaller wealth. In the model, all households invest less in capital stocks in a typical sudden stop because aggregate productivity decreases and the interest rate rises. For the bottom 50% of households with little capital stock before a sudden stop, they mainly borrow up to the limit or reduce savings to compensate for consumption losses due to the drop in income. This drags down their net wealth at the end of the sudden stop episode.

Richer households change their assets due to a combination of two effects. First, these households are endowed with larger asset buffers and income during a sudden stop episode. Though they slightly disinvest as TFP goes down today, the degree of disinvestment is smaller than that of households in the 51-94% percentile range of the net wealth distribution. Second, the top 5% of households hold larger debt in normal times to invest in capital. When a sudden stop occurs, they become constrained and immediately cut their debt to accommodate lower borrowing limits. Hence, they significantly reduce their debt, which prevents their net assets from plummeting.

The consumption response of households is a mirror image of the deleveraging that occurs across the net wealth distribution, as shown in panel (B) of Figure 11. The bottom 50% of households drastically reduce consumption as they suffer from drops in income and borrowing limits. The top 5% of households also significantly reduce consumption to pay back debt immediately during a sudden stop. Households in the middle, however, reduce consumption by a lesser amount because they have held almost no debt in normal times, and their consumption reduction

stems from drops in income and the market value of assets.

7 Normative Analysis

Individual households take the price of capital as given, but they do not internalize that their capital demand collectively determines that price in equilibrium and affects the collateral capacity of every household in the economy. The pecuniary externality creates an opportunity for the government to raise welfare by taxing the positions that households carry into the next period. I consider a utilitarian government that recognizes this effect and chooses taxes on capital holdings and debt issuance under commitment. The wealth tax τ_t^w is levied on the market value of the capital stock at the beginning of each period.²² The debt tax τ_t^b is levied on debt issuance. All tax revenues are returned through the lump-sum transfer T_t .

The government's problem is

$$\max_{\{\tau_t^w, \tau_t^b, T_t, q_t\}_{t \geq 0}, \{c_{j,t}, n_{j,t}, k_{j,t+1}, b_{j,t+1}\}_{j,t \geq 0}} \mathbb{E}_0 \sum_{t \geq 0} \beta^t \int u(\tilde{c}_{j,t}) dj \quad (53)$$

$$\begin{aligned} \text{s.t. } c_{j,t} + q_t k_{j,t+1} + \frac{\zeta}{2} (k_{j,t+1} - k_{j,t})^2 + (1 - \tau_t^b) \frac{b_{j,t+1}}{R_t} \\ = z_t \omega_j f(k_{j,t}, n_{j,t}) + q_t (1 - \tau_t^w) (1 - \delta) k_{j,t} + b_{j,t} \\ + \frac{\phi}{2} \frac{(\int [k_{i,t+1} - (1 - \delta) k_{i,t}] di)^2}{\int k_{i,t} di} + T_t, \end{aligned} \quad (54)$$

$$g_{j,t} \equiv \frac{b_{j,t+1}}{R_t} + \theta_t q_t k_{j,t+1} \geq 0, \quad (55)$$

$$q_t = 1 + \phi \frac{\int [k_{j,t+1} - (1 - \delta) k_{j,t}] dj}{\int k_{j,t} dj}, \quad (56)$$

$$T_t = \tau_t^w q_t (1 - \delta) \int k_{j,t} dj - \tau_t^b \int \frac{b_{j,t+1}}{R_t} dj, \quad (57)$$

$$z_t \omega_j f_n(k_{j,t}, n_{j,t}) = v'(n_{j,t}), \quad (58)$$

$$\left[q_t (1 - \theta_t (1 - \tau_t^b)) + \zeta (k_{j,t+1} - k_{j,t}) \right] u'(\tilde{c}_{j,t}) + \theta_t q_t \Omega_{j,t} - \Psi_{j,t} = 0, \quad (59)$$

$$M_{j,t} \equiv (1 - \tau_t^b) u'(\tilde{c}_{j,t}) - \Omega_{j,t} \geq 0, \quad M_{j,t} g_{j,t} = 0. \quad (60)$$

Here $\tilde{c}_{j,t} \equiv c_{j,t} - v(n_{j,t})$ and

$$\Omega_{j,t} \equiv \beta R_t \mathbb{E}_t u'(\tilde{c}_{j,t+1}), \quad (61)$$

²²Without commitment, the government would impose a confiscatory wealth tax.

$$\Psi_{j,t} \equiv \beta \mathbb{E}_t \left[u'(\tilde{c}_{j,t+1}) X_{j,t+1}^{\zeta} \right], \quad (62)$$

$$X_{j,t}^{\zeta} \equiv z_t \omega_j f_k(k_{j,t}, n_{j,t}) + q_t(1 - \tau_t^w)(1 - \delta) + \zeta(k_{j,t+1} - k_{j,t}). \quad (63)$$

Constraints (59) and (60) require the government to respect the investment and saving decisions induced by its policy. The debt tax changes the current financing cost, while the future wealth tax changes the return to capital in $\Psi_{j,t}$. The multipliers $\beta^t \gamma_{j,t}$ and $\beta^t \eta_{j,t}$ on these two conditions measure the cost of changing household investment and saving decisions. Policy commitments made before the initial date remain binding, so the government cannot unexpectedly tax the return on capital chosen in the previous period. The Lagrangian and the full set of first-order conditions are reported in Appendix C.5.

Although the government does not set the capital price q_t directly, it internalizes that the taxes affect q_t . The effect of taxation on q_t enters the government's optimal condition for capital as

$$\mathcal{E}_t^q = q_{K',t}(\mathcal{D}_t + \mathcal{K}_t), \quad (64)$$

where $\mathcal{D}_t$ and $\mathcal{K}_t$ denote distributive and collateral externalities associated with the price changes, as studied by Dávila and Korinek (2018). The collateral externality implies that a higher price relaxes borrowing limits by raising the value of pledgeable capital. The distributive externality implies that a higher price benefits households selling capital at the expense of households purchasing it. The government takes both effects into account when choosing the allocation implemented by the two taxes. The signs of the two externalities are theoretically ambiguous as in Dávila and Korinek (2018).

The two tax instruments operate on different household decisions. Let $\lambda_{j,t}$ denote the social marginal value of wealth that the government assigns to an additional unit of household j 's wealth. The government's optimal condition for the wealth tax is

$$\underbrace{-\text{Cov}_j(\lambda_t, k_t)}_{\text{redistributive motive}} + \underbrace{\int \gamma_{j,t-1} u'(\tilde{c}_{j,t}) dj}_{\text{return on capital chosen in } t-1} = 0. \quad (65)$$

The first term signifies the redistribution motive. Capital is concentrated among wealthy households whose marginal value of wealth is relatively low, and taxation on wealth transfers resources toward households for whom additional consumption is more valuable. This motive gives the government a reason to impose a positive wealth tax. The second term acts in the opposite direc-

tion, as a higher wealth tax discourages investment by lowering the return on investment. The optimal wealth tax balances two motives.

The government's optimal condition with respect to the debt tax is obtained as

$$\underbrace{\frac{1}{R_t} \text{Cov}_j(\lambda_t, b_{t+1})}_{\text{redistributive motive}} + \underbrace{\theta_t q_t \int \gamma_{j,t} u'(\tilde{c}_{j,t}) dj}_{\text{higher investment costs}} - \underbrace{\int \eta_{j,t} u'(\tilde{c}_{j,t}) dj}_{\text{higher saving returns}} = 0. \quad (66)$$

A positive debt tax reduces the proceeds received by debt issuers and returns the revenue through the government budget. This creates a redistribution motive for taxing debt when aggregate debt is concentrated among capital-rich households, as shown in the first term of (66). At the same time, a higher debt tax makes capital purchases today more expensive for those whose borrowing is constrained and increases the return on saving in bonds. The optimal debt tax balances the redistributive motive with its effects on the returns on investment and saving.

Since the signs of the optimal taxes are theoretically ambiguous, I numerically solve for the optimal taxes under commitment and examine how the taxes shape macroeconomic dynamics during sudden stop episodes. I follow the approach of Le Grand and Ragot (2025) to solve for the government's optimal taxes with commitment.²³ I approximate the policy functions to the first order around the ergodic state associated with each value of θ , as described in Appendix E.2.

I compare three allocations to separate the roles of the two tax instruments. The first allocation is the laissez faire allocation without any tax. The second allocation is under the government's optimal debt tax with commitment but without the wealth tax, which is a common policy instrument studied in the literature on sudden stops. The third allocation is under the government's optimal taxation on both wealth and external debt. I simulate all three allocations under the same sequence of aggregate shocks. I identify sudden stop dates from the laissez faire allocation using the definition in Section 6 and compare macroeconomic dynamics in the three economies around the sudden stops.²⁴

Optimal tax rates on wealth and external debt around sudden stop episodes are illustrated in Figure 12. The government's optimal wealth tax rate is 16.6% one period before a sudden stop. It falls to 11.5% in the sudden stop and rises to 20% in the following period. The positive wealth tax

²³I impose a simplifying assumption that the government takes the first moments of aggregate capital and bonds as state variables rather than the entire distribution.

²⁴I solve the three economies following the same algorithm to make the dynamics comparable, which makes sudden stop dynamics in the laissez faire equilibrium numerically different from the equilibrium solved with the Krusell-Smith method, but the dynamics are qualitatively the same.

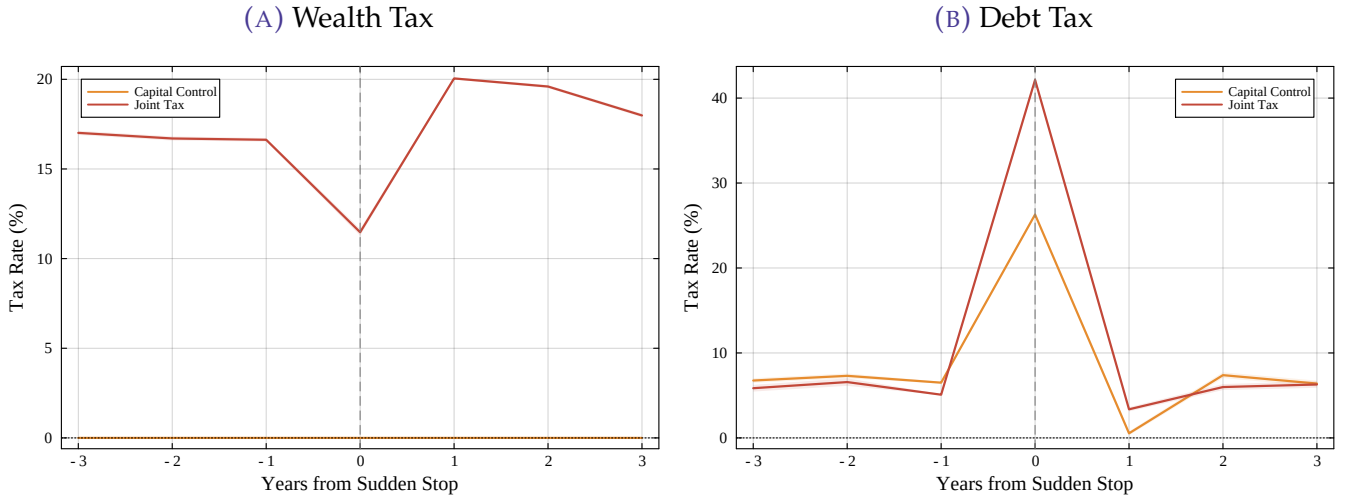


FIGURE 12. Optimal Tax Rates at Baseline Sudden-Stop Dates

Notes: The two panels plot the government's optimal tax rates around the same sudden stop dates generated from the laissez faire allocation. Capital Control denotes the allocation in which the wealth tax is fixed at zero, and Joint Tax denotes the allocation in which the government chooses both taxes.

reflects the redistributive motive in equation (65), since capital is concentrated among households with a lower social marginal value of wealth. Its decline in the sudden stop reflects the opposing effect on the return to capital. TFP and the borrowing limit are low and borrowing costs are high in a typical sudden stop. Committing to a high wealth tax in this state would further weaken the incentive to carry capital into the crisis. The government consequently lowers the wealth tax when the adverse state occurs and raises it after TFP and the borrowing limit recover.

The optimal debt tax moves in the opposite direction. It rises from 5.1% before the sudden stop to 42.1% in the sudden stop, and falls to 3.4% in the following period. Similarly, the debt tax rises during the sudden stop under optimal capital controls without wealth taxes, from 6.5% to 26.3% before falling to 0.5% in the following period. Thus, both optimal policies commit to a high debt tax in the adverse state, but the increase is larger when the government jointly chooses the two taxes. A household entering the sudden stop with $b_{j,t} < 0$ must service this debt when the borrowing limit is tight, while the high tax makes it costly to issue new debt to roll over the existing position. Under commitment, households anticipate this limited ability to roll over debt and carry less debt into states with a low θ . The ex ante benefit of reducing leverage preemptively and the associated collateral externality can therefore outweigh the ex post cost of making consumption smoothing more difficult during the sudden stop. This is in stark contrast to the optimal time-consistent capital control tax without commitment studied by Bianchi and Mendoza

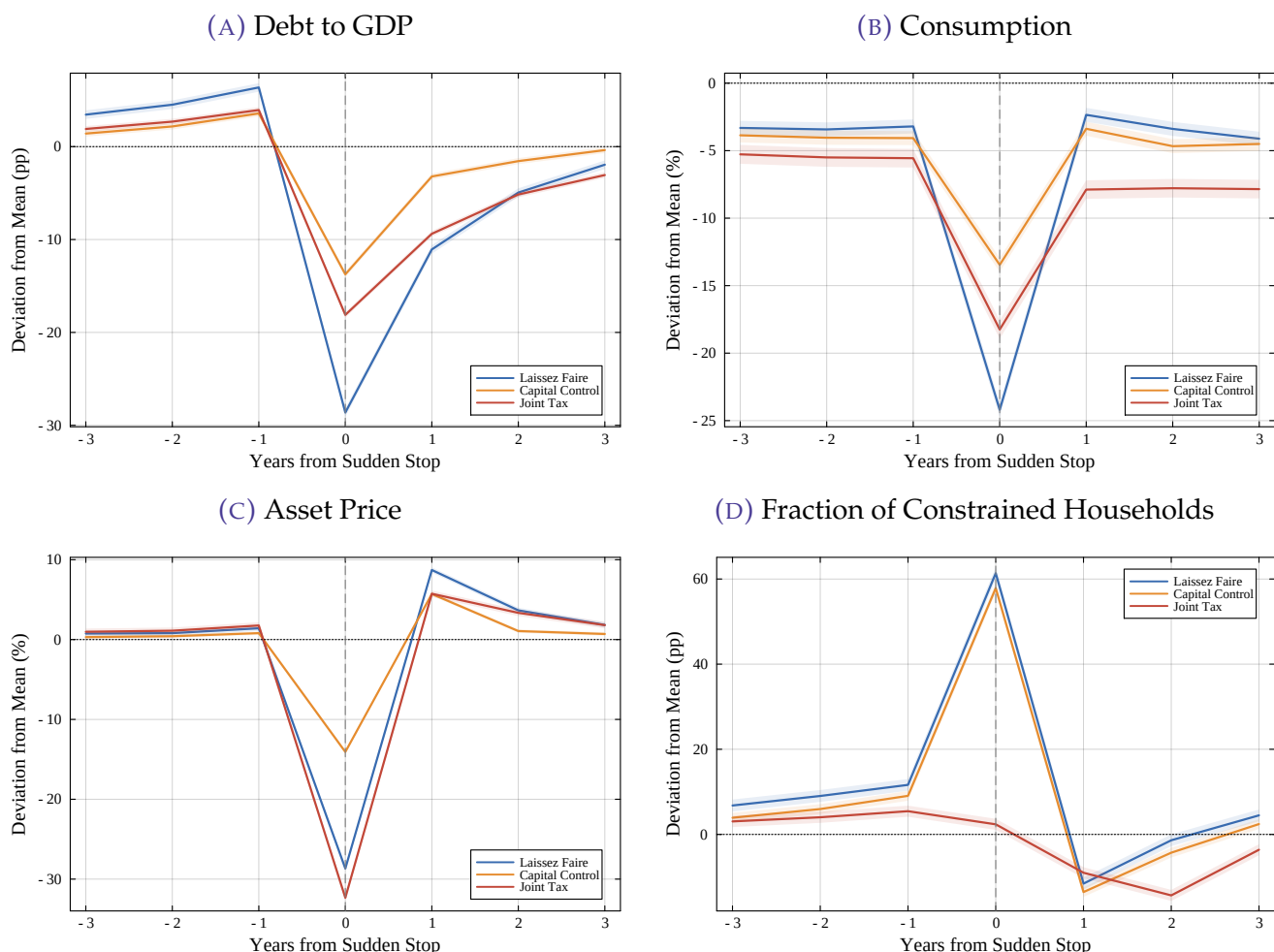


FIGURE 13. Capital Controls and Joint Taxation at Baseline Sudden-Stop Dates

Notes: The plots illustrate percentage and percentage-point deviations from the mean around the sudden stop dates identified from the laissez faire allocation. All three allocations are simulated under the same aggregate shocks.

(2018), under which the government reduces the tax to allow for more borrowing in a bad state.

The comparison with the capital control clarifies the role of the wealth tax in the joint policy. When the debt tax is the only instrument, it must both regulate external borrowing and account for redistribution across bond positions. When both instruments are available, the wealth tax addresses redistribution across capital positions, while the debt tax is used more intensively to prevent carrying debt into a state with a tight borrowing limit. The wealth tax aims to preserve the return to capital, whereas the debt tax rises to limit debt rollover and strengthen the incentive to enter the state with a smaller debt position.

I compare the macroeconomic dynamics around sudden stops in Figure 13. I first compare the laissez faire allocation with optimal joint taxation. One period before a sudden stop, debt is 47.3% of output in the laissez faire allocation and 27.8% under joint taxation. The smaller debt position

reduces the balance-sheet adjustment when the adverse state occurs. In the sudden stop, the debt-to-GDP ratio falls by 28.6 percentage points relative to its mean in the *laissez faire* allocation and by 18.1 percentage points under joint taxation. The current-account reversal similarly falls from 35.8 to 22.5 percentage points. Consumption falls by 24.2% relative to its mean under *laissez faire* and by 18.2% under joint taxation. Most notably, the fraction of constrained households rises by 61.4 percentage points in the *laissez faire* allocation but by only 2.4 percentage points under joint taxation. The lower initial debt position and the redistribution generated by the two taxes therefore prevent the financial shock from pushing a large additional fraction of households against the borrowing limit.

The government does not stabilize the capital price in the joint taxation equilibrium. The capital price falls by 28.7% in the *laissez faire* allocation and by 32.3% under joint taxation, and the investment contraction is also larger under the joint policy. This difference reflects the distributive externality. The government is willing to accept a lower capital price because a lower price makes capital cheaper for households with a high social marginal value of wealth. Joint taxation, therefore, reduces external debt, the consumption contraction, and the increase in constrained households relative to *laissez faire*, but it does not achieve these gains by stabilizing the capital price.

I next compare joint taxation with optimal capital controls. Under optimal capital controls, the government reduces debt to 32.1% of output one period before the sudden stop, which is higher than the 27.8% under joint taxation but well below the *laissez faire* level. Despite entering the sudden stop with more debt than under joint taxation, the capital-control allocation exhibits a smaller contemporaneous adjustment. The debt-to-GDP ratio falls by 13.7 percentage points under capital control, compared with 18.1 percentage points under joint taxation, and the current-account reversal is 17.9 percentage points rather than 22.5 percentage points. Consumption falls by 13.5% and the capital price by 14.1% under capital control, compared with 18.2% and 32.3% under joint taxation. The capital control alone, therefore, provides greater stabilization of external debt, consumption, and the capital price.

This does not mean that the fraction of constrained households in sudden stops is much lower under optimal capital control. The fraction of constrained households rises by 57.9 percentage points under capital control, but by only 2.4 percentage points under jointly optimal taxation.

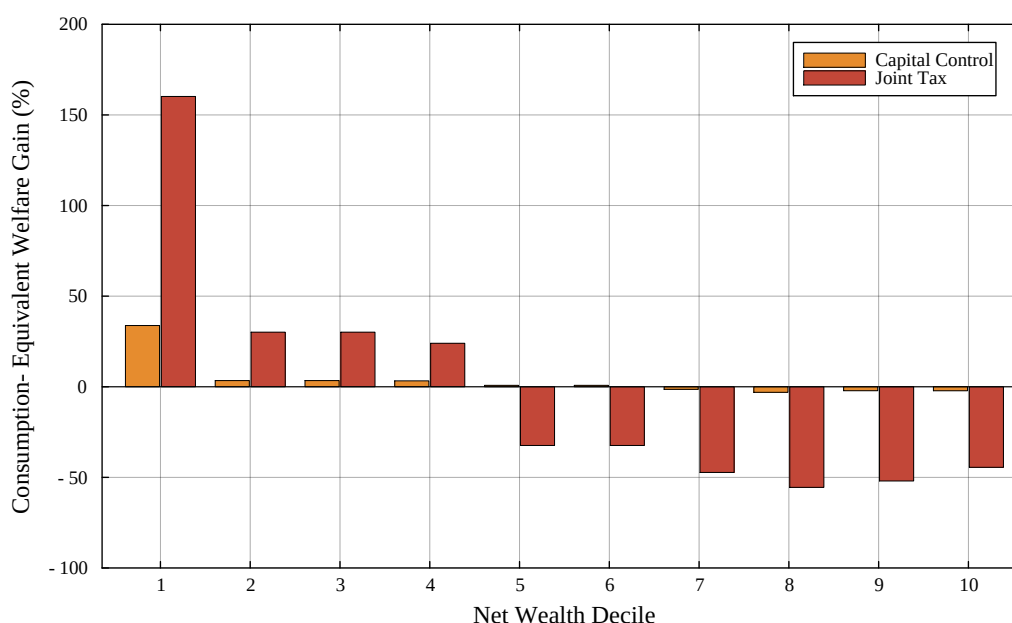


FIGURE 14. Welfare Gains from Capital Controls and Joint Taxation by Net Wealth Decile

Notes: The figure reports consumption-equivalent welfare gains relative to the laissez faire allocation. Households are assigned to net wealth deciles using their rank in the laissez faire allocation.

Adding the wealth tax allows the government to redistribute resources across capital positions and insulate households against the tightening of the borrowing limit. The joint policy accepts a larger adjustment in capital prices and aggregate quantities than capital control alone, but prevents the sudden stop from spreading across household balance sheets.

I conclude the normative analysis by evaluating welfare gains and losses under the two optimal policies across the wealth distribution. Relative to the laissez faire allocation, optimal joint taxation raises social welfare by 17.38% when measured as a consumption-equivalent variation. The optimal capital control raises social welfare by 10.03%. Thus, adding the wealth tax raises the social welfare gain by 7.35 percentage points, even though the capital control alone provides greater stabilization of consumption and the capital price.

Neither policy is a Pareto improvement, as shown in Figure 14. I first compare joint taxation with the laissez faire allocation. Households in the bottom four wealth deciles gain between 24.0% and 160.2%, whereas households in the upper six deciles lose between 32.4% and 55.5%. The distributional consequences are more muted under capital control. Relative to the laissez faire allocation, households in the bottom six wealth deciles gain between 0.8% and 33.8%, while those in the upper four deciles lose between 1.4% and 3.1%. Compared with capital control, joint

taxation generates much larger gains at the bottom and much larger losses across the rest of the wealth distribution. Capital control improves welfare mainly by reducing external debt and the severity of the sudden stop. The joint optimal taxes add redistribution through the wealth tax, which raises utilitarian welfare further but makes its gains and losses substantially more unequal across households.

8 Concluding Remarks

This paper shows that wealth inequality amplifies sudden stops. Cross-country data show that capital outflows of the same size generate larger contractions in more unequal economies, while European household data trace this pattern to leveraged households that reduce both debt and assets together. In a tractable model with households' borrowing limits tied to market values of wealth, greater wealth inequality raises the mass of constrained sellers and deepens declines in asset prices and aggregate consumption. The quantitative model reproduces this amplification and shows that optimal taxes on wealth and external debt improve social welfare by reducing external debt before sudden stops and distributing resources to poorer households. Wealth inequality is central to the propagation of sudden stops.

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A Data Appendix

A.1 List of Sudden Stop Crises

I classify a sudden stop as a country-year observation by applying two filters as in Korinek and Mendoza (2014): (i) when the current account to GDP is 2 standard deviations above a country's mean over time and (ii) when the VIX and JP Morgan EMBI spreads are 1 standard deviation above the mean for advanced economies and emerging market economies, respectively. Table A1 summarizes the years in which sudden stops occurred in 47 of the 54 countries in the sample. Current account reversal episodes that are classified for regression A11 correspond to the country-year observations that satisfy the criterion for the current account to GDP ratio but do not meet the criterion for the VIX and EMBI spreads.

A.2 List of Countries in ECB HFCS and Data Adjustments

I analyze 15 countries in the first and second waves of the ECB HFCS. Table A2 summarizes the reference years of each country in the two waves of the survey and the adjustment factors provided by Household Finance and Consumption Network (2013, 2016) to make the raw data values comparable across countries and time.

Since price levels differ across countries, the HFCS provides adjustment factors to make the nominal values in the data comparable across countries. Using this information, I convert the raw data values of all countries to comparable 2014 euro values as

$$x_{i,1}^{\text{Comp. 2014 Eur.}} = x_{i,1} \times IAF_i \times PPP_{i,2}, \quad (\text{A1})$$

$$x_{i,2}^{\text{Comp. 2014 Eur.}} = x_{i,2} \times PPP_{i,2}, \quad (\text{A2})$$

where $x_{i,t}$ is the raw data value of an observation in country i in wave t , PPP_i is the purchasing power parity adjustment factor in Table A2, and IAF_i is the inflation adjustment factor of country i . I perform this adjustment for all countries except Spain. HFCS revised the data for Spain in 2017 and provided the purchasing power parity adjustment factor of 0.945 for Spain in wave 1, which converts the Spanish data from 2011 values to the comparable 2014 euro value. This implies that the purchasing power parity adjustment factor for Spanish data in 2014 should be 0.908, given the inflation adjustment factor of 1.049.

TABLE A1: List of Sudden Stop Episodes

Country	Year		Country	Year		
Algeria	1991	1999	Malaysia	1994		
Argentina	1990	2002	Mexico	1983	1995	
Australia	2008		Morocco	1983	2003	
Bulgaria	1994	2009	Netherlands	2003	2009	
Brazil	1984	2003	Nigeria	1992	1995	2000
Canada	1982		Norway	1990	1999	2008
Chile	1982		Pakistan	2002	2009	
Colombia	1989	1999	Panama	1995	1998	
Croatia	1998		Peru	1985		
Denmark	1999	2010	Philippines	1984	1998	2009
Dominican Republic	1981	1994	Poland	1999		
Ecuador	1999		Portugal	1982	2012	
Egypt	1991		Russia	1999		
El Salvador	1999		South Africa	1983	1985	
Finland	2000		Spain	2009	2012	
France	1997		Sweden	2001		
Germany	2009		Switzerland	2010		
Greece	2012		Thailand	1998		
Hungary	1995	2009	Türkiye	1994	2001	
Iceland	2001	2009	Ukraine	1999		
Indonesia	1984	1998	United Kingdom	1991		
Italy	1983	2012	Uruguay	1982	2009	
Japan	2015		Venezuela	1983		
Korea	1998					

Notes: Sudden Stop episodes without the household wealth Gini coefficients are excluded from the sample.

TABLE A2: Reference Year by Countries in ECB HFCS

Country	Wave 1		Wave 2		Inflation Adjustment Factor between Waves 1 and 2
	Reference Year	PPP	Reference Year	PPP	
Belgium	2010	0.850	2014	0.802	1.079
Germany	2010	0.907	2014	0.859	1.072
Greece	2009	1.041	2014	1.021	1.066
Spain	2011	0.945	2014	0.908	1.049
France	2009	0.842	2014	0.810	1.081
Italy	2010	0.914	2014	0.848	1.079
Cyprus	2010	1.072	2014	0.968	1.068
Luxembourg	2010	0.796	2014	0.724	1.093
Malta	2010	1.305	2013	1.078	1.069
Netherlands	2009	0.875	2013	0.794	1.091
Austria	2010	0.862	2014	0.824	1.101
Portugal	2010	1.150	2013	1.066	1.079
Slovenia	2010	1.145	2014	1.068	1.074
Slovakia	2010	1.413	2014	1.286	1.094
Finland	2009	0.801	2013	0.713	1.108

Notes: I discarded Estonia, Hungary, Ireland, Latvia, and Poland from the sample because they did not conduct the first wave of the survey. PPP denotes the purchasing power parity adjustment factor provided by Household Finance and Consumption Network (2013, 2016) and is calculated based on the Harmonized Index of Consumer Prices (HICP).

TABLE A3: Population Share by Net Wealth Decile and Housing Tenure (%)

Group	Net wealth decile										Total
	1	2	3	4	5	6	7	8	9	10	
2011											
Renter / non-owner	8.89	7.63	4.72	2.69	1.52	0.75	0.34	0.17	0.13	0.12	27.0
Outright owner	0.18	1.48	3.20	4.33	5.27	5.97	6.44	6.49	6.73	6.60	46.7
Mortgagor	1.05	0.89	2.04	2.98	3.23	3.29	3.20	3.35	3.14	3.24	26.4
2014											
Renter / non-owner	8.57	8.52	4.94	2.76	1.60	0.72	0.30	0.22	0.12	0.12	27.9
Outright owner	0.16	0.64	2.85	4.15	5.34	6.06	6.62	6.92	6.88	6.69	46.3
Mortgagor	1.38	0.78	2.21	3.10	3.06	3.22	3.08	2.87	3.02	3.17	25.9

Notes: Each cell is the average share of households falling in a given net-wealth-decile $\times$ tenure cell. Net wealth deciles are from the ECB HFCS series DNTOP10. A household is a renter/non-owner if it holds no real estate (DA1110= 0 and DA1120= 0), an outright owner if it holds real estate with no mortgage (DL1100= 0), and a mortgagor if it holds real estate financed with a mortgage (DL1100> 0). Shares are computed within each country-year, weighted by the household weight HW0010, and then averaged across the 15 countries separately for each wave; the five implicates are combined by Rubin's rules. Columns are net wealth deciles; the three groups sum to 100 within each wave.

TABLE A4: Gross Asset Share by Net Wealth Decile and Housing Tenure (%)

Group	Net wealth decile										Total
	1	2	3	4	5	6	7	8	9	10	
2011											
Renter / non-owner	0.12	0.27	0.36	0.35	0.37	0.31	0.23	0.18	0.21	0.50	2.9
Outright owner	0.03	0.25	0.82	1.59	2.56	3.82	5.25	7.13	10.42	27.12	59.0
Mortgagor	0.73	0.42	0.99	1.74	2.31	2.80	3.33	4.33	5.69	15.85	38.2
2014											
Renter / non-owner	0.09	0.24	0.36	0.38	0.40	0.32	0.20	0.22	0.17	0.69	3.1
Outright owner	0.03	0.08	0.64	1.40	2.46	3.69	5.27	7.26	10.35	27.19	58.4
Mortgagor	1.00	0.42	1.18	1.92	2.28	2.87	3.33	3.87	5.52	16.22	38.6

Notes: Each cell is the average share of aggregate household gross assets (DA3001) held by a given net-wealth-decile $\times$ tenure cell. Net wealth deciles are from the ECB HFCS series DNTOP10. A household is a renter/non-owner if it holds no real estate (DA1110= 0 and DA1120= 0), an outright owner if it holds real estate with no mortgage (DL1100= 0), and a mortgagor if it holds real estate financed with a mortgage (DL1100> 0). Shares are computed within each country-year, weighted by the household weight HW0010, and then averaged across the 15 countries separately for each wave; the five implicates are combined by Rubin's rules. Columns are net wealth deciles; the three groups sum to 100 within each wave.

A.3 Decomposing the Change in Wealth Gini Coefficients

Wealth inequality increases when either the top households become wealthier or the bottom households become poorer. In light of this, I examine whether the variation in the wealth Gini coefficients is driven by a specific group of households within the distribution or if it signifies a larger dispersion of wealth in general. I decompose the cumulative change in each country's wealth Gini coefficient across the sample period into the parts attributable to the wealth growth of each wealth decile. To be specific, I follow the Shapley value decomposition approach of Shorrocks (2013).

For each country c , I denote the average real net wealth of a household decile $d \in \{1, \dots, 10\}$ in the years 1979 and 2016 as $w_d^{c,0}$ and $w_d^{c,T}$, respectively. Let γ^c be the gross cumulative growth rate of average wealth of country c throughout the sample periods, and $\gamma_c = \mu^{c,T} / \mu^{c,0}$ the gross growth rate of mean wealth. Let $G(\mathbf{w}^{c,0})$ be the Gini coefficient calculated with a vector of wealth levels of all decile groups in country c at the beginning of the sample. If every decile's wealth grew at the common rate γ_c , the whole distribution would simply be rescaled, and the Gini would remain unchanged with $G(\gamma_c \mathbf{w}^{c,0}) = G(\mathbf{w}^{c,0})$. The change in the Gini thus arises because deciles grow at different rates. To attribute it to each individual decile, consider the case when a decile's wealth is the value realized in 2016, while the others are held at the no-change benchmark $\gamma_c w_d^{c,0}$.

The contribution of decile d 's wealth growth to the change in the Gini coefficient is calculated as the weighted average of the difference in Gini by switching decile from the no-change benchmark to its realized value for all possible combinations of the other deciles' wealth values. Following the standard formula of Shapley value, a decile d 's contribution ϕ_d^c is calculated as

$$\phi_d^c = \sum_{S \subseteq \mathcal{N} \setminus \{d\}} \frac{|S|! (N - |S| - 1)!}{N!} \left[G(\mathbf{w}^{c, S \cup \{d\}}) - G(\mathbf{w}^{c, S}) \right], \quad (\text{A3})$$

where $\mathcal{N} \equiv \{1, 2, \dots, 10\}$. $G(\mathbf{w}^{c, S \cup \{d\}})$ is the Gini when decile d and a group of other deciles S is switched to their realized wealth values, while $G(\mathbf{w}^{c, S})$ is the Gini when that group S is switched to their realized wealth values, but decile d remains at its no-change benchmark. Shorrocks (2013) shows that the efficiency property of Shapley values also holds for Gini coefficients, and summing up ϕ_d^c for all deciles exactly yields the difference between Gini at time 0 and T .

I express the contributions ϕ_d^c of a decile d in country c as a fraction of the total change in the Gini coefficient of country c , and take the median of the values across all countries for each

TABLE A5: Contribution of Wealth Deciles to Change in Wealth Gini

Decile	Contribution of Decile (%)
Bottom 10%	−0.0
10–20%	−0.6
20–30%	3.6
30–40%	8.9
40–50%	13.1
50–60%	15.2
60–70%	14.8
70–80%	10.1
80–90%	0.2
Top 10%	32.0
Bottom 50%	25.0
Middle 40%	40.3
Top 10%	32.0

decile. Table A5 summarizes the median contribution of each decile to the change in the wealth Gini. A positive number means that the wealth of the decile changed in a way that increased the Gini, i.e., an upper decile pulled ahead or a lower decile slipped behind. A negative entry means that it lowered the Gini. I observe that the wealth growth of the top 10% contributed the most to the change in wealth inequality over time. However, the upper middle deciles also contributed significantly to the change in wealth inequality. Considering that these groups' wealth levels are typically below average, this implies that the lower part of the distribution significantly lagged behind in terms of wealth growth. Hence, the variation in wealth inequality in the data stems not only from top wealth concentration but also from middle groups becoming poorer. This implies an increase in the dispersion of wealth from both ends.

B Additional Results

B.1 Change in Real Estate Subcategories in Sudden Stops

Figure A1 plots the change in real estate subcategories in sudden stop countries relative to other countries, estimated as (6). Real estate values of mortgagors dropped more in the sudden stop countries than in the comparison countries, whereas there is no significant differences in the real estate growth of outright owners across the two groups of countries. Among the subcategories of real estate, holdings of real estate other than household main residences and the mortgages associated with them dropped the most, consistent with the finding in Panel (A) of Figure 4.

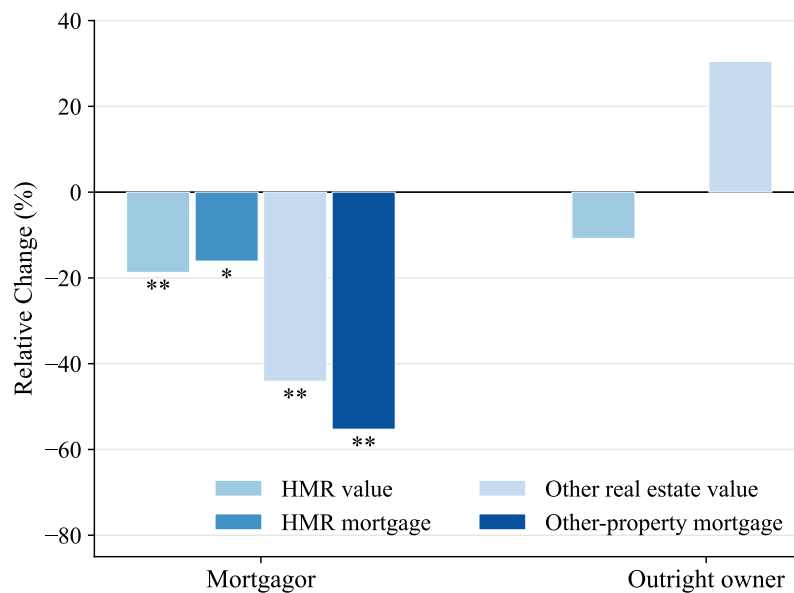


FIGURE A1. Values of Real Estate and Mortgages

Notes: The plot shows changes of the value of the main residence (DA1110), the mortgage on the main residence (DL1110), the value of other real estate (DA1120), and the mortgage secured by other property (DL1100–DL1110). Stars denote one-sided permutation p -values for the predicted direction, with ** and * indicating significance at the 5% and 10% levels.

B.2 Local Projection of the Differential Effects of Wealth Inequality

Wealth inequality incurs deeper drops in aggregate demand and asset prices for a prolonged period after a sudden stop. I estimate local projection versions of the baseline regressions (2) and (3) specified as

$$\Delta y_{i,t+h} = \alpha + \beta \text{Net Flow}_{it}^{SS} + \gamma \text{Net Flow}_{it}^{SS} \times v_{i,t-1} + \Gamma X_{it} + \mu_i + \eta_t + \varepsilon_{it}, \quad (\text{A4})$$

where $\Delta y_{i,t+h}$ is the cumulative growth of the natural logs of GDP, consumption, investment, and stock prices from time t to $t + h$, respectively. Figure A2 shows that the recovery of aggregate demand and asset price takes up to 3 years and the drops are deeper under larger wealth inequality.

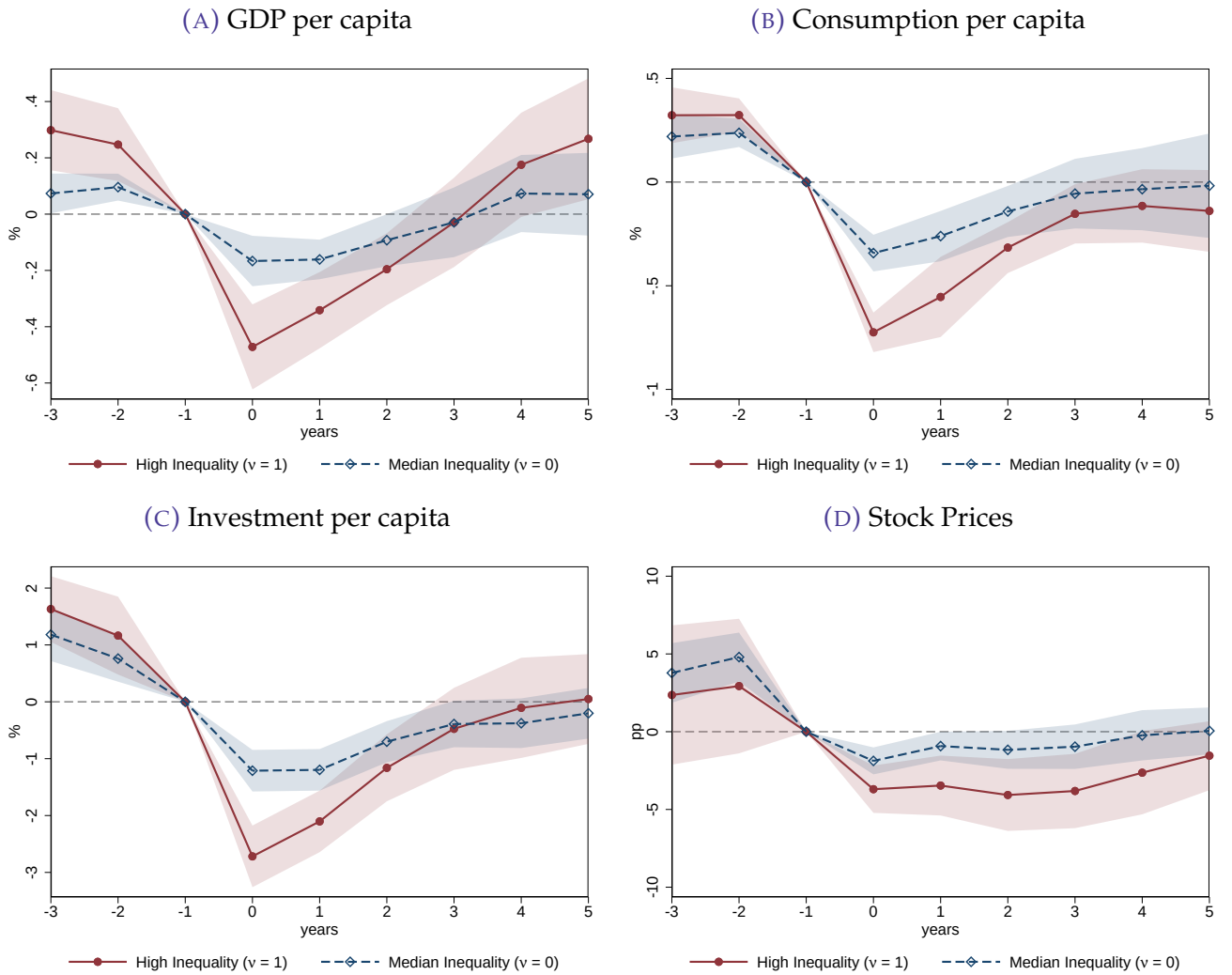


FIGURE A2. Dynamic Effects of Capital Outflows in Sudden Stops

B.3 Regression Controlling for Wealth Gini

I estimate regressions on GDP, consumption, investment, and stock price growth rates as

$$y_{it} = \alpha + \beta \text{Net Flow}_{it}^{SS} + \gamma \text{Net Flow}_{it}^{SS} \times \log \text{Gini}_{i,t-1} + \Gamma X_{it} + \mu_i + \eta_t + \varepsilon_{it}, \quad (\text{A5})$$

$$\Delta \log q_{it} = \alpha + \beta \text{Net Flow}_{it}^{SS} + \gamma \text{Net Flow}_{it}^{SS} \times \log \text{Gini}_{i,t-1} + \Gamma X_{it} + \mu_i + \varepsilon_{it}, \quad (\text{A6})$$

where I replace the inequality measure ν with the natural logarithm of the wealth Gini coefficients.

I confirm that the results are consistent with the baseline results in Table 1. The coefficients of the interaction term become larger in magnitude, and their statistical significance is similar to the baseline regression result.

TABLE A6: Controlling for Wealth Gini Coefficients

	GDP	Consumption	Investment	Stock Price
Net Flow _t ^{SS}	-1.245*** (0.208)	-1.215*** (0.135)	-6.167*** (0.591)	-8.121** (3.166)
Net Flow _t ^{SS} × log Gini _{t-1}	-3.728*** (0.678)	-3.084*** (0.507)	-17.382*** (2.098)	-21.316* (10.747)
log Gini _{t-1}	0.017 (0.022)	-0.030 (0.025)	-0.008 (0.093)	-0.414 (0.289)
GDP _{t-1}	0.690*** (0.046)	0.636*** (0.063)	2.037*** (0.174)	-1.275 (0.972)
Real Exchange Rate _{t-1}	0.002 (0.012)	0.043** (0.018)	0.063 (0.063)	-0.181 (0.133)
Trade Openness _{t-1}	0.000** (0.000)	0.000 (0.000)	0.000 (0.001)	-0.000 (0.001)
Financial Development _{t-1}	0.022 (0.022)	0.052** (0.024)	0.107 (0.112)	-0.773*** (0.264)
Financial Openness _{t-1}	-0.001 (0.001)	-0.002 (0.002)	0.003 (0.012)	0.039 (0.029)
Debt/GDP _{t-1}	-0.001 (0.001)	-0.000 (0.002)	-0.004 (0.005)	-0.009 (0.015)
Reserves/GDP _{t-1}	0.011 (0.019)	0.016 (0.025)	0.041 (0.100)	0.501** (0.245)
Observations	864	864	847	782
Within R ²	0.530	0.432	0.357	0.085

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

B.4 Regression Using the Top-to-Bottom Wealth Ratio

Using a different measure of wealth inequality to compare the sudden stop dynamics across countries does not undermine the baseline result. In Table A7, I replace the wealth Gini coefficients with the ratio of wealth of the top 10% and the bottom 50%, collected from the World Inequality Database, to construct the inequality measure ν in regressions (2) and (3). The regression results are consistent with the baseline results, but the differential effects on stock price growth are insignificant. The contractionary movements of aggregate demand in sudden stops are robust to the choice of measure of wealth inequality.

TABLE A7: Controlling for the Top 10% to Bottom 50% Wealth Ratio

	GDP	Consumption	Investment	Stock Price
Net Flow $_t^{SS}$	-0.083 (0.094)	-0.251*** (0.056)	-0.820*** (0.301)	-1.105* (0.580)
Net Flow $_t^{SS} \times \nu_{t-1}^{T10/B50}$	-0.001*** (0.000)	-0.001*** (0.000)	-0.004*** (0.001)	-0.008** (0.004)
$\nu_{t-1}^{T10/B50}$	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)
GDP $_{t-1}$	0.699*** (0.046)	0.615*** (0.065)	2.035*** (0.208)	-1.333 (0.966)
Real Exchange Rate $_{t-1}$	0.010 (0.012)	0.053*** (0.016)	0.108* (0.056)	-0.194 (0.154)
Trade Openness $_{t-1}$	0.000** (0.000)	0.000 (0.000)	0.001 (0.001)	-0.001 (0.001)
Financial Development $_{t-1}$	0.022 (0.021)	0.058** (0.023)	0.115 (0.099)	-0.869*** (0.288)
Financial Openness $_{t-1}$	-0.001 (0.002)	-0.001 (0.002)	-0.014** (0.006)	0.044 (0.028)
Debt/GDP $_{t-1}$	-0.001 (0.001)	0.000 (0.002)	-0.006 (0.006)	-0.011 (0.016)
Reserves/GDP $_{t-1}$	0.006 (0.017)	0.011 (0.026)	-0.021 (0.082)	0.551** (0.258)
Observations	853	853	836	768
Within R ²	0.548	0.451	0.375	0.096

Notes: The inequality measure $\nu_{t-1}^{T10/B50}$ is constructed exactly as the baseline measure ν_{t-1} in (2) and (3), but applied to the ratio of the top-10% to the bottom-50% household net-wealth share from the World Inequality Database instead of the wealth Gini coefficient. ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

B.5 Marginal Effect of Wealth Inequality

I examine whether wealth inequality explains the fluctuations in aggregate demand and stock prices that are not explained by the control variables used in (2). I orthogonalize each country's wealth inequality measure $v_{i,t-1}$ with respect to the control variables in (2) by estimating

$$v_{i,t-1} = \zeta + \kappa \text{Net Flow}_{it}^{SS} + \Theta X_{it} + \gamma_i + \iota_t + \varepsilon_{it}, \quad (\text{A7})$$

and obtaining $v_{i,t-1}^\perp$ as the regression residuals. Then, I replace $v_{i,t-1}$ with $v_{i,t-1}^\perp$ in (2) as

$$y_{it} = \alpha + \beta \text{Net Flow}_{it}^{SS} + \gamma \text{Net Flow}_{it}^{SS} \times v_{i,t-1}^\perp + \Gamma X_{it} + \mu_i + \eta_t + \varepsilon_{it}, \quad (\text{A8})$$

from which I obtain results that are consistent with the baseline regression estimates.

TABLE A8: Marginal Effect of Wealth Inequality on Sudden Stops

	GDP	Consumption	Investment	Stock Price
Net Flow _t ^{SS}	-0.192*** (0.048)	-0.345*** (0.028)	-1.255*** (0.167)	-2.095*** (0.506)
Net Flow _t ^{SS} × v _{t-1} [⊥]	-0.309*** (0.062)	-0.252*** (0.045)	-1.414*** (0.176)	-1.821** (0.818)
v _{t-1} [⊥]	0.000 (0.002)	-0.003 (0.003)	-0.006 (0.009)	-0.039 (0.024)
GDP _{t-1}	0.684*** (0.045)	0.631*** (0.063)	2.010*** (0.179)	-1.247 (0.974)
Real Exchange Rate _{t-1}	0.006 (0.012)	0.047** (0.018)	0.080 (0.063)	-0.167 (0.130)
Trade Openness _{t-1}	0.000** (0.000)	0.000 (0.000)	0.000 (0.001)	-0.000 (0.001)
Financial Development _{t-1}	0.024 (0.022)	0.052** (0.025)	0.116 (0.112)	-0.783*** (0.258)
Financial Openness _{t-1}	-0.002 (0.001)	-0.002 (0.002)	0.001 (0.012)	0.038 (0.029)
Debt/GDP _{t-1}	-0.001 (0.001)	-0.000 (0.002)	-0.004 (0.005)	-0.009 (0.015)
Reserves/GDP _{t-1}	0.015 (0.018)	0.019 (0.025)	0.059 (0.099)	0.507** (0.241)
Observations	864	864	847	782
Within R ²	0.531	0.432	0.357	0.085

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

B.6 Subsample Regressions

Regressions (2) and (3) exploit within-country variation in wealth inequality to identify its amplifying effect on recessions during sudden stops. However, wealth inequality is highly persistent within each country, which limits the scope of within-country identification. To examine whether the amplifying effect also operates through cross-country differences in wealth inequality levels, I split the sample into two groups based on the cross-country median of average wealth Gini coefficients: countries with above-median inequality and those with below-median inequality. I then estimate the following regressions separately for each group:

$$y_{it} = \alpha + \beta \text{Net Flow}_{it} + \Gamma X_{it} + \mu_i + \eta_t + \varepsilon_{it}, \quad (\text{A9})$$

$$\Delta \log q_{it} = \alpha + \beta \text{Net Flow}_{it} + \Gamma X_{it} + \mu_i + \varepsilon_{it}, \quad (\text{A10})$$

where X_{it} includes the same country characteristics as in (2) and (3). The results show that recessions in sudden stops are more pronounced in the high-inequality subsample than in the low-inequality subsample.

TABLE A9: Subsample Regressions

	GDP		Consumption		Investment		Stock Price	
	High	Low	High	Low	High	Low	High	Low
Net Flow _{<i>t</i>}	-0.710*** (0.166)	-0.057*** (0.016)	-0.739*** (0.125)	-0.232*** (0.048)	-3.739*** (0.438)	-0.616** (0.253)	-4.401*** (1.614)	-1.381** (0.525)
v_{t-1}	0.002 (0.003)	-0.000 (0.001)	0.000 (0.009)	-0.007*** (0.002)	-0.012 (0.014)	-0.009 (0.008)	-0.055 (0.070)	-0.039 (0.031)
GDP _{<i>t-1</i>}	0.706*** (0.044)	0.641*** (0.073)	0.672*** (0.072)	0.573*** (0.082)	2.307*** (0.206)	1.637*** (0.229)	-0.778 (0.788)	-2.306 (1.457)
Real Exchange Rate _{<i>t-1</i>}	-0.017 (0.018)	0.013 (0.015)	0.022 (0.027)	0.056** (0.022)	-0.088 (0.076)	0.167* (0.087)	-0.225 (0.162)	-0.149 (0.194)
Trade Openness _{<i>t-1</i>}	0.000* (0.000)	0.000** (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.001)	0.001 (0.001)	0.000 (0.001)	-0.001 (0.002)
Financial Development _{<i>t-1</i>}	0.020 (0.039)	0.004 (0.017)	0.039 (0.047)	0.049* (0.024)	-0.006 (0.203)	0.118 (0.086)	-0.845* (0.433)	-0.806*** (0.203)
Financial Openness _{<i>t-1</i>}	-0.001 (0.002)	0.002 (0.002)	-0.002 (0.003)	-0.001 (0.002)	0.009 (0.015)	-0.004 (0.008)	0.028 (0.036)	0.066 (0.040)
Debt/GDP _{<i>t-1</i>}	-0.000 (0.001)	-0.001 (0.001)	0.002 (0.001)	-0.002* (0.001)	0.002 (0.003)	-0.011 (0.006)	-0.003 (0.009)	-0.014 (0.028)
Reserves/GDP _{<i>t-1</i>}	0.022 (0.044)	0.009 (0.018)	0.052 (0.060)	0.008 (0.017)	0.224 (0.196)	-0.038 (0.066)	0.530 (0.412)	0.545* (0.265)
Observations	391	471	391	471	374	471	375	407
Within R ²	0.555	0.467	0.456	0.387	0.415	0.287	0.060	0.137

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

B.7 Sudden Stops in Emerging Markets

I show that the baseline results indicate that wealth inequality amplifies contractions and asset market corrections in sudden stops in emerging market economies where asset markets are less developed. I estimate the baseline regressions separately for the emerging market economies in Table A10, while discarding South Korea from the sample because it switched from an emerging market to an advanced economy. The results for emerging market economies are qualitatively consistent with the baseline results.

TABLE A10: Sudden Stops in Emerging Markets and Advanced Economies

	GDP	C	I	Δq
Net Flow _{<i>t</i>}	-0.514** (0.109)	-0.524*** (0.109)	-2.677*** (0.665)	-2.229 (1.757)
Net Flow _{<i>t</i>} ^{SS} × ν_{t-1}	-0.148 (0.132)	-0.182** (0.083)	-0.668* (0.333)	-2.449 (1.456)
ν_{t-1}	-0.001 (0.002)	-0.001 (0.005)	-0.014 (0.009)	-0.037 (0.039)
GDP _{<i>t-1</i>}	0.666*** (0.052)	0.710*** (0.090)	2.167*** (0.195)	-0.803 (1.004)
Real Exchange Rate _{<i>t-1</i>}	-0.011 (0.017)	0.026 (0.026)	-0.052 (0.077)	-0.237 (0.160)
Trade Openness _{<i>t-1</i>}	0.000 (0.000)	0.000 (0.000)	0.000 (0.001)	-0.000 (0.001)
Financial Development _{<i>t-1</i>}	0.024 (0.034)	0.017 (0.036)	0.066 (0.174)	-0.541 (0.355)
Financial Openness _{<i>t-1</i>}	-0.000 (0.002)	-0.001 (0.003)	0.015 (0.016)	0.026 (0.040)
Debt/GDP _{<i>t-1</i>}	-0.008 (0.006)	-0.006 (0.007)	-0.055** (0.024)	-0.023 (0.034)
Reserves/GDP _{<i>t-1</i>}	0.036 (0.036)	0.052 (0.049)	0.324 (0.217)	0.437 (0.447)
Observations	366	366	349	332
Within R ²	0.537	0.441	0.377	0.061

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

B.8 Controlling for Interaction Terms with Other Country Characteristics

The differential effects of wealth inequality on contractionary dynamics in sudden stops in Table 1 are robust to controlling for the interaction terms of the Net Flow^{SS} with the other country characteristics, including the lagged values of GDP per capita to account for differences in average income across countries before a sudden stop, as shown in Table A11.

TABLE A11: Including Interaction Terms with Other Control Variables

	GDP	C	I	Stock Price	GDP	C	I	Stock Price
Net Flow _{<i>t</i>} ^{SS}	-1.166* (0.656)	-0.959*** (0.197)	-4.392*** (1.376)	6.504 (6.482)	0.851 (0.657)	-0.321 (0.444)	2.376 (1.799)	21.610* (10.609)
Net Flow _{<i>t</i>} ^{SS} × ν_{t-1}	-0.512*** (0.093)	-0.329*** (0.057)	-1.587* (0.818)	-5.063** (1.851)	-0.546*** (0.114)	-0.340*** (0.070)	-1.701*** (0.477)	-5.674*** (2.004)
Net Flow _{<i>t</i>} ^{SS} × Debt/GDP _{<i>t-1</i>}	-0.105 (0.065)	-0.054* (0.030)	-0.092 (0.342)	-0.619 (1.037)	-0.014 (0.068)	-0.025 (0.048)	0.214 (0.190)	-0.082 (0.993)
Net Flow _{<i>t</i>} ^{SS} × Reserves/GDP _{<i>t-1</i>}	2.349*** (0.608)	2.091*** (0.594)	13.631*** (2.696)	20.027 (12.657)	5.431*** (1.168)	3.066*** (0.941)	23.975*** (3.103)	42.809** (16.025)
Net Flow _{<i>t</i>} ^{SS} × Financial Development _{<i>t-1</i>}	0.609 (0.770)	0.483* (0.279)	0.764 (1.769)	-16.323 (11.112)	-0.022 (0.435)	0.284 (0.353)	-1.353 (1.361)	-22.080** (9.270)
Net Flow _{<i>t</i>} ^{SS} × Financial Openness _{<i>t-1</i>}	-0.135 (0.125)	0.024 (0.048)	0.448 (0.511)	-0.045 (1.774)	-0.041 (0.095)	0.054 (0.070)	0.763** (0.278)	0.586 (1.347)
Net Flow _{<i>t</i>} ^{SS} × Trade Openness _{<i>t-1</i>}	0.010* (0.005)	0.004 (0.003)	0.015 (0.020)	0.017 (0.062)	0.002 (0.005)	0.001 (0.004)	-0.012 (0.012)	-0.039 (0.064)
Net Flow _{<i>t</i>} ^{SS} × log GDP per capita _{<i>t-1</i>}					-0.136*** (0.037)	-0.043* (0.025)	-0.457*** (0.072)	-0.951* (0.475)
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	864	864	847	782	864	864	847	782
Within R ²	0.543	0.437	0.371	0.094	0.551	0.438	0.377	0.097

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

B.9 Effects of Different Types of Capital Outflows

The baseline results in Table 1 are primarily driven by capital flows other than direct investments, which tend to be long-term investments on physical capital. Table A12 summarizes the estimates from regressions in which I replace the Net Flow^{SS} variable with disaggregated types of capital flows. Debt-like flows are the sum of portfolio debt and other investment flows, which mainly consist of bank loans. Non-DI flows are the sum of portfolio debt, portfolio equity, and other investment flows.

TABLE A12: Effects of Different Types of Capital Outflows

	Total	Debt-like	Non-DI	DI	4-Way
Net Flow _t ^{SS}	-0.154*** (0.047)				
Net Flow _t ^{SS} × ν_{t-1}	-0.309*** (0.064)				
Debt-like _t ^{SS}		-0.171*** (0.054)			
Debt-like _t ^{SS} × ν_{t-1}		-0.212*** (0.075)			
Non-DI _t ^{SS}			-0.210*** (0.066)		
Non-DI _t ^{SS} × ν_{t-1}			-0.215*** (0.074)		
Direct Investment _t ^{SS}				0.011 (0.067)	-0.056 (0.059)
Direct Investment _t ^{SS} × ν_{t-1}				0.251** (0.103)	0.004 (0.109)
Portfolio Equity _t ^{SS}					-0.350** (0.140)
Portfolio Equity _t ^{SS} × ν_{t-1}					-2.071* (1.122)
Portfolio Debt _t ^{SS}					-0.207*** (0.070)
Portfolio Debt _t ^{SS} × ν_{t-1}					-0.412** (0.180)
Other _t ^{SS}					-0.187*** (0.055)
Other _t ^{SS} × ν_{t-1}					-0.215*** (0.067)
ν_{t-1}	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)	0.000 (0.002)	0.003 (0.002)
GDP _{t-1}	0.690*** (0.045)	0.689*** (0.044)	0.692*** (0.043)	0.678*** (0.047)	0.696*** (0.043)
Real Exchange Rate _{t-1}	0.002 (0.012)	0.003 (0.014)	0.002 (0.013)	0.008 (0.016)	0.001 (0.011)
Trade Openness _{t-1}	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
Financial Development _{t-1}	0.022 (0.022)	0.021 (0.022)	0.024 (0.021)	0.026 (0.021)	0.029 (0.020)
Financial Openness _{t-1}	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
Debt/GDP _{t-1}	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
Reserves/GDP _{t-1}	0.011 (0.019)	0.011 (0.018)	0.010 (0.018)	0.013 (0.021)	0.008 (0.017)
Observations	864	864	864	864	864
Within R ²	0.531	0.527	0.533	0.500	0.551

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

B.10 Comparison to Other Crisis Episodes

Crises that did not involve capital outflows did not incur significant declines in GDP, and wealth inequality did not have any differential effects. To show this in Table A13, I estimate the regression

$$y_{it} = \alpha + \beta D_{it} + \gamma D_{it} \times v_{i,t-1} + \Gamma X_{it} + \mu_i + \eta_t + \varepsilon_{it}, \quad (\text{A11})$$

where X_{it} is the set of controls used in regression (2). D_{it} is a dummy variable representing either a sudden stop, a bank-run defined by Jamilov, König, Müller, and Saidi (2025) as a banking crisis without a macroeconomic cause, or a banking crisis classified by Laeven and Valencia (2020). Capital did not flow out of a country in these two alternative crisis episodes on average; the average capital inflow-to-GDP ratio in these episodes is 3.34% for both of them.

TABLE A13: Different Types of Crisis

	Sudden Stop		Bank Run (JKMS)		Banking Crisis (LV)	
	GDP	Stock Price	GDP	Stock Price	GDP	Stock Price
Crisis_t	-0.013** (0.006)	-0.081 (0.056)	-0.010 (0.007)	-0.489* (0.254)	-0.005 (0.006)	-0.590*** (0.117)
$\text{Crisis}_t \times v_{t-1}$	-0.011* (0.006)	-0.133* (0.078)	-0.001 (0.006)	0.052 (0.126)	-0.002 (0.006)	0.065 (0.148)
v_{t-1}	-0.000 (0.002)	-0.062** (0.030)	-0.001 (0.002)	-0.059* (0.029)	-0.001 (0.002)	-0.068** (0.025)
GDP_{t-1}	0.696*** (0.038)	-0.885 (1.087)	0.695*** (0.044)	-0.656 (0.963)	0.693*** (0.046)	-0.553 (0.872)
Real Exchange Rate $_{t-1}$	0.004 (0.006)	0.033 (0.186)	0.007 (0.006)	0.037 (0.170)	0.007 (0.005)	0.086 (0.149)
Trade Openness $_{t-1}$	0.000** (0.000)	0.000 (0.001)	0.000*** (0.000)	-0.000 (0.001)	0.000*** (0.000)	-0.000 (0.001)
Financial Development $_{t-1}$	0.022 (0.016)	-0.875*** (0.301)	0.024 (0.015)	-0.889*** (0.294)	0.025 (0.016)	-0.735*** (0.181)
Financial Openness $_{t-1}$	-0.001 (0.002)	0.026 (0.028)	-0.001 (0.002)	0.028 (0.025)	-0.001 (0.002)	0.023 (0.029)
Debt/GDP $_{t-1}$	-0.001 (0.001)	-0.012 (0.016)	-0.001 (0.001)	-0.006 (0.012)	-0.001 (0.001)	-0.004 (0.010)
Reserves/GDP $_{t-1}$	0.013 (0.013)	0.446* (0.260)	0.015 (0.014)	0.433 (0.275)	0.015 (0.014)	0.274 (0.310)
Constant	-0.036*** (0.011)	0.432 (0.272)	-0.039*** (0.011)	0.470 (0.277)	-0.040*** (0.011)	0.354* (0.206)
Observations	1128	941	1128	941	1128	941
Within R ²	0.527	0.064	0.518	0.112	0.515	0.154

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

B.11 Regressions with Both Wealth and Income Inequality Measures

I estimate the baseline regressions while controlling for the interaction terms and the inequality measures for both the wealth and income Gini coefficients. Table A14 shows that wealth inequality has significant differential effects on sudden stops, even when I control for the income inequality measure and the associated interaction term. As indicated in Table 3, income inequality does not amplify the recessions in sudden stops.

TABLE A14: Regressions with Both Inequality Measures

	GDP	Consumption	Investment	Stock Price
Net Flow $_t^{SS}$	-0.125*** (0.043)	-0.277*** (0.038)	-0.888*** (0.261)	-1.775** (0.720)
Net Flow $_t^{SS} \times \nu_{t-1}$	-0.332*** (0.086)	-0.281*** (0.055)	-1.571*** (0.233)	-1.925* (1.050)
ν_{t-1}	0.001 (0.002)	-0.002 (0.003)	-0.003 (0.009)	-0.051* (0.029)
Net Flow $_t^{SS} \times \nu_{t-1}^{Income}$	0.049 (0.095)	0.060* (0.034)	0.323 (0.209)	0.137 (0.673)
ν_{t-1}^{Income}	-0.000 (0.002)	0.001 (0.002)	0.003 (0.005)	0.043* (0.021)
GDP $_{t-1}$	0.692*** (0.044)	0.639*** (0.062)	2.056*** (0.169)	-1.238 (0.981)
Real Exchange Rate $_{t-1}$	0.002 (0.012)	0.043** (0.019)	0.063 (0.063)	-0.182 (0.130)
Trade Openness $_{t-1}$	0.000** (0.000)	0.000 (0.000)	0.000 (0.001)	-0.001 (0.001)
Financial Development $_{t-1}$	0.024 (0.021)	0.052** (0.025)	0.119 (0.108)	-0.789*** (0.255)
Financial Openness $_{t-1}$	-0.001 (0.001)	-0.002 (0.002)	0.002 (0.012)	0.039 (0.030)
Debt/GDP $_{t-1}$	-0.001 (0.001)	-0.000 (0.002)	-0.004 (0.005)	-0.011 (0.013)
Reserves/GDP $_{t-1}$	0.012 (0.019)	0.016 (0.024)	0.044 (0.103)	0.511** (0.239)
Observations	864	864	847	782
Within R ²	0.531	0.433	0.360	0.088

Notes: ***, **, and * denote significance at 1%, 5%, and 10% levels using the Cameron et al. (2011) two-way clustered standard errors.

C Derivations and Proofs

C.1 Proof of Lemma 1

The first-order conditions of an unconstrained agent i in period 1 satisfy

$$q_1 u'(c_{i1}) = \beta(f'(k_{i2}) + 1), \quad (\text{A12})$$

$$u'(c_{i1}) = 1, \quad (\text{A13})$$

which yields $q_1 R - 1 = f'(k_{i2}) < f'(0)$. Since $f'(0) \rightarrow \infty$, k_{i2} must be positive. Let $(f')^{-1}(q_1 R - 1) = \bar{k}_2$. The unconstrained household's period-1 wealth is $\tilde{w}_i = z_1 f(k_{i1}) + q_1 k_{i1} + b_{i1}$, and under log utility and $\beta R = 1$, its consumption is $c_{i1} = 1$. Hence, its debt choice b_{i2} satisfies

$$\begin{aligned} u' \left(\tilde{w}_i - q_1 \bar{k}_2 - \frac{b_{i2}}{R} \right) &= 1, \\ \iff b_{i2} &= R(\tilde{w}_i - q_1 \bar{k}_2 - 1). \end{aligned} \quad (\text{A14})$$

Hence, b_{i2} is increasing in $\tilde{w}_i$. Therefore, there exists a lower bound for $\tilde{w}_i$ such that $b_{i2} = -\theta q_1 R \bar{k}_2$. The lower bound of $\tilde{w}_i$ for the unconstrained households satisfies

$$\bar{w} = 1 + (1 - \theta) q_1 \bar{k}_2. \quad (\text{A15})$$

Now denote $\bar{k}_1$ as the level of k_{i1} that corresponds to $\tilde{w}_i = 1 + (1 - \theta) q_1 \bar{k}_2$, which exists because both k_{i1} and $\tilde{w}_i$ are monotonic in the initial wealth w_i . To see how productivity z_1 and the asset price q_1 affect $\bar{k}_2$ and $\bar{w}$, I differentiate $\bar{k}_2$ and $\bar{w}$.

1. Unconstrained agents' asset holdings $\bar{k}_2$

Recall that $\bar{k}_2 = (f')^{-1}(q_1 R - 1)$, which does not depend on z_1 . I have

$$\frac{\partial \bar{k}_2}{\partial q_1} = \frac{R}{f''(\bar{k}_2)} = \frac{R \bar{k}_2}{(\alpha - 1) f'(\bar{k}_2)} = -\frac{R \bar{k}_2}{(1 - \alpha)(q_1 R - 1)} < 0, \quad (\text{A16})$$

and the unconstrained agents reduce (increase) asset holdings when the price goes up (down).

2. Threshold level of wealth $\bar{w}$

Since z_1 does not enter the threshold $\bar{w} = 1 + (1 - \theta) q_1 \bar{k}_2$, the productivity shock does not move the threshold directly, and

$$\frac{\partial \bar{w}}{\partial z_1} = 0. \quad (\text{A17})$$

The productivity shock instead shifts the wealth distribution through each household's output, as $\partial \tilde{w}_i / \partial z_1 = f(k_{i1}) > 0$. For completeness, the direct effect of the borrowing tightness θ on $\bar{w}$ is

$$\frac{\partial \bar{w}}{\partial \theta} = -q_1 \bar{k}_2 < 0, \quad (\text{A18})$$

and the effect of q_1 on $\bar{w}$ is obtained as

$$\frac{\partial \bar{w}}{\partial q_1} = -\frac{\alpha q_1 R + 1 - \alpha}{(1 - \alpha)(q_1 R - 1)} (1 - \theta) \bar{k}_2 < 0, \quad (\text{A19})$$

which implies that fewer (more) households are constrained as the wealth threshold goes down (up) when the asset price goes up (down).

C.2 Proof of Lemma 2

Totally differentiating the constrained household's consolidated Euler equation (23) with respect to q_1 and k_{i2} , the price effect of a constrained household's asset demand is²⁵

$$\frac{\partial k_{i2}}{\partial q_1} = \frac{\theta + (1 - \theta)(q_1 k_{i1} - (1 - \theta)q_1 k_{i2} - c_{i1})u''(c_{i1})}{\beta f''(k_{i2}) + (1 - \theta)^2 q_1^2 u''(c_{i1})} < 0. \quad (\text{A20})$$

The elasticity of asset demand for constrained households can be expressed as

$$\begin{aligned} \left| \frac{dk_{i2}}{dq_1} \frac{q_1}{k_{i2}} \right| &= \left| \frac{\theta + (1 - \theta)(q_1 k_{i1} - (1 - \theta)q_1 k_{i2} - c_{i1})u''(c_{i1})}{\beta f''(k_{i2}) + (1 - \theta)^2 q_1^2 u''(c_{i1})} \cdot \frac{q_1}{k_{i2}} \right| \\ &= \frac{\theta c_{i1} + 1 - \theta - (1 - \theta)(q_1 k_{i1} - (1 - \theta)q_1 k_{i2})u'(c_{i1})}{(1 - \alpha)((1 - \theta) - \frac{1}{q_1 R} c_{i1} + \theta c_{i1}) + (1 - \theta)^2 q_1 k_{i2} u'(c_{i1})}, \end{aligned} \quad (\text{A21})$$

where I substitute (23) into the price effect (A20) to obtain the second equality. Dropping the nonpositive term $-(1 - \theta)q_1 k_{i1} u'(c_{i1})$ from the numerator and rewriting the denominator using the identity $(1 - \theta) - \frac{c_{i1}}{q_1 R} + \theta c_{i1} = (1 - \theta)(1 - c_{i1}) + \left(1 - \frac{1}{q_1 R}\right) c_{i1}$, I have

$$\left| \frac{dk_{i2}}{dq_1} \frac{q_1}{k_{i2}} \right| \leq \frac{\theta c_{i1} + 1 - \theta + (1 - \theta)^2 q_1 k_{i2} u'(c_{i1})}{(1 - \alpha) \left[(1 - \theta)(1 - c_{i1}) + \left(1 - \frac{1}{q_1 R}\right) c_{i1} \right] + (1 - \theta)^2 q_1 k_{i2} u'(c_{i1})}. \quad (\text{A22})$$

The denominator strictly exceeds $(1 - \alpha) \left(1 - \frac{1}{q_1 R}\right)$ times the numerator, term by term:

$$(1 - \alpha) \left[(1 - \theta)(1 - c_{i1}) + \left(1 - \frac{1}{q_1 R}\right) c_{i1} \right] - (1 - \alpha) \left(1 - \frac{1}{q_1 R}\right) (\theta c_{i1} + 1 - \theta)$$

²⁵Throughout, I maintain that every constrained household satisfies $\theta c_{i1}^2 + (1 - \theta)(z_1 f(k_{i1}) + b_{i1}) > 0$, which, by the budget constraint, equals c_{i1}^2 times the numerator of (A20) and therefore delivers its sign. Equivalently, $q_1 k_{i1} < \tilde{w}_i + \frac{\theta c_{i1}^2}{1 - \theta}$: no constrained household's capital valuation exceeds its net worth by more than $\frac{\theta c_{i1}^2}{1 - \theta}$. Since $c_{i1} \leq 1$, the condition implies $z_1 f(k_{i1}) + b_{i1} > -\frac{\theta}{1 - \theta}$.

$$\begin{aligned}
&= \frac{(1-\alpha)(1-\theta)(1-c_{i1})}{q_1 R} \geq 0, \\
&(1-\theta)^2 q_1 k_{i2} u'(c_{i1}) \left[1 - (1-\alpha) \left(1 - \frac{1}{q_1 R} \right) \right] > 0,
\end{aligned} \tag{A23}$$

where the first line uses $c_{i1} \leq 1$ for the constrained households and the second uses $(1-\alpha) \left(1 - \frac{1}{q_1 R} \right) <$

1. Therefore

$$\left| \frac{dk_{i2}}{dq_1} \frac{q_1}{k_{i2}} \right| < \frac{q_1 R}{(1-\alpha)(q_1 R - 1)} = \left| \frac{\partial \bar{k}_2}{\partial q_1} \frac{q_1}{\bar{k}_2} \right|, \tag{A24}$$

where the price elasticity of asset demand for the unconstrained households is obtained from Appendix C.1. Since $k_{i2} < \bar{k}_2$ for the constrained households, (A24) implies

$$\left| \frac{\partial k_{i2}}{\partial q_1} \right| < \left| \frac{\partial \bar{k}_2}{\partial q_1} \right|. \tag{A25}$$

C.3 Proof of Proposition 1

I first prove the effects of the shock on asset prices and aggregate consumption, and then I proceed to examine its effect on capital outflows.

Effect on asset price: This is immediate from (27) and any negative shock on z_1 lowers q_1 .

Effect on aggregate consumption: The partial effect of z_1 on a constrained household's consumption is positive by (29). For the general equilibrium term, differentiating $c_{i1} = \tilde{w}_i - (1-\theta)q_1 k_{i2}$ with respect to q_1 yields

$$\frac{\partial c_{i1}}{\partial q_1} = k_{i1} - (1-\theta)k_{i2} - (1-\theta)q_1 \frac{\partial k_{i2}}{\partial q_1}. \tag{A26}$$

With the price effect (A20) and the consolidated Euler equation (23), I obtain

$$\left(\beta f''(k_{i2}) + (1-\theta)^2 q_1^2 u''(c_{i1}) \right) \frac{\partial c_{i1}}{\partial q_1} = \beta f''(k_{i2}) k_{i1} - \alpha(1-\theta)\beta f'(k_{i2}) - (1-\theta)\beta. \tag{A27}$$

Every term on the right-hand side is strictly negative, and the coefficient on the left-hand side is also negative. Therefore, I have

$$\frac{dC_1}{dz_1} = \int_{i \in \mathcal{C}} \left. \frac{\partial c_{i1}}{\partial z_1} \right|_{q_1} di + \int_{i \in \mathcal{C}} \frac{\partial c_{i1}}{\partial q_1} \frac{dq_1}{dz_1} di > 0, \tag{A28}$$

so a negative shock on z_1 decreases aggregate consumption.

Effect on capital outflows: Capital outflow is defined as $\frac{B_2}{R} - B_1$. The balance of payments $C_1 + B_2/R = z_1\bar{F} + B_1$ gives

$$\frac{d}{dz_1} \left(\frac{B_2 - B_1}{R} \right) = \bar{F} - \frac{dC_1}{dz_1}, \quad (\text{A29})$$

so a negative shock generates a capital outflow, $\frac{d}{dz_1} (B_2/R) < 0$, if and only if the fall in aggregate consumption exceeds the fall in output and $dC_1/dz_1 > \bar{F}$.

Let $|\mathcal{C}|$ denote the mass of the constrained set. It is continuous and strictly decreasing in z_1 . A higher realization raises every household's wealth relative to the threshold $\bar{w}$ so $|\mathcal{C}| \rightarrow 0$ as $z_1 \rightarrow \infty$ while $|\mathcal{C}| \rightarrow 1$ as $z_1 \rightarrow 0$ when the wealth distribution is bounded so that no household remains rich enough to stay unconstrained at vanishing productivity. The map $z_1 \mapsto dC_1/dz_1 - \bar{F}$ is continuous since aggregate consumption is continuous in z_1 and across $\bar{w}$, where $c_{i1} \rightarrow 1$. As $|\mathcal{C}| \rightarrow 0$, unconstrained households hold $c_{i1} = 1$ fixed. Thus, dC_1/dz_1 in (30) integrates over the constrained set, and the price response dq_1/dz_1 in (27) has its numerator over that set; both tend to zero, and $dC_1/dz_1 - \bar{F} \rightarrow -\bar{F} < 0$. As $|\mathcal{C}| \rightarrow 1$, the constraint binds for every household. Hence, market clearing gives $\int k_{i2} di = \int k_{i1} di = \bar{K}$ and $C_1 = z_1\bar{F} + B_1 + \theta q_1\bar{K}$, whence $dC_1/dz_1 - \bar{F} = \theta\bar{K}dq_1/dz_1 > 0$. By the intermediate value theorem, the set $\{z_1 : dC_1/dz_1 \leq \bar{F}\}$ is nonempty; let z^* be its infimum. Then $z^* > 0$, $dC_1/dz_1|_{z^*} = \bar{F}$ by continuity and $dC_1/dz_1 > \bar{F}$ for every $z_1 < z^*$ by minimality with no single-crossing assumption. An economy whose productivity is already low $z_1 < z^*$ therefore runs a capital outflow in response to a further negative shock. The fully constrained limit only serves to indicate $dC_1/dz_1 - \bar{F}$ as $|\mathcal{C}| \rightarrow 1$; the threshold z^* lies above it, where households are only partially constrained.

C.4 Proof of Proposition 2

I begin with the sketch of the proof. By assumption, the threshold $\bar{w}$ lies below the mean of $\Phi(q_1^F, F)$, and since the threshold falls and every household's wealth rises as the price rises, the same ranking extends to every price above q_1^F . A larger spread then places more mass below the threshold at any such price, which lowers aggregate asset demand. Hence, the asset price must fall to clear the market: $q_1^F < q_1^G$. The fraction of constrained households is even larger under F . More constrained households raise the overall direct effect of the productivity shock in the numerator of (27) and, by Lemma 2, lower the average price elasticity of asset demand

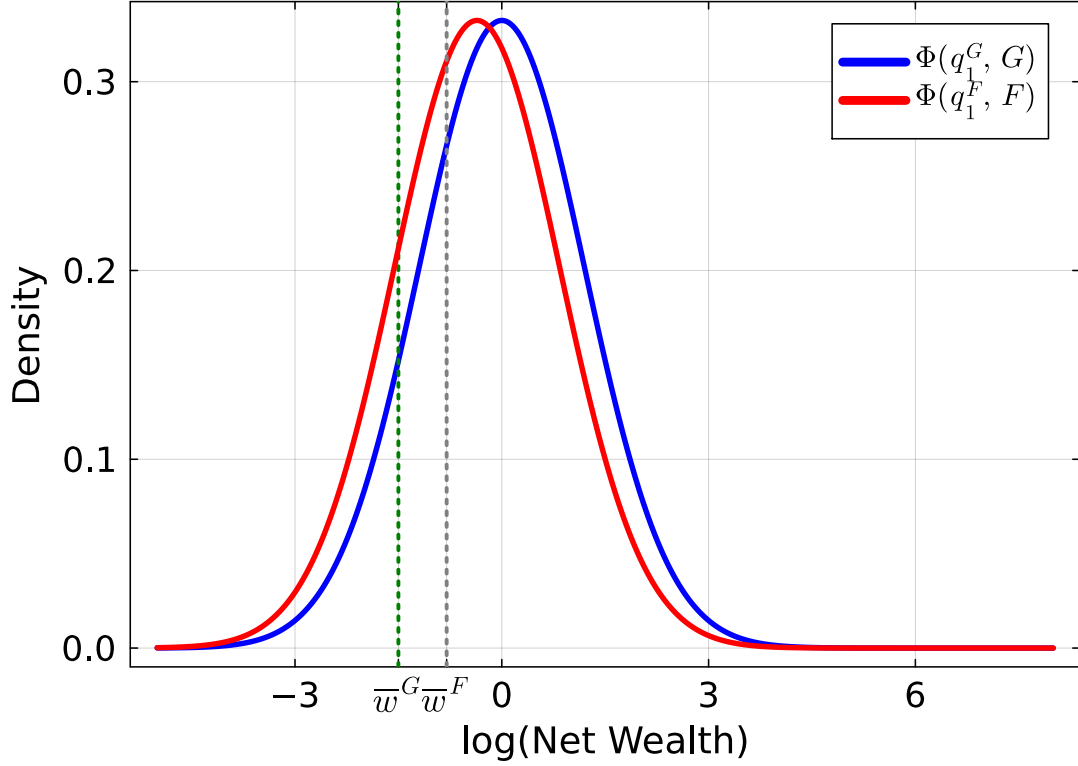


FIGURE A3. Proof of Proposition 2

in the denominator. Hence, the drop in the asset price is larger when wealth is more unequally distributed at the beginning of period 1. Figure A3 graphically illustrates the proof. Note that the distribution of $\Phi(q_1^F, F)$ shifts to the left relative to $\Phi(q_1^G, F)$ as the asset price decreases.

Formally, the proposition makes three claims as follows:

- (i) the equilibrium asset price is lower under the more unequal distribution: $q_1^F < q_1^G$;
- (ii) the price response to the shock is larger: $\left. \frac{dq_1}{dz_1} \right|_{\Phi(q_1^G, G)} < \left. \frac{dq_1}{dz_1} \right|_{\Phi(q_1^F, F)}$;
- (iii) the consumption response to the shock is larger: $\left. \frac{dC_1}{dz_1} \right|_{\Phi(q_1^G, G)} < \left. \frac{dC_1}{dz_1} \right|_{\Phi(q_1^F, F)}$.

By assumption, G and F are members of the family of continuous distributions Φ_λ with $\lambda \in [1, \lambda_F]$, where $\lambda = 1$ corresponds to G and λ_F corresponds to F . I write the equilibrium price that clears the market under Φ_λ as $q_1(\lambda)$, so $q_1(1) = q_1^G$ and $q_1(\lambda_F) = q_1^F$. At every price q_1 between q_1^F and q_1^G , the induced wealth distributions satisfy $\Phi_\lambda(w) = \lambda \Phi_1(w)$ at every wealth level w below the mean, with the mean held fixed. Households are indexed by their position on the period-0 policy schedule, along which (k_{i1}, b_{i1}) and hence $\tilde{w}_i$ are functions of the index. $\tilde{w}_i$ is strictly increasing in the index at every price in $[q_1^F, q_1^G]$; I therefore index households equivalently

by $\tilde{w}_i$, and $|\mathcal{C}| = \Phi_\lambda(\bar{w})$ denotes the fraction of constrained households. Since k_{i1} is a function of the index, the proportionality applies not only to the wealth distribution but also to the integral of any household-level response over a region below the mean, which is how it is used below: every integrand appearing in (27) and (30) depends on the household only through $(k_{i1}, \tilde{w}_i)$ and hence through the index alone.

Claim (i): The mean of the wealth distribution, $z_1 \int f(k_{i1}) d\Phi_\lambda + \int b_{i1} d\Phi_\lambda + q_1 \bar{K}$, is common across λ and is strictly increasing in q_1 , while the threshold $\bar{w} = 1 + (1 - \theta)q_1 \bar{k}_2(q_1)$ is strictly decreasing in q_1 by Lemma 1. Since $\bar{w}$ lies below the mean at q_1^F by assumption, the threshold lies strictly below the mean at every price $q_1 \geq q_1^F$. At a given price q_1 , aggregate asset demand is $\int_{\tilde{w}_i < \bar{w}} k_{i2}(\tilde{w}_i; q_1) d\Phi_\lambda + \bar{k}_2(q_1) (1 - \Phi_\lambda(\bar{w}))$, where the individual demand $k_{i2}(\tilde{w}_i; q_1)$ is bounded, nondecreasing in $\tilde{w}_i$, constant above $\bar{w}$, and continuous at $\bar{w}$. For any price q_1 between q_1^F and q_1^G and $\lambda' > \lambda$, the region below the threshold lies below the mean, and the proportionality of the family applies; thus, integrating by parts, aggregate demand under $\Phi_{\lambda'}$ falls short of aggregate demand under Φ_λ by

$$(\lambda' - \lambda) \int_{\tilde{w}_i < \bar{w}} \frac{\partial k_{i2}}{\partial \tilde{w}_i} \Phi_1(\tilde{w}_i) d\tilde{w}_i > 0, \quad (\text{A30})$$

so a larger λ creates an excess supply at any such price. Differentiating aggregate demand with respect to q_1 yields

$$\frac{d}{dq_1} \left[\int_{\tilde{w}_i < \bar{w}} k_{i2} d\Phi_\lambda + \bar{k}_2 (1 - \Phi_\lambda(\bar{w})) \right] = - \int_{\tilde{w}_i < \bar{w}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| d\Phi_\lambda - (1 - |\mathcal{C}|) \left| \frac{\partial \bar{k}_2}{\partial q_1} \right| < 0, \quad (\text{A31})$$

the negative of the denominator of (27), since $\frac{\partial k_{i2}}{\partial q_1} < 0$ by (A20), and the boundary terms cancel due to the continuity of asset demand at $\bar{w}$. Three consequences follow. First, applying (A30) at q_1^F with $\lambda' = \lambda_F$ shows that aggregate demand at q_1^F under Φ_λ exceeds its value under Φ_{λ_F} , which is $\bar{K}$; with (A31), $q_1(\lambda) > q_1^F$ for every $\lambda < \lambda_F$, and in particular $q_1^F < q_1^G$. Second, applying (A30) at q_1^G shows that aggregate demand at q_1^G under Φ_λ falls short of $\bar{K}$ for every $\lambda > 1$, so $q_1(\lambda) < q_1^G$. Hence, every equilibrium price along the family lies between q_1^F and q_1^G , the price region on which the proportionality of the family applies. Third, applying (A30) at $q_1(\lambda)$ gives $q_1(\lambda') < q_1(\lambda)$ for $\lambda' > \lambda$, so the equilibrium price is strictly decreasing in λ . Denoting the excess supply generated per unit of λ by $\chi \equiv \int_{\tilde{w}_i < \bar{w}} \frac{\partial k_{i2}}{\partial \tilde{w}_i} \Phi_1 d\tilde{w}_i$, implicit differentiation of the market clearing condition with (A31) shows that $\frac{dq_1}{d\lambda} < 0$ along the family, with magnitude $\left| \frac{dq_1}{d\lambda} \right|$ equal to χ divided by the

denominator of (27).

The fraction of constrained households along the family is bounded by its value under F . For $q_1 \geq q_1^F$, every household's wealth is weakly higher, and the threshold is weakly lower than at q_1^F , so $\Phi_1(\bar{w}(q_1); q_1) \leq \Phi_1(\bar{w}(q_1^F); q_1^F)$. Hence, since $\bar{w}$ lies strictly below the median of $\Phi(q_1^F, F)$,

$$|\mathcal{C}|(\lambda) = \lambda \Phi_1(\bar{w}(q_1(\lambda)); q_1(\lambda)) \leq \lambda_F \Phi_1(\bar{w}(q_1^F); q_1^F) = \Phi(\bar{w}; q_1^F, F) < \frac{1}{2}, \quad (\text{A32})$$

which is the only place where the median condition enters the argument.

Household-level responses: Claims (ii) and (iii) rest on the same household-level objects, which I record once. Every constrained household satisfies

$$\frac{\partial k_{i2}}{\partial z_1} = \frac{(1-\theta)q_1 f(k_{i1})}{\beta |f''(k_{i2})| c_{i1}^2 + (1-\theta)^2 q_1^2}, \quad (\text{A33})$$

$$\left| \frac{\partial k_{i2}}{\partial q_1} \right| = \frac{\theta c_{i1}^2 + (1-\theta)c_{i1} - (1-\theta)q_1 (k_{i1} - (1-\theta)k_{i2})}{\beta |f''(k_{i2})| c_{i1}^2 + (1-\theta)^2 q_1^2}, \quad (\text{A34})$$

$$\left. \frac{\partial c_{i1}}{\partial z_1} \right|_{q_1} = f(k_{i1}) \cdot \frac{\beta |f''(k_{i2})| c_{i1}^2}{\beta |f''(k_{i2})| c_{i1}^2 + (1-\theta)^2 q_1^2}, \quad (\text{A35})$$

$$\begin{aligned} \frac{\partial c_{i1}}{\partial q_1} &= k_{i1} \cdot \frac{\beta |f''(k_{i2})| c_{i1}^2}{\beta |f''(k_{i2})| c_{i1}^2 + (1-\theta)^2 q_1^2} + \alpha(1-\theta) \cdot \frac{\beta f'(k_{i2}) c_{i1}^2}{\beta |f''(k_{i2})| c_{i1}^2 + (1-\theta)^2 q_1^2} \\ &\quad + (1-\theta) \cdot \frac{\beta c_{i1}^2}{\beta |f''(k_{i2})| c_{i1}^2 + (1-\theta)^2 q_1^2} > 0. \end{aligned} \quad (\text{A36})$$

Substituting $|\partial k_{i2}/\partial q_1|$ into the budget constraint gives

$$q_1 \frac{\partial c_{i1}}{\partial q_1} = \frac{q_1 (k_{i1} - (1-\theta)k_{i2}) \beta |f''(k_{i2})| c_{i1}^2 + (\theta c_{i1}^2 + (1-\theta)c_{i1}) (1-\theta) q_1^2}{\beta |f''(k_{i2})| c_{i1}^2 + (1-\theta)^2 q_1^2}.$$

Since the numerator of (A34) is positive by (A20), $(1-\theta)q_1 (k_{i1} - (1-\theta)k_{i2}) < \theta c_{i1}^2 + (1-\theta)c_{i1}$, so the first summand is at most $\frac{c_{i1}(\theta c_{i1} + 1 - \theta)}{1 - \theta}$ times the share of $\beta |f''(k_{i2})| c_{i1}^2$ in the denominator, while the second equals $\frac{c_{i1}(\theta c_{i1} + 1 - \theta)}{1 - \theta}$ times the share of $(1-\theta)^2 q_1^2$. Because the two shares sum to one,

$$\frac{q_1}{c_{i1}} \frac{\partial c_{i1}}{\partial q_1} \leq \frac{\theta c_{i1} + 1 - \theta}{1 - \theta} \leq \frac{1}{1 - \theta}, \quad \text{and} \quad \frac{q_1}{k_{i2}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| < \frac{q_1 R}{(1-\alpha)(q_1 R - 1)}, \quad (\text{A37})$$

and the second inequality holds by Lemma 2. The denominator common to (A33)–(A36) is the sum of two positive terms whose logarithmic derivatives along the household's re-optimization are $(2-\alpha) \frac{q_1}{k_{i2}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| + 2 \frac{q_1}{c_{i1}} \frac{\partial c_{i1}}{\partial q_1}$ and 2, both nonnegative and, by (A37), both at most $\frac{(2-\alpha)q_1 R}{(1-\alpha)(q_1 R - 1)} + \frac{2}{1-\theta}$. Since the logarithmic derivative of a sum of positive terms is a convex combination of those

of its summands, we have

$$0 \leq \frac{d \log (\beta |f''(k_{i2})| c_{i1}^2 + (1-\theta)^2 q_1^2)}{d \log q_1} \leq \frac{(2-\alpha) q_1 R}{(1-\alpha)(q_1 R - 1)} + \frac{2}{1-\theta}. \quad (\text{A38})$$

Two consequences follow. The direct effect (A33) is $(1-\theta) q_1 f(k_{i1})$ over that denominator, so $\frac{d \log (\partial k_{i2} / \partial z_1)}{d \log q_1} \leq 1$ by the lower half of (A38). The numerator of (A34) rises with the price at rate $2\theta c_{i1} \frac{\partial c_{i1}}{\partial q_1} > 0$ by the budget constraint, so $\frac{d \log |\partial k_{i2} / \partial q_1|}{d \log q_1}$ is at least the negative of the right-hand side of (A38).

Condition (19) converts these into numbers. Every constrained household demands less than $\bar{k}_2$, so market clearing gives $\bar{k}_2 > \bar{K}$ and hence $1 - \frac{1}{q_1 R} = \frac{\alpha \bar{k}_2^{\alpha-1}}{1 + \alpha \bar{k}_2^{\alpha-1}} < \frac{\alpha \bar{K}^{\alpha-1}}{1 + \alpha \bar{K}^{\alpha-1}}$, ensuring that (19) delivers

$$\left(3 + \frac{4}{1-\theta}\right) (1-\alpha) \left(1 - \frac{1}{q_1 R}\right) < \alpha, \quad (\text{A39})$$

and in particular $(1-\alpha) \left(1 - \frac{1}{q_1 R}\right) < \frac{\alpha}{7}$ and $\frac{2(1-\alpha)}{1-\theta} \left(1 - \frac{1}{q_1 R}\right) < \frac{\alpha}{2}$. Moreover, by Bernoulli's inequality, (19) gives $\bar{K} \geq \left(1 + (1-\alpha) \left(3 + \frac{4}{1-\theta}\right)\right)^{\frac{1}{1-\alpha}} \geq 4 + \frac{4}{1-\theta} \geq \frac{4}{1-\theta}$, so that with $q_1 > \frac{1}{R} \geq \frac{1}{2}$,

$$\frac{(1-\alpha) \left(1 - \frac{1}{q_1 R}\right)}{(1-\theta)^2 q_1 \bar{K}} \leq \frac{1-\alpha}{1-\theta} \left(1 - \frac{1}{q_1 R}\right) \cdot \frac{1}{4 q_1} < \frac{\alpha}{8}. \quad (\text{A40})$$

Claim (ii): Three forces act on $\frac{dq_1}{dz_1}$ as λ rises. At the going price, the family puts proportionally more households on the constrained side; the price falls, which pushes more households across the threshold; and the price fall re-evaluates the response of every household already on each side. Only the third can work against the result, and I show that the first dominates it. Two facts prepare the comparison. By the level comparison (A25), the denominator of (27) is squeezed between the unconstrained households' slope and its own unconstrained part,

$$(1 - |\mathcal{C}|) \left| \frac{\partial \bar{k}_2}{\partial q_1} \right| \leq \int_{\tilde{w}_i < \bar{w}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| d\Phi_\lambda + (1 - |\mathcal{C}|) \left| \frac{\partial \bar{k}_2}{\partial q_1} \right| \leq \left| \frac{\partial \bar{k}_2}{\partial q_1} \right|, \quad (\text{A41})$$

and, since $\chi = \frac{1}{\lambda} \int_{\tilde{w}_i < \bar{w}} (\bar{k}_2 - k_{i2}) d\Phi_\lambda$ with $0 < k_{i2} < \bar{k}_2$ gives $\lambda \chi \leq |\mathcal{C}| \bar{k}_2$, the closed form $\left| \frac{\partial \bar{k}_2}{\partial q_1} \right| = \frac{R \bar{k}_2}{(1-\alpha)(q_1 R - 1)}$, which rearranges to $\frac{\bar{k}_2}{q_1} = (1-\alpha) \left(1 - \frac{1}{q_1 R}\right) \left| \frac{\partial \bar{k}_2}{\partial q_1} \right|$, bounds the speed of the price along the family,

$$\frac{\lambda}{q_1} \left| \frac{dq_1}{d\lambda} \right| \leq \frac{|\mathcal{C}|}{1 - |\mathcal{C}|} (1-\alpha) \left(1 - \frac{1}{q_1 R}\right) < (1-\alpha) \left(1 - \frac{1}{q_1 R}\right), \quad (\text{A42})$$

using (A32) in the last step. Let $\varepsilon \equiv g_\lambda(\bar{w}) \left[k_{i1}(\bar{w}) + \left| \frac{\partial \bar{w}}{\partial q_1} \right| \right] \left| \frac{dq_1}{d\lambda} \right| \geq 0$ be the mass of households crossing the threshold per unit increase of λ . By Leibniz's rule, using $\Phi_\lambda = \lambda \Phi_1$ below the mean at the price $q_1(\lambda)$, which lies between q_1^F and q_1^G by Claim (i),

$$\frac{d}{d\lambda} \int_{\bar{w}_i < \bar{w}} \frac{\partial k_{i2}}{\partial z_1} d\Phi_\lambda = \frac{1}{\lambda} \int_{\bar{w}_i < \bar{w}} \frac{\partial k_{i2}}{\partial z_1} d\Phi_\lambda + \varepsilon \frac{\partial k_{i2}}{\partial z_1} \Big|_{\bar{w}^-} + \frac{dq_1}{d\lambda} \int_{\bar{w}_i < \bar{w}} \frac{d}{dq_1} \left(\frac{\partial k_{i2}}{\partial z_1} \right) d\Phi_\lambda, \quad (\text{A43})$$

$$\begin{aligned} \frac{d}{d\lambda} \left[\int_{\bar{w}_i < \bar{w}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| d\Phi_\lambda + (1 - |\mathcal{C}|) \left| \frac{\partial \bar{k}_2}{\partial q_1} \right| \right] &= \frac{1}{\lambda} \left[\int_{\bar{w}_i < \bar{w}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| d\Phi_\lambda - |\mathcal{C}| \left| \frac{\partial \bar{k}_2}{\partial q_1} \right| \right] + \varepsilon \left(\left| \frac{\partial k_{i2}}{\partial q_1} \right|_{\bar{w}^-} - \left| \frac{\partial \bar{k}_2}{\partial q_1} \right| \right) \\ &\quad + \frac{dq_1}{d\lambda} \left[\int_{\bar{w}_i < \bar{w}} \frac{d}{dq_1} \left| \frac{\partial k_{i2}}{\partial q_1} \right| d\Phi_\lambda + (1 - |\mathcal{C}|) \frac{d}{dq_1} \left| \frac{\partial \bar{k}_2}{\partial q_1} \right| \right], \end{aligned} \quad (\text{A44})$$

the three terms in each line are the three forces in the order stated above. I take $\frac{d}{d\lambda} \log \frac{dq_1}{dz_1}$, the difference between the logarithmic λ -derivatives of the numerator and the denominator of (27), term by term, and measure every term against the first.

The first term of (A43) is $\frac{1}{\lambda}$ times the numerator of (27), while the first terms of (A44) equal $\frac{1}{\lambda}$ times the denominator of (27) minus $\frac{1}{\lambda} \left| \frac{\partial \bar{k}_2}{\partial q_1} \right|$, by adding and subtracting $\frac{1}{\lambda} (1 - |\mathcal{C}|) \left| \frac{\partial \bar{k}_2}{\partial q_1} \right|$. Their combined contribution is

$$\frac{1}{\lambda} \cdot \frac{\left| \frac{\partial \bar{k}_2}{\partial q_1} \right|}{\int_{\bar{w}_i < \bar{w}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| d\Phi_\lambda + (1 - |\mathcal{C}|) \left| \frac{\partial \bar{k}_2}{\partial q_1} \right|} \geq \frac{1}{\lambda}, \quad (\text{A45})$$

by the upper half of (A41). The ε -terms contribute nonnegatively: a household crossing the threshold begins to sell assets in response to the shock and raises the numerator of (27), while its demand becomes less price-sensitive than that of an unconstrained household's by (A25) and lowers the denominator.

For the third term, one further bound is needed since the constrained households' aggregate price slope enters the denominator feedback. Because the common denominator in (A34) is at least $(1 - \theta)^2 q_1^2$, $k_{i1} \geq 0$, and $c_{i1} \leq 1$, each constrained household satisfies $\left| \frac{\partial k_{i2}}{\partial q_1} \right| \leq \frac{1}{(1 - \theta)^2 q_1^2} + \frac{k_{i2}}{q_1}$. Integrating with $\int_{\bar{w}_i < \bar{w}} k_{i2} d\Phi_\lambda \leq \bar{K}$ from the market clearing condition (26) yields

$$\int_{\bar{w}_i < \bar{w}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| d\Phi_\lambda \leq \frac{|\mathcal{C}|}{(1 - \theta)^2 q_1^2} + \frac{\bar{K}}{q_1} \leq (1 - \alpha) \left(1 - \frac{1}{q_1 \bar{R}} \right) \left[1 + \frac{|\mathcal{C}|}{(1 - \theta)^2 q_1 \bar{K}} \right] \left| \frac{\partial \bar{k}_2}{\partial q_1} \right|, \quad (\text{A46})$$

the second inequality uses $\frac{\bar{K}}{q_1} \leq \frac{\bar{k}_2}{q_1} = (1 - \alpha) \left(1 - \frac{1}{q_1 \bar{R}} \right) \left| \frac{\partial \bar{k}_2}{\partial q_1} \right|$. Now measure the third terms

against (A45). For the unconstrained households, the closed form of $\left| \frac{\partial \bar{k}_2}{\partial q_1} \right|$ gives $\frac{d}{dq_1} \left| \frac{\partial \bar{k}_2}{\partial q_1} \right| = -(2 - \alpha) \left| \frac{\partial \bar{k}_2}{\partial q_1} \right|^2 \frac{1}{\bar{k}_2}$. So, with $\lambda \chi \leq |C| \bar{k}_2$ and the lower half of (A41), their piece is at least $-(2 - \alpha)|C|$ times (A45). For the constrained households, the numerator side uses $\frac{d \log(\partial k_{i2}/\partial z_1)}{d \log q_1} \leq 1$ pointwise with (A42) and contributes at least $-(1 - \alpha) \left(1 - \frac{1}{q_1^R}\right) |C|$ times (A45). The denominator side uses the lower bound on $\frac{d \log|\partial k_{i2}/\partial q_1|}{d \log q_1}$ with (A42) and (A46) and contributes at least

$$- \left[(2 - \alpha) + \frac{2(1 - \alpha)}{1 - \theta} \left(1 - \frac{1}{q_1^R}\right) \right] (1 - \alpha) \left(1 - \frac{1}{q_1^R}\right) \left[1 + \frac{|C|}{(1 - \theta)^2 q_1^R} \right]$$

times (A45), in which the first bracket is at most 2 by (A39). Collecting the three forces with $|C| < \frac{1}{2}$ from (A32), (A39), and (A40),

$$\frac{d}{d\lambda} \log \frac{dq_1}{dz_1} \geq \frac{1}{\lambda} \left[\frac{\alpha}{2} - \frac{5}{2}(1 - \alpha) \left(1 - \frac{1}{q_1^R}\right) - \frac{\alpha}{8} \right] > \frac{1}{\lambda} \left[\frac{\alpha}{2} - \frac{5\alpha}{14} - \frac{\alpha}{8} \right] = \frac{\alpha}{56\lambda} > 0. \quad (\text{A47})$$

Integrating over $[1, \lambda_F]$ yields

$$\left. \frac{dq_1}{dz_1} \right|_{\Phi(q_1^G, G)} < \left. \frac{dq_1}{dz_1} \right|_{\Phi(q_1^F, F)}.$$

Claim (iii): By (30),

$$\frac{d}{d\lambda} \left(\frac{dC_1}{dz_1} \right) = \frac{d}{d\lambda} \int_{i \in C} \frac{\partial c_{i1}}{\partial z_1} \bigg|_{q_1} di + \frac{dq_1}{dz_1} \frac{d}{d\lambda} \int_{i \in C} \frac{\partial c_{i1}}{\partial q_1} di + \left(\int_{i \in C} \frac{\partial c_{i1}}{\partial q_1} di \right) \frac{d}{d\lambda} \left(\frac{dq_1}{dz_1} \right),$$

whose third term is positive since the integral is positive by (A36) and $\frac{d}{d\lambda} \frac{dq_1}{dz_1} > 0$ by Claim (ii). It therefore suffices that the two integrals rise in λ . Both integrands are built, up to factors predetermined at the beginning of period 1, from the same three ratios displayed in (A35) and (A36): the consumption response to the shock is $f(k_{i1})$ times the first ratio, and the consumption response to the price is k_{i1} times the first, $\alpha(1 - \theta)$ times the second, and $(1 - \theta)$ times the third. Each of the two integrals therefore obeys the decomposition of (A43), with no unconstrained boundary term because unconstrained households do not change their consumption, and with the crossing term nonnegative because both responses are positive at the threshold. Only the price-feedback term can be adverse, and by (A42) its magnitude is at most $(1 - \alpha) \left(1 - \frac{1}{q_1^R}\right)$ times the largest positive logarithmic price derivative of the three ratios, relative to the fixed-price term. It is thus enough that each of these three products is smaller than one.

The three ratios share the denominator of (A33)–(A36), whose logarithmic price derivative is nonnegative by (A38) and hence only lowers the ratios' logarithmic derivatives. For the third ratio, the numerator βc_{i1}^2 has a logarithmic derivative $2 \frac{q_1}{c_{i1}} \frac{\partial c_{i1}}{\partial q_1}$, so its product with $(1 - \alpha) \left(1 - \frac{1}{q_1^R}\right)$

is below $\frac{\alpha}{2}$ by (A37) and (A39). For the second ratio, the numerator $\beta f'(k_{i2}) c_{i1}^2$ adds $(1 - \alpha) \frac{q_1}{k_{i2}} \left| \frac{\partial k_{i2}}{\partial q_1} \right|$, whose product with $(1 - \alpha) \left(1 - \frac{1}{q_1 R}\right)$ is below $1 - \alpha$ by (A37), so the product for the second ratio is below $(1 - \alpha) + \frac{\alpha}{2} < 1$.

The first ratio is the one that requires care because its numerator $\beta |f''(k_{i2})| c_{i1}^2$ carries the largest logarithmic derivative. Writing the share of $(1 - \theta)^2 q_1^2$ in the common denominator as the complement of the share of $\beta |f''(k_{i2})| c_{i1}^2$, the ratio's logarithmic price derivative is the former share times $(2 - \alpha) \frac{q_1}{k_{i2}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| + 2 \frac{q_1}{c_{i1}} \frac{\partial c_{i1}}{\partial q_1} - 2$. Bounding the numerator of (A34) by $c_{i1} (\theta c_{i1} + 1 - \theta) + (1 - \theta)^2 q_1 k_{i2}$ splits the first term into a piece proportional to that share squared, which the subtraction of twice the share absorbs since $2 - \alpha < 2$, and a remainder. Hence, the ratio's logarithmic price derivative is at most

$$(2 - \alpha) \frac{(1 - \theta)^2 q_1^3 c_{i1} (\theta c_{i1} + 1 - \theta)}{k_{i2} (\beta |f''(k_{i2})| c_{i1}^2 + (1 - \theta)^2 q_1^2)^2} + 2 \frac{q_1}{c_{i1}} \frac{\partial c_{i1}}{\partial q_1}. \quad (\text{A48})$$

Using $(\beta |f''(k_{i2})| c_{i1}^2 + (1 - \theta)^2 q_1^2)^2 \geq 4 \beta |f''(k_{i2})| c_{i1}^2 (1 - \theta)^2 q_1^2$, then $|f''(k_{i2})| k_{i2} = (1 - \alpha) f'(k_{i2})$ and finally the identity obtained by multiplying the consolidated Euler equation (23) by $c_{i1} / u'(c_{i1})$ and using $\beta = 1/R$,

$$\beta f'(k_{i2}) c_{i1} = q_1 \left[(1 - \theta)(1 - c_{i1}) + \left(1 - \frac{1}{q_1 R}\right) c_{i1} \right],$$

the first term of (A48) is at most $\frac{(2 - \alpha)(\theta c_{i1} + 1 - \theta)}{4(1 - \alpha) \left[(1 - \theta)(1 - c_{i1}) + \left(1 - \frac{1}{q_1 R}\right) c_{i1} \right]}$. For every $c_{i1} \in [0, 1]$, both sides of

$$\left(1 - \frac{1}{q_1 R}\right) (\theta c_{i1} + 1 - \theta) \leq (1 - \theta)(1 - c_{i1}) + \left(1 - \frac{1}{q_1 R}\right) c_{i1} \quad (\text{A49})$$

are affine in c_{i1} , and the inequality holds at $c_{i1} = 0$ and with equality at $c_{i1} = 1$, so it holds throughout. Multiplying (A48) by $(1 - \alpha) \left(1 - \frac{1}{q_1 R}\right)$ and applying (A49) to the first term and (A37) with (A39) to the second, the product for the first ratio is below $\frac{2 - \alpha}{4} + \frac{\alpha}{2} < 1$.

All three products are therefore smaller than one, both integrals rise in λ , and with the positive third term, $\frac{d}{d\lambda} \frac{dC_1}{dz_1} > 0$. Integrating over $[1, \lambda_F]$,

$$\frac{dC_1}{dz_1} \Big|_{\Phi(q_1^G, G)} < \frac{dC_1}{dz_1} \Big|_{\Phi(q_1^F, F)}.$$

This completes the proof.

C.5 First-Order Conditions of the Commitment Taxation Problem

This appendix derives the first-order conditions used in Section 7. I retain the calibrated adjustment cost $\zeta > 0$ and define

$$K_t \equiv \int k_{j,t} dj, \quad (\text{A50})$$

$$B_{t+1} \equiv \int b_{j,t+1} dj, \quad (\text{A51})$$

$$I_t \equiv K_{t+1} - (1 - \delta)K_t. \quad (\text{A52})$$

The capital price and the rebated profit of the capital producers are

$$q_t = 1 + \phi \frac{I_t}{K_t}, \quad (\text{A53})$$

$$\pi_t = \frac{\phi I_t^2}{2K_t} = \frac{(q_t - 1)I_t}{2}, \quad (\text{A54})$$

and

$$q_{K',t} \equiv \frac{\partial q_t}{\partial K_{t+1}} = \frac{\phi}{K_t}. \quad (\text{A55})$$

For brevity, define

$$g_{j,t} \equiv \frac{b_{j,t+1}}{R_t} + \theta_t q_t k_{j,t+1}, \quad (\text{A56})$$

$$M_{j,t} \equiv (1 - \tau_t^b) u'(\tilde{c}_{j,t}) - \Omega_{j,t}, \quad (\text{A57})$$

$$A_t \equiv q_t [1 - \theta_t (1 - \tau_t^b)], \quad (\text{A58})$$

$$X_{j,t}^\zeta \equiv z_t \omega_j f_k(k_{j,t}, n_{j,t}) + q_t (1 - \tau_t^w) (1 - \delta) + \zeta (k_{j,t+1} - k_{j,t}), \quad (\text{A59})$$

$$\Omega_{j,t} \equiv \beta R_t \mathbb{E}_t u'(\tilde{c}_{j,t+1}), \quad (\text{A60})$$

$$\Psi_{j,t} \equiv \beta \mathbb{E}_t \left[u'(\tilde{c}_{j,t+1}) X_{j,t+1}^\zeta \right]. \quad (\text{A61})$$

Let $\lambda_{j,t}$ be the multiplier on household j 's budget constraint, $\gamma_{j,t}$ the multiplier on the investment Euler equation, and $\eta_{j,t}$ the multiplier on the saving Euler equation. Let $\varphi_{j,t}$ be the multiplier associated with a binding collateral constraint, $\kappa_{j,t}$ the multiplier on the intratemporal labor condition, ξ_t the multiplier on the capital-price equation, and $\kappa_{T,t}$ the multiplier on the government budget constraint.

The government's Lagrangian is

$$\mathcal{L} = \mathbb{E}_0 \sum_{t \geq 0} \beta^t \left\{ \int u(\tilde{c}_{j,t}) dj \right.$$

$$\begin{aligned}
& + \int \lambda_{j,t} \left[z_t \omega_j f(k_{j,t}, n_{j,t}) + q_t (1 - \tau_t^w) (1 - \delta) k_{j,t} + b_{j,t} + \pi_t + T_t \right. \\
& \quad \left. - c_{j,t} - q_t k_{j,t+1} - \frac{\zeta}{2} (k_{j,t+1} - k_{j,t})^2 - (1 - \tau_t^b) \frac{b_{j,t+1}}{R_t} \right] dj \\
& + \int \varphi_{j,t} g_{j,t} dj \\
& + \int \gamma_{j,t} \left[(A_t + \zeta(k_{j,t+1} - k_{j,t})) u'(\tilde{c}_{j,t}) + \theta_t q_t \Omega_{j,t} - \Psi_{j,t} \right] dj \\
& + \int \eta_{j,t} M_{j,t} dj \\
& + \int \kappa_{j,t} [z_t \omega_j f_n(k_{j,t}, n_{j,t}) - v'(n_{j,t})] dj \\
& + \zeta_t \left[q_t - 1 - \phi \frac{I_t}{K_t} \right] \\
& + \kappa_{T,t} \left[\tau_t^w q_t (1 - \delta) K_t - \tau_t^b \frac{B_{t+1}}{R_t} - T_t \right] \}. \tag{A62}
\end{aligned}$$

Let $\mathcal{C}_t$ and $\mathcal{U}_t$ denote the sets of constrained and unconstrained households. For $j \in \mathcal{C}_t$, the collateral constraint satisfies $g_{j,t} = 0$. For $j \in \mathcal{U}_t$, the saving Euler equation satisfies $M_{j,t} = 0$. Accordingly, $\varphi_{j,t} = 0$ applies to unconstrained households, and $\eta_{j,t} = 0$ applies to constrained households. The remaining inequalities,

$$M_{j,t} \geq 0 \quad \text{for } j \in \mathcal{C}_t, \quad g_{j,t} \geq 0 \quad \text{for } j \in \mathcal{U}_t, \tag{A63}$$

hold at the solution.

The objects $\Omega_{j,t}$ and $\Psi_{j,t}$ contain date- $t + 1$ marginal utilities. By the law of iterated expectations, the date- t terms involving these objects contribute to the date- $t + 1$ Lagrangian as

$$\int \left[\gamma_{j,t} \left(\theta_t q_t R_t - X_{j,t+1}^\zeta \right) - R_t \eta_{j,t} \right] u'(\tilde{c}_{j,t+1}) dj. \tag{A64}$$

The date- t first-order conditions therefore include the realized effects of the previous period's multipliers.

The first-order condition with respect to consumption is

$$\begin{aligned}
\lambda_{j,t} = u'(\tilde{c}_{j,t}) & + \left[(A_t + \zeta(k_{j,t+1} - k_{j,t})) \gamma_{j,t} + (1 - \tau_t^b) \eta_{j,t} \right. \\
& \left. + \left(\theta_{t-1} q_{t-1} R_{t-1} - X_{j,t}^\zeta \right) \gamma_{j,t-1} - R_{t-1} \eta_{j,t-1} \right] u''(\tilde{c}_{j,t}). \tag{A65}
\end{aligned}$$

Thus, $\lambda_{j,t}$ incorporates both the current value of household wealth and its effects on investment and saving decisions made in adjacent periods.

The first-order condition with respect to labor is

$$\begin{aligned}
0 = & \lambda_{j,t} [z_t \omega_j f_n(k_{j,t}, n_{j,t}) - v'(n_{j,t})] \\
& + \kappa_{j,t} [z_t \omega_j f_{nn}(k_{j,t}, n_{j,t}) - v''(n_{j,t})] \\
& - \gamma_{j,t-1} u'(\tilde{c}_{j,t}) z_t \omega_j f_{kn}(k_{j,t}, n_{j,t}).
\end{aligned} \tag{A66}$$

Because the intratemporal labor condition is imposed as a constraint, its first bracket is zero at the solution. Hence,

$$\kappa_{j,t} = \frac{\gamma_{j,t-1} u'(\tilde{c}_{j,t}) z_t \omega_j f_{kn}(k_{j,t}, n_{j,t})}{z_t \omega_j f_{nn}(k_{j,t}, n_{j,t}) - v''(n_{j,t})}. \tag{A67}$$

The first-order condition with respect to the lump-sum transfer is

$$\kappa_{T,t} = \int \lambda_{j,t} dj. \tag{A68}$$

The value of public funds is therefore the average value that the government assigns to additional household wealth.

The first-order condition for the wealth tax is

$$\int \lambda_{j,t} (K_t - k_{j,t}) dj + \int \gamma_{j,t-1} u'(\tilde{c}_{j,t}) dj = 0. \tag{A69}$$

Because the household measure is one, we have

$$\int \lambda_{j,t} (K_t - k_{j,t}) dj = -\text{Cov}_j(\lambda_t, k_t). \tag{A70}$$

The wealth-tax condition can therefore be written as

$$\underbrace{-\text{Cov}_j(\lambda_t, k_t)}_{\text{redistribution from capital holders}} + \underbrace{\int \gamma_{j,t-1} u'(\tilde{c}_{j,t}) dj}_{\text{return on capital chosen in } t-1} = 0, \tag{A71}$$

which is equation (65) in the main text.

The first-order condition for the debt tax is

$$\frac{1}{R_t} \int \lambda_{j,t} (b_{j,t+1} - B_{t+1}) dj + \theta_t q_t \int \gamma_{j,t} u'(\tilde{c}_{j,t}) dj - \int \eta_{j,t} u'(\tilde{c}_{j,t}) dj = 0. \tag{A72}$$

The first integral is the covariance between the value of household wealth and the household bond position. Hence, we have

$$\underbrace{\frac{1}{R_t} \text{Cov}_j(\lambda_t, b_{t+1})}_{\text{redistribution across bond positions}} + \underbrace{\theta_t q_t \int \gamma_{j,t} u'(\tilde{c}_{j,t}) dj}_{\text{current investment}} - \underbrace{\int \eta_{j,t} u'(\tilde{c}_{j,t}) dj}_{\text{current saving}} = 0, \tag{A73}$$

which gives equation (66) in the main text.

The first-order condition with respect to household bond holdings is

$$(1 - \tau_t^b)\lambda_{j,t} + \tau_t^b \int \lambda_{i,t} di = \varphi_{j,t} + \beta R_t \mathbb{E}_t \lambda_{j,t+1}. \quad (\text{A74})$$

Changing one household's bond position affects its current resources, future wealth, collateral constraint, and the tax revenue returned to all households.

The first-order condition with respect to the capital price is

$$\xi_t = I_t \int \lambda_{i,t} di - (\mathcal{D}_t + \mathcal{K}_t), \quad (\text{A75})$$

where

$$\begin{aligned} \mathcal{D}_t \equiv & (1 - \tau_t^w)(1 - \delta) \text{Cov}_i(\lambda_t, k_t) - \text{Cov}_i(\lambda_t, k_{t+1}) \\ & + \int \gamma_{i,t} u'(\tilde{c}_{i,t}) di - (1 - \tau_t^w)(1 - \delta) \int \gamma_{i,t-1} u'(\tilde{c}_{i,t}) di, \end{aligned} \quad (\text{A76})$$

$$\mathcal{K}_t \equiv \theta_t \int \varphi_{i,t} k_{i,t+1} di - \theta_t \int \gamma_{i,t} M_{i,t} di. \quad (\text{A77})$$

The first two terms of $\mathcal{D}_t$ capture the redistribution generated by revaluing current and future capital positions. The remaining terms account for the investment decisions that the government must implement. The component $\mathcal{K}_t$ captures how a change in the capital price affects collateral constraints and the implementation of household investment.

The first-order condition with respect to household capital is

$$\begin{aligned} & (q_t + \zeta(k_{j,t+1} - k_{j,t}))\lambda_{j,t} \\ = & \theta_t q_t \varphi_{j,t} + \beta \mathbb{E}_t \left[\lambda_{j,t+1} X_{j,t+1}^\zeta \right] \\ & + \zeta(\gamma_{j,t} - \gamma_{j,t-1})u'(\tilde{c}_{j,t}) \\ & + \beta \gamma_{j,t} \mathbb{E}_t \left[u'(\tilde{c}_{j,t+1}) (\zeta - z_{t+1} \omega_j f_{kk}(k_{j,t+1}, n_{j,t+1})) \right] \\ & - \beta \zeta \mathbb{E}_t \left[\gamma_{j,t+1} u'(\tilde{c}_{j,t+1}) \right] \\ & + \beta \mathbb{E}_t \left[\kappa_{j,t+1} z_{t+1} \omega_j f_{nk}(k_{j,t+1}, n_{j,t+1}) \right] \\ & + q_{K',t} (\mathcal{D}_t + \mathcal{K}_t) \\ & + \beta \mathbb{E}_t \left[\xi_{t+1} \frac{(q_{t+1} - 1) + (1 - \delta)\phi}{K_{t+1}} \right] \\ & - \beta \mathbb{E}_t \left[\kappa_{T,t+1} \left((1 - \delta) [q_{t+1} (1 - \tau_{t+1}^w) - 1] + \frac{(q_{t+1} - 1)^2}{2\phi} \right) \right]. \end{aligned} \quad (\text{A78})$$

The term

$$\mathcal{E}_t^q \equiv q_{K',t} [\mathcal{D}_t + \mathcal{K}_t] \quad (\text{A79})$$

is the current price effect reported in equation (64) of the main text. It is absent from an individual household's capital condition because the household takes q_t as given. The last two expectation terms in (A78) collect the effects of current capital on the next period's price equation, capital-producer rebate, and wealth-tax base.

D Effects of Other Shocks in the Tractable Model

D.1 Effects of Financial Shocks

I consider a financial shock that lowers the tightness of the borrowing constraint θ in period 1. I set aggregate productivity to $z_1 = 1$ so that $f(k) = k^\alpha$ in every period and let the period-1 tightness θ be stochastic from the perspective of period 0. The environment is otherwise identical to the one in Section 4, and the existence of the wealth threshold that distinguishes constrained and unconstrained households holds the same as for the productivity shock. I show that a negative financial shock has effects analogous to those of a negative productivity shock, with the difference that its direct effect operates through the households' debt capacity rather than through their output.

Lemma 1. *The unconstrained households demand an identical amount of capital*

$$\bar{k}_2 = (f')^{-1}(q_1 R - 1), \quad \frac{\partial \bar{k}_2}{\partial q_1} < 0. \quad (\text{A80})$$

Households with $\tilde{w}_i < \bar{w}$ are constrained, where

$$\bar{w} = 1 + (1 - \theta)q_1 \bar{k}_2, \quad (\text{A81})$$

and

$$\frac{\partial \bar{w}}{\partial \theta} = -q_1 \bar{k}_2 < 0, \quad \frac{\partial \bar{w}}{\partial q_1} = -\frac{\alpha q_1 R + 1 - \alpha}{(1 - \alpha)(q_1 R - 1)}(1 - \theta)\bar{k}_2 < 0. \quad (\text{A82})$$

Proof. The unconstrained first-order conditions in period 1 are $q_1 u'(c_{i1}) = \beta(f'(k_{i2}) + 1)$ and $u'(c_{i1}) = 1$, which give $\bar{k}_2 = (f')^{-1}(q_1 R - 1)$ and $\partial \bar{k}_2 / \partial q_1 = R / f''(\bar{k}_2) < 0$. Under log utility and $\beta R = 1$, an unconstrained household consumes $c_{i1} = 1$, so its bond choice $b_{i2} = R(\tilde{w}_i - q_1 \bar{k}_2 - 1)$ is increasing in $\tilde{w}_i$. The threshold $\bar{w}$ is the wealth at which this coincides with the binding limit $b_{i2} = -\theta q_1 R \bar{k}_2$, which yields (A81). Differentiating in θ and q_1 gives (A82). $\square$

Unlike the productivity shock, the financial shock raises the threshold wealth directly through $\partial \bar{w} / \partial \theta < 0$, pushing marginal households into the constrained region without any change in output. Combining the first-order conditions with respect to k_{i2} and b_{i2} for a constrained household and totally differentiating with respect to θ , q_1 , and k_{i2} , the direct effect of the financial shock on

asset demand is obtained as

$$\frac{\partial k_{i2}}{\partial \theta} = \frac{q_1(1 - u'(c_{i1})) + (1 - \theta)q_1^2 k_{i2} u''(c_{i1})}{\beta f''(k_{i2}) + (1 - \theta)^2 q_1^2 u''(c_{i1})} > 0. \quad (\text{A83})$$

A negative financial shock lowers constrained households' asset demand because it reduces their debt capacity, generating asset sales. The price elasticity of asset demand is again smaller for constrained households, exactly as in Lemma 2, since the price effect $\partial k_{i2}/\partial q_1$ does not depend on the source of the shock.

The effect of the financial shock on the asset price is obtained from the market clearing condition as

$$\frac{dq_1}{d\theta} = \frac{\int_C \left| \frac{\partial k_{i2}}{\partial \theta} \right| di}{\int_C \left| \frac{\partial k_{i2}}{\partial q_1} \right| di + \int_U \left| \frac{\partial \bar{k}_2}{\partial q_1} \right| di} > 0, \quad (\text{A84})$$

so a negative shock to θ lowers q_1 . Combining this with the partial derivatives of $\bar{w}$, I obtain the effects of a negative financial shock.

Proposition A1. *A negative shock to θ has the following effects.*

1. *The asset price q_1 falls.*
2. *A larger fraction of households is constrained.*
3. *Capital outflow increases and aggregate consumption decreases.*

Proof. The first effect is immediate from (A84). For the threshold, combining the partial derivatives of (A81) with (A84) gives

$$\frac{d\bar{w}}{d\theta} = \frac{\partial \bar{w}}{\partial \theta} + \frac{\partial \bar{w}}{\partial q_1} \frac{dq_1}{d\theta} < 0, \quad (\text{A85})$$

so the fraction of constrained households rises as θ falls through both the direct threshold effect and the general equilibrium price effect. For the third effect, only constrained households change consumption and

$$\frac{dC_1}{d\theta} = \underbrace{\int_C \frac{\alpha q_1 f'(k_{i2}) - q_1^2 R + q_1}{f''(k_{i2}) + (1 - \theta)^2 q_1^2 R u''(c_{i1})} di}_{\text{Partial equilibrium effect}} + \underbrace{\int_C \frac{f''(k_{i2}) k_{i1} - \alpha(1 - \theta) f'(k_{i2}) - (1 - \theta) \frac{dq_1}{d\theta}}{f''(k_{i2}) + (1 - \theta)^2 q_1^2 R u''(c_{i1})} di}_{\text{General equilibrium effect } > 0} > 0. \quad (\text{A86})$$

Unlike the productivity shock, period-1 output does not depend on θ , so the balance-of-payments

identity gives

$$\frac{B_2 - B_1}{R} = -C_1 + Y_1 + \frac{R-1}{R}B_1 \implies \frac{d}{d\theta} \left(\frac{B_2 - B_1}{R} \right) = -\frac{dC_1}{d\theta} < 0. \quad (\text{A87})$$

Hence, a negative financial shock increases capital outflow, in contrast to the ambiguous sign under a productivity shock. $\square$

Finally, wealth inequality amplifies the effect of the financial shock. Let $\Phi(q_1, G)$ denote the distribution of $\tilde{w}_i$ induced by G at price q_1 , and connect G and F by the same proportional family indexed by $\lambda \in [1, \lambda_F]$, as in Proposition 2.

Proposition A2. *Under the hypotheses of Proposition 2, with $\partial k_{i2}/\partial z_1$ replaced by $\partial k_{i2}/\partial \theta$ and condition (19) strengthened to*

$$\bar{K}^{1-\alpha} \geq 1 + (1-\alpha) \left(3 + \frac{8}{1-\theta} \right), \quad (\text{A88})$$

greater wealth inequality amplifies the response of q_1 to the financial shock, and the equilibrium price q_1^F under the more unequal distribution F is lower than q_1^G .

Proof. The argument follows that of Appendix C.4, with the direct effect $\partial k_{i2}/\partial z_1$ replaced by $\partial k_{i2}/\partial \theta$ from (A83). Claim (i) is unchanged since it uses only the shape of asset demand and the family, so $q_1^F < q_1^G$ and the bound (A32) on the constrained mass continue to hold. In Claim (ii), every object built from the denominator of (A84) is unchanged, and $\partial k_{i2}/\partial \theta > 0$ holds throughout the constrained region. Thus, the fixed-price term, the threshold-crossing term, and the unconstrained part of the price-feedback term are signed exactly as before. The one change is the direct-effect bound in the constrained part of the price-feedback term. Multiplying the numerator and denominator of (A83) by $-c_{i1}^2$, we get

$$\frac{\partial k_{i2}}{\partial \theta} = \frac{q_1 c_{i1} (1 - c_{i1}) + (1 - \theta) q_1^2 k_{i2}}{\beta |f''(k_{i2})| c_{i1}^2 + (1 - \theta)^2 q_1^2},$$

a ratio of sums of positive terms since $c_{i1} < 1$. The logarithmic price derivatives of the numerator summands along the household's re-optimization are $1 + \frac{q_1}{c_{i1}} \frac{\partial c_{i1}}{\partial q_1} \frac{1-2c_{i1}}{1-c_{i1}} < 1 + \frac{2}{1-\theta}$ for $q_1 c_{i1} (1 - c_{i1})$, by (A37) and $(1 - 2c_{i1})/(1 - c_{i1}) < 1$, and $2 - \frac{q_1}{k_{i2}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| < 2$ for $(1 - \theta) q_1^2 k_{i2}$. By the convex-combination property behind (A38), $\frac{d \log(\partial k_{i2}/\partial \theta)}{d \log q_1} < 1 + \frac{2}{1-\theta}$ replaces the bound of 1 for the productivity shock, which adds $\frac{2(1-\alpha)}{1-\theta} \left(1 - \frac{1}{q_1 R} \right) |C|$ to the coefficients collected in (A47). Condition

(A88) implies (19) and, by the argument behind (A39) and (A40), delivers $(1 - \alpha) \left(1 - \frac{1}{q_1 \bar{K}}\right) < \frac{\alpha}{11}$, $\frac{1-\alpha}{1-\theta} \left(1 - \frac{1}{q_1 \bar{K}}\right) < \frac{\alpha}{8}$, and $\bar{K} \geq 4 + \frac{8}{1-\theta}$. With $|\mathcal{C}| < \frac{1}{2}$ from (A32), the collection step reads

$$\frac{\alpha}{2} - \frac{5}{2}(1 - \alpha) \left(1 - \frac{1}{q_1 \bar{K}}\right) - \frac{1 - \alpha}{1 - \theta} \left(1 - \frac{1}{q_1 \bar{K}}\right) - \frac{(1 - \alpha) \left(1 - \frac{1}{q_1 \bar{K}}\right)}{(1 - \theta)^2 q_1 \bar{K}} > \alpha \left(\frac{1}{2} - \frac{5}{22} - \frac{1}{8} - \frac{1}{32}\right) > 0,$$

so $\frac{d}{d\lambda} \log \frac{dq_1}{d\theta} > 0$ and the price response is larger under the more unequal distribution. $\square$

As with the productivity shock, wealth inequality deepens the price and consumption contraction under a financial shock by raising the fraction of constrained households before the shock and intensifying the fire sale pressure on the asset market. Table A15 shows the effect of a negative shock on θ_1 when θ_1 decreases from 0.8 to 0.7 in period 1 in two economies with different levels of wealth inequality.

TABLE A15: Numerical Results from the Model with Financial Shock

	Baseline	Alternative
Initial Wealth Gini	0.604	0.695
Period 1 Wealth Gini ($\theta_1 = \theta_H$)	0.570	0.644
Constrained Fraction at Period 1 ($\theta_1 = \theta_H$)	37.44%	49.36%
Constrained Fraction at Period 1 ($\theta_1 = \theta_L$)	45.52%	57.60%
$C_1^L - C_1^H$	-2.70%	-3.46%
$q_1^L - q_1^H$	-1.22%	-1.66%

D.2 Effects of Interest Rate Shocks

I consider an alternative version of the tractable model in Section 4, where I set aggregate productivity to $z_1 = 1$ and impose that the borrowing constraint tightness θ is fixed over time, while the interest rate R_1 in period 1 is stochastic and follows a binary distribution. To be specific, I consider the $R_1 \in \{R_h, R_l\}$ with $R_h > R_l = 1/\beta$. R_1 is R_h with probability p . The rest of the model is identical to the one presented in Section 4. To be specific, I analyze the crisis dynamics in period 1 when the interest rate rises.

An individual household is either constrained or unconstrained in equilibrium. As in the main section, there exists a cutoff level of net wealth across the wealth distribution G that separates constrained and unconstrained households. I denote $\bar{w} \equiv \frac{1}{\beta R_1} + (1 - \theta)q_1 \bar{k}_2$ as the wealth threshold where $\bar{k}_2 = (f')^{-1}(q_1 R_1 - 1)$. For the constrained agents with $\bar{w}_i < \bar{w}$, the binding collateral

constraint implies $b_{i2} = -R_1\theta q_1 k_{i2}$. Combining the first-order conditions with respect to k_{i2} and b_{i2} of the constrained households, I obtain

$$(1 - \theta)q_1 u'(\tilde{w}_i - (1 - \theta)q_1 k_{i2}) = \beta(f'(k_{i2}) + 1 - \theta q_1 R_1), \quad (\text{A89})$$

and capital demand k_{i2} increases with wealth $\tilde{w}_i$. I can decompose the effect of the interest rate shock on the constrained households' asset demand as

$$\frac{dk_{i2}}{dR_1} = \underbrace{\frac{\beta\theta q_1}{\beta f''(k_{i2}) + (1 - \theta)^2 q_1^2 u''(c_{i1})}}_{\text{Direct effect} \equiv \frac{\partial k_{i2}}{\partial R_1} < 0} + \underbrace{\frac{\beta\theta R_1 + (1 - \theta)(q_1 k_{i1} - \tilde{w}_i)u''(c_{i1})}{\beta f''(k_{i2}) + (1 - \theta)^2 q_1^2 u''(c_{i1})}}_{\text{Price effect} \equiv \frac{\partial k_{i2}}{\partial q_1} < 0} \frac{dq_1}{dR_1}, \quad (\text{A90})$$

which is obtained from the total differentiation of (A89) with respect to R_1 , q_1 , and k_{i2} . Lemma 2 still holds, and the asset demand of constrained households is less elastic than that of unconstrained households. Additionally, the partial effect of the interest rate shock on constrained agents' capital demand is smaller in magnitude than its effect on unconstrained agents' capital demand. I can demonstrate this by

$$\left| \frac{\partial k_{i2}}{\partial R_1} \right| = \left| \frac{\beta\theta q_1}{\beta f''(k_{i2}) + (1 - \theta)^2 q_1^2 u''(c_{i1})} \right| < \left| \frac{\theta q_1}{f''(k_{i2})} \right| < \left| \frac{q_1}{f''(k_{i2})} \right| < \left| \frac{\partial \bar{k}_2}{\partial R_1} \right|. \quad (\text{A91})$$

Taking the total differentiation of the asset market clearing condition with respect to R_1 and q_1 , I can characterize the effect of the interest rate shock on the price of assets as

$$\frac{dq_1}{dR_1} = \frac{\int \frac{\partial k_{i2}}{\partial R_1} di}{\int \left| \frac{\partial k_{i2}}{\partial q_1} \right| di} < 0, \quad (\text{A92})$$

Equation (A92) shows that a rise in the interest rate lowers the asset price. Its effect on the fraction of constrained households runs through two opposing channels. The lower q_1 raises the threshold $\bar{w}$ (Lemma 1, $\partial \bar{w} / \partial q_1 < 0$), while the direct effect $\partial \bar{w} / \partial R_1|_{q_1} = -\frac{1}{\beta R_1^2} + (1 - \theta)q_1 \frac{\partial \bar{k}_2}{\partial R_1} < 0$ lowers it, since a higher interest rate reduces both unconstrained consumption $1/(\beta R_1)$ and unconstrained capital $\bar{k}_2$. The net change in the constrained fraction is therefore ambiguous and is resolved numerically in Table A16, where it rises.

The partial derivatives of k_{i2} with respect to q_1 and R_1 are smaller than those of $\bar{k}_2$ with respect to q_1 and R_1 , respectively. Hence, the absolute values of both the numerator and the denominator in (A92) decrease as the fraction of constrained households increases. Thus, whether the asset price drop due to an interest rate shock increases or decreases under greater inequality depends on which one decreases more in absolute terms. While this is ambiguous theoretically, I provide a

TABLE A16: Numerical Results from the Model with Interest Rate Shock

	Baseline	Alternative
Initial Wealth Gini	0.604	0.695
Period 1 Wealth Gini ($R_1 = R_l$)	0.570	0.644
Constrained Fraction at Period 1 ($R_1 = R_l$)	37.71%	58.35%
Constrained Fraction at Period 1 ($R_1 = R_h$)	38.93%	61.63%
$C_1^{R_h} - C_1^{R_l}$	-2.35%	-2.43%
$q_1^{R_h} - q_1^{R_l}$	-1.98%	-2.00%

numerical example under the interest rate shock in Table A16. The drops in aggregate consumption and capital price are larger when the initial wealth distribution is more unequal, resonating with the numerical example in Section 4. The difference between the two initial wealth distributions is smaller than in the productivity-shock example in Table 5 because I am imposing a 2 percentage-point rise of R_1 , which is a smaller shock than a 5% decrease in z_1 in Table 5.

E Computation Algorithms

E.1 Algorithm for the Competitive Equilibrium

I solve for the recursive competitive equilibrium using the Krusell and Smith (1998) algorithm. The aggregate state is (z, R, θ, K) , where z is TFP, R is the world interest rate, $\theta \in \{\theta_H, \theta_L\}$ is the tightness of the borrowing constraint, and K is aggregate capital. The individual state variables are the agent's permanent productivity ω , capital k , and bond holdings b passed on from the previous period. I specify the law of motion for aggregate capital as

$$\log K' = a + b \log K + c \log z + d \log R + e \mathbb{I}(\theta = \theta_L), \quad (\text{A93})$$

and solve for the policy functions that are consistent with the law of motion.

1. Generate discrete grids for the aggregate states and individual states.
2. Initialize the coefficients $\beta^{(0)} = (a, b, c, d, e)$ in (A93).
3. Given $\beta^{(j)}$, solve the household's Bellman equation using value function iteration over the full state space.
 - (a) For each aggregate state, compute K' from (A93), aggregate investment using $I = K' - (1 - \delta)K$, and the price of capital using $q = 1 + \phi I / K$.
 - (b) For each individual state, solve for labor using the intratemporal condition.
 - (c) For each individual state, maximize over (k', b') subject to the budget constraint and the borrowing constraint until convergence.
4. Using the converged policy functions, simulate a panel of N agents for T periods. In each period, compute aggregate capital as the cross-sectional mean, forecast K' from (A93), and update each agent's individual states using the interpolated policy functions. Aggregate states (z', R', θ') are drawn jointly.
5. Estimate (A93) using the simulated data. Update the coefficient to $\beta^{(j+1)}$.
6. If $\|\beta^{(j+1)} - \beta^{(j)}\| < \varepsilon$, the algorithm has converged. Otherwise, return to step 3.

E.2 Algorithm for the Optimal Commitment Tax

Following an approach similar to Le Grand and Ragot (2025), I approximate the government's optimal policy functions to the first order around the ergodic state associated with each value of θ . The state variables include household's capital and bond positions, as well as the multipliers inherited from the previous period, which encode the government's past commitments.

1. Discretize the permanent productivity distribution and the aggregate shocks z , R , and θ .
2. Solve for the competitive equilibrium, and use its allocations and first-order policy functions to initialize the government's problem.
3. Given the current first-order policy functions, solve the government's first-order conditions in Appendix C.5 at the ergodic state associated with each current value of θ .
 - (a) For each realization of future aggregate shocks, use the first-order policy function associated with the future value of θ to compute continuation allocations and multipliers, and construct the corresponding conditional expectations.
 - (b) Solve the full nonlinear system for the current allocation and multipliers, treating the debt tax, wealth tax, and transfers as policy choices.
4. Linearize the government's first-order conditions around the ergodic state associated with each value of θ . Use the resulting policy functions to update the two ergodic states using their implied conditional first moments.
5. Using the converged first-order policy functions, simulate the economy under the sequence of aggregate shocks.